Influences on and consequences of wildlife vehicle collisions and roadkill in Australia

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A thesis submitted in fulfilment of the requirements for the degree of

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Declaration

I hereby declare that this thesis is the product of my own work, and that, to the best of my knowledge, it is original. It contains no material previously published or written by another person except where due acknowledgment has been made in the text. Any help received in preparing this thesis, and all sources used, have been acknowledged. I certify that this submission has not been previously submitted for any degree or qualification at the University of Sydney or other institute of higher learning.

Signed: Bruce Englefield Dated: 29 April 2020

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The four years since commencing my post-graduate studies as a 75 year old student living in Tasmania, remote from the University of Sydney, have proved to be an exciting, challenging and at times frustrating experience, but also very rewarding. I would have been unable to achieve what has been achieved without the support, encouragement and professionalism of many people.

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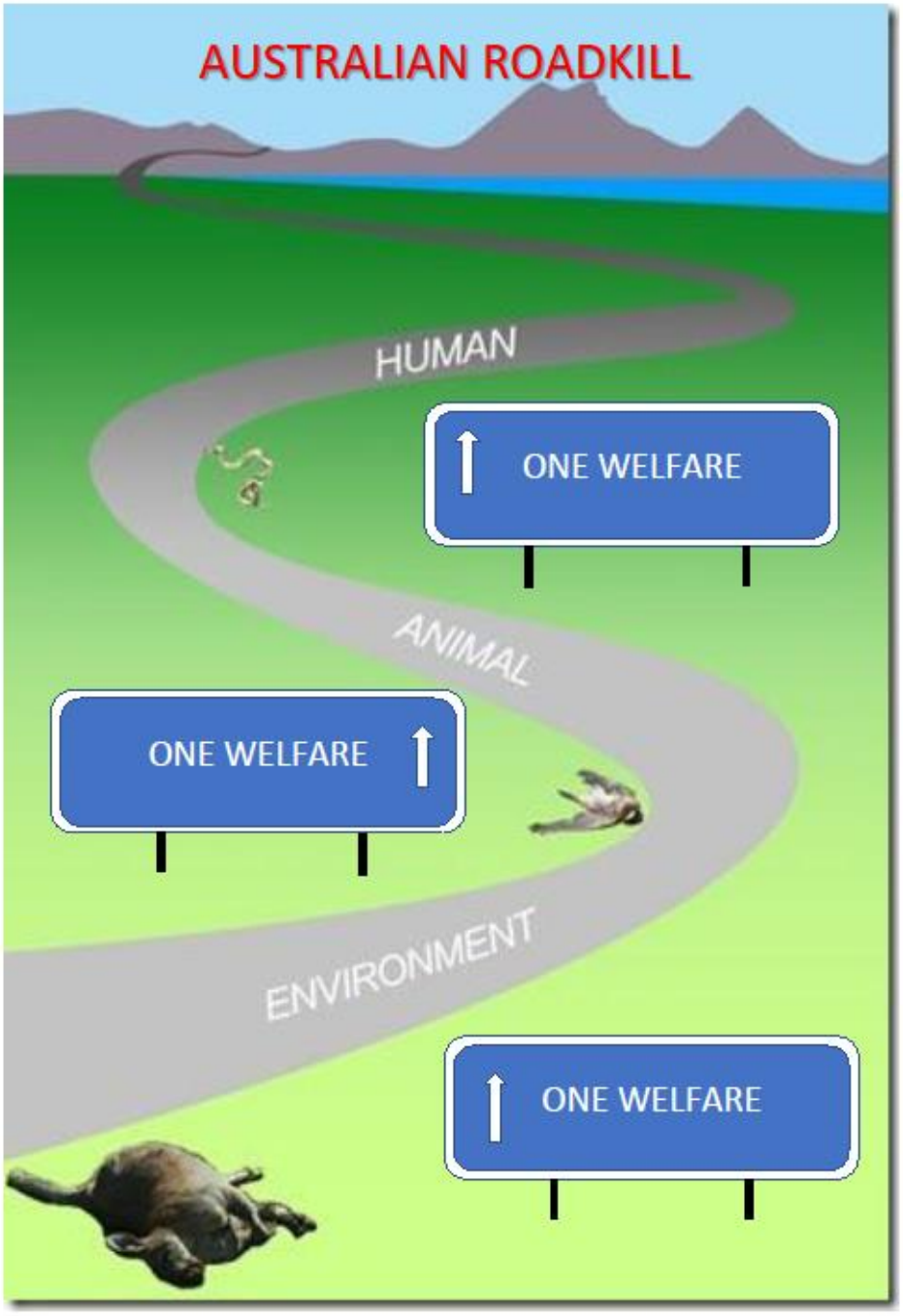
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Figure A1. One Welfare. Roadkill and interaction between humans, animals and the environment



Abstract

Wildlife vehicle collisions (WVC) are an increasing global problem as human habitation and related infrastructure progressively encroaches upon wildlife habitat. They can cause human and animal death, injury and suffering, drive threatened animal species to extinction and deplete established animal populations. They require considerable financial resources to manage and can disrupt transport systems. In Australia, the responsibility for mitigating roadkill is at both a state and local council level. The presence of dead animals resulting from WVC, colloquially called roadkill in most English-speaking locations, that remain strewn across roads and verges, is distressing and unsightly. In areas reliant on tourism income this can have deleterious financial implications. The management of the consequential roadkill and injured and orphaned animals from WVC is complex.

The rescue, rehabilitation, and release or euthanasia of injured and orphaned animals in Australia is managed by a mainly volunteer workforce of approximately 20,000 wildlife carers. They are self-funded within a state system of legislation, regulation and codes of practice. However, their needs and concerns are poorly understood. There are no national data on animal roadkill victims, the long-term viability of the wildlife carer system, the physical, financial and mental wellbeing of wildlife carers, the effect on conservation and animal welfare of the release of rehabilitated and hand-reared animals back to the wild, or the efficacy of technical innovation applied to roadkill mitigation systems. This presents a gap in the knowledge required for focussed and comprehensive decision-making on support for wildlife carer networks, roadkill mitigation measures and changes in wildlife regulations.

One Welfare is an emerging term. The concept refers to animal welfare, human welfare and environmental sustainability. It serves to highlight the interconnections between animal welfare, human wellbeing and the environment, and promotes direct

and indirect links. Roadkill can be placed in the concept of One Welfare as it affects humans, animals and the environment.

The current thesis investigates these effects and the management of Australian roadkill, and injured and orphaned animals across five studies.

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Abbreviations

AAWS	Australian Animal Welfare Strategy
ACT	Australian Capital Territory
AIC	Akaike Information Criterion
AKF	Australian Koala Foundation
AO	Order of Australia
API	Application programming interface
AUD	Australian dollars
AVA	Australian Veterinary Association
CF	Compassion fatigue
CSIRO	Commonwealth Scientific and Industrial Research
Dr	Doctor
DSM	Diagnostic and Statistical Manual of Mental Health
GAM	Generalised Additive Model
GDI	Grief diagnostic instrument
GPS	Global Positioning System
ISAZ	International Society for Anthrozoology
LED	Light emitting diode
LMM	Linear mixed model
MBACI	Multiple Before-After-Control-Impact
MCMC	Markov Chain Monte Carlo
MSC	Master of Science
NAN	Native Animal Network Inc.
NPWS	National Parks and Wildlife Service
NSW	New South Wales
NT	Northern Territory
NWRA	National Wildlife Rehabilitators Association
OIE	Office International des Epizooties
PIT	Passive integrated transponder
QLD	Queensland
RAC	Royal Automobile Club
REDC	Research Electronic Data Capture
RICC	Rescue and Immediate Care Course
RRApp	Roadkill Reporter Application
RSPCA	Royal Society for the Prevention of Cruelty to
SA	South Australia
SE	Standard error
TAS	Tasmania
USA	United Sates of America
UK	United Kingdom
US	Unconditioned stimulus
VF	Virtual fence
VIC	Victoria
WA	Western Australia
WIRES	Wildlife Information, Rescue and Education Service
WVC	Wildlife Vehicle Collisions

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Published papers

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Additional publications completed during candidature

Englefield, B, McGreevy, P (2019) Simply returning rescued wildlife-back-to-the-wild-may-not-be-in-their-best-interest. *The Conversation.* 11th June, 2019. https://theconversation.com/simply-returning-rescued-wildlife-back-to-the-wild-may-not-be-in-their-best-interest-118521

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I was responsible for designing the study and survey, collecting, collating and interpreting data and statistical analysis, writing the drafts of the manuscript, responding to reviewers' comments and coordinating the submission and publication of the manuscript. Steve Candy undertook statistical analysis. My co-

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I was responsible for designing the study, the field work, collecting, collating and interpreting data and assisting the statistical analysis, writing the drafts of the manuscript, responding to reviewers' comments and coordinating the submission and publication of the manuscript. Steve Candy undertook assisting with the study design and statistical analysis. My co-authors, Steve Candy, Melissa Starling and Paul McGreevy, contributed to reviewing and editing the manuscript.

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I was responsible for designing the study and the Roadkill Reporter App, supervising the construction of the App, collecting, collating and interpreting data and assisting the statistical analysis, writing the drafts of the manuscript, and coordinating the submission of the manuscript. Bethany Wilson undertook statistical analysis and Caidryn Roder collated data to produce graphical and text input. My co-authors, Bethany Wilson, Caidryn Roder, Melissa Starling and Paul McGreevy, contributed to reviewing and editing the manuscript.

As supervisor for the candidature upon which this thesis is based, I can confirm that the authorship attribution statements above are correct.

Paul McGreevy 1st March 2020

Chapter 1: Introduction

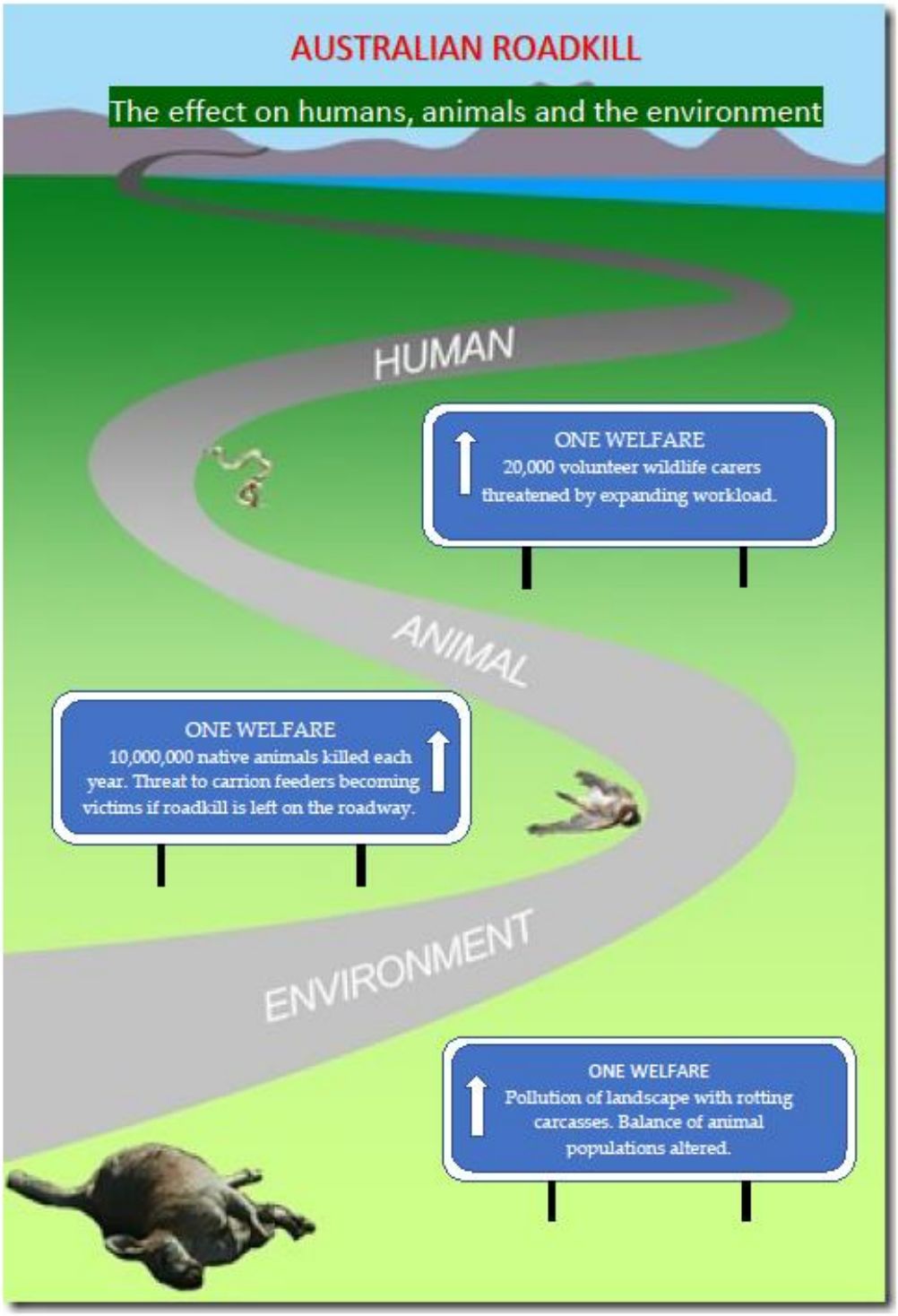
1.1 General introduction

Wildlife vehicle collisions (WVC) and the consequent human and wildlife death and injury are a global phenomenon. A 2008 report to the United States Congress recognised that WVCs are a growing problem and represent an increasing percentage of the accidents on North American roads [1]. Many factors contribute to the increasing burden of WVC. An increasing human population, and the concomitant expansion of urbanisation and subsequent increased travel for work and leisure significantly increase the number of vehicles on the roads. The US Census Bureau International Data Base suggests that the world human population has increased from an estimated 5,278,000,000 in 1990 to a predicted 7,585,000,000 in 2020 [2]. Vehicle numbers increased from 654,000,000 in 2005 to 947,000,000 in 2015 [3]. Infrastructure is required to service the needs of this increase in population and vehicle numbers. The world road network increased by 7% annually from an approximated 18,015,713 km in 2013 [4] to 64,285,009 km in 2019 [5]. In developing countries, for example China and India, the total length of all public roads increased by 570,000 km and 926,455 km respectively from 2010 to 2015 [6, 7]. Even in developed countries, the road networks are still increasing. For example, in Australia, the road network is increasing at approximately 2,000 kilometres per year [8]. Land clearing and habitat destruction to build these roads as well as for industry, housing and farming causes loss of habitat and forces animals to relocate. Collectively these factors feed into an increasing number of WVC and hence animal and human death and injury. Approximately 200 humans die from WVC each year in the USA [1] and an average of 12 human fatalities and 550 injuries per year occurred in the UK between 2000 and 2005 as a result of collisions between vehicles and ungulates [9]. There were more than 5,000 road accidents involving animals in New South Wales in the decade between 1996 and 2005, with 17,000 people injured and 22 people killed [10, 11]. Attewell and Glase (2000) examined Australian

national transport agency data on WVC and reported 1,392 injuries and 94 fatalities to humans from 1990 to 1997.

Roadkill in Australia has been estimated at four million marsupials and six million birds, reptiles and other species each year [12-16]. As well as the effect on human and animal welfare, roadkill also affects the environment, and particularly endangered and threatened species. In North America the Florida panther (*Puma concolor coryi*) [17, 18], in India the tiger *(Panthera tigris tigris)* [19], in Europe the European mink *(Mustela lutreola)* [20] and in Australia the Tasmanian devil *(Sarcophilus harrisii),* eastern quoll *(Dasyurus viverrinus)* [16, 21] and cassowary *(Casuarius* spp.*)* are threatened by the roadkill toll [22]. WVC affect humans, animals and the environment and as such merit consideration under a One Welfare framework.

Figure A1. One Welfare. Roadkill and interaction between humans, animals and the environment



The One Welfare concept considers the welfare of animals, humans and the environment, and proposes optimal outcomes for all. That said, imbalances can occur, such that one sector may gain more than another and so optimal outcomes for all may be elusive. Indeed, in some instances, one sector wins while others lose. For example, to put current habitat destruction into perspective in Australia, the total state-wide clearing of woody vegetation in Queensland rose from approximately 78,000 hectares in 2010 to 400,000 hectares in 2015. This most current annual rate of clearance is an area equivalent to the whole of the state of Tasmania being cleared in a period of 17 years – a clear win, lose, lose situation for humans, animals and the environment respectively. Other situations where humans win while animals and the environment lose can be seen in farming practices that change the environment in ways that encourage wild animals to proliferate. This necessitates the culling of wildlife if a balance is disrupted: another win, lose, lose situation. However, with WVC and roadkill, the current situation represents wildlife loss, human loss and environmental loss. According to the One Welfare philosophy, optimal functionality could be achieved through a major programme of roadkill reduction measures; financial, training and mental health support for wildlife carers; wildlife reintroduction biology research programmes and monitoring roadkill changes through a national database.

In bringing humans, animals and the environment together in a One Welfare concept, a paradigm shift is occurring in no longer regarding non-human animals as property and the environment as a possession. The culture of the indigenous peoples of Australia with regard to the environment is explained in an Aboriginal context. Cassandra Lawton, a Gungarri woman from south-west Queensland, explains 'We belong to the land, the land does not belong to us' and Professor Mick Dodson AM states that 'Territory is defined by spiritual as well as physical links, when we say country we are talking about the whole of the landscape, not just places on it' [23].

In this context the environment could be seen to have rights. A similar argument can be made with non-human animals.

Animal sentience is the ability of an animal to experience feelings. In humans these might be exemplified by contentment, joy, fear, pain or frustration. In animals the concept of sentience relies upon animals being aware of the environment they are in and to learn from experiences. It is only in the last 50 years that sentience in animals has been explored scientifically [24-29] and even more recently enshrined in policy and law. In 2017 the Victorian government recognised animal sentience in its Animal Welfare Action Plan and in 2019 the ACT was the first Australian jurisdiction to change the legal status of animals from being purely 'property' to sentient beings [30]. This could mean a new model will be needed for the way animals and the environment are managed with reference to rescuing, rehabilitating and releasing wildlife to the wild. The mental state of the animals will need to be considered. Also destroying habitat to produce more roads could be seen as an abuse of the right of the environment to have protection. Thus, using a One Welfare approach to exploring aspects of human, animal and environmental welfare involved in the complex issue of roadkill, the current thesis produces an understanding not previously available. It links the following aspects relating to deaths of animals on the roads: effects of roadkill on wildlife carers, the legislation under which the carers operate, roadkill mitigation measures, animal behaviour modification and how native wildlife animals and the environment are affected.

1.2 Central research questions

a. How much yearly roadkill is there in Australia and where are the hotspots?

b. How does roadkill affect the welfare of native wildlife and those caring for rescued animals?

c. Are there constraints, placed on wildlife carers by legislation, that compromise carers' wellbeing and/or that of the animals for which they care?

d. Is it possible to modify native animal behaviour to mitigate roadkill?

e. How sustainable is the current system of managing roadkill?

1.3 Aims of thesis

The overarching aim of this thesis is to examine the complex issues involved in the management of Australian roadkill through a One Welfare lens. Specifically, the objectives of the thesis are to:

a. obtain an understanding of the size of the roadkill problem in Australia
b. estimate the number of wildlife animals rescued, rehabilitated and returned to the wild
c. determine whether a new model based on animal sentience should be applied to the process of rescue and release, where in the One Welfare concept mental needs are accepted as being equally important as physical needs
d. investigate how animal deaths during rehabilitation, working hours, financial commitments and legislation affect the wellbeing of wildlife carers
e. investigate the effectiveness of new technology designed to modify animal behaviour and mitigate roadkill
f. produce a baseline roadkill map of Australia and identify roadkill hotspots.

1.4 Thesis outline

The layout of this thesis is displayed in Figures A1-A6. Study One (Chapter Two) investigated roadkill rescue and its effect on wildlife carers. It concluded that over 4 million Australian mammals become roadkill per year, producing an estimated 560,000 orphans of which up to 50,000 are rescued, rehabilitated and released. In accepting total responsibility for rescued animals, wildlife carers face many physical, financial and psychological stressors.

It postulated that they could be subject to 17 of the 20 recognised types of grief; that they provided, on average, up to the equivalent of AUD31,000 per year of time input and up to AUD2,000 per animal reared. Coupled with the physical demands of caring for the animals, such factors make carers susceptible to burnout and compassion fatigue.

Study Two (Chapter Three) examined the evolution, and explained the consequences of a fragmented, complex, contradictory, inconsistent system of Australian animal welfare regulatory management, as it applies to stakeholders caring for wildlife. It demonstrated how, in most jurisdictions, it is illegal to microchip, band, or mark an animal, meaning that no reliable method is available to monitor animals during captivity or after release. It, therefore, suggests there may be moral, ethical, and practical reasons for not releasing hand-reared orphan native animals to the wild and that the regulatory framework that provides this mandated requirement should be reviewed.

Study Three (Chapter Four) canvassed the opinions of Australian wildlife rehabilitators who cared for rescued marsupials, via an online survey. This contained questions about demographics, motivation, financial and physical input, animal rearing, euthanasia, and release strategy, and a diagnostic instrument that measured grief. Analysis revealed 86% of rehabilitators are female, 70% are over the age of 46 years and their prime motivation for becoming a wildlife carer was having an affinity with animals. On average, length of time in the sector is 11.4 years, the working week is 31.6 hours, caring for 15 animals per year of which 2.6 die. Among respondent carers, 28% were experiencing moderate to severe grief. The average financial commitment is AUD5,300 per year. When measuring grief experienced by wildlife carers, the lower their age, the longer their time caring, the greater their financial input and the greater the number of joeys that died while in their care, all correlated with the severity of the grief experienced.

Study Four (Chapter 5) was a field trial of a commercial roadkill mitigation device that forms a so-called 'virtual fence' (VF) and is reported to reduce roadkill by up to 90%. A total of 126 days of monitoring of roadkill was undertaken along a 4.5 kilometre segment of a Tasmanian highway. Each of a Crossover, Multiple Before-After-Control-Impact and simple On versus Off comparison failed to detect a significant VF effect in reducing roadkill. Adjustment for spatial and temporal trends using a Generalised Additive Model with Poisson had a similar finding. This study failed to confirm previously reported estimates of reduction in roadkill rates claimed for this VF of 50–90%, despite having adequate power to do so.

Study Five (Chapter 6) involved the design and construction of a smartphone app, the Roadkill Reporter (RRApp) to record roadkill. The main criterion for the app was that it should produce reliable and accurate data that did not rely on professional scientific expertise or IT skills to undertake recordings. The RRApp can be installed free of charge into iPhone or androids. It enables both professional and citizen scientists to take a photograph of roadkill that is GPS-, time- and date-stamped before being uploaded to a website. Citizen scientists were engaged to record Australian roadkill in a project with both short-term and extended aims. The short-term aims have been realised with baseline data from three months' observations showing the extent of roadkill across Australia. Also, numerous roadkill hotspots have been identified, which means that local authorities can source this information and maximise financial resources by concentrating their efforts to mitigate roadkill at these hotspots. The national and local publicity that accompanied the launch of the RRApp successfully achieved another aim of the project, which was to bring the problem of roadkill further into the public domain. The extended aim is to be able to monitor changes in roadkill that may give indications of possible wildlife decline, particularly in threatened species, and provide information on animal migration and efficacy of roadkill mitigation measures, and continue to promote public awareness.

The final chapter of the thesis summarises and discusses all the information and research presented and suggests avenues of future research.

1.5 Avenues of approach to research

Using the One Welfare framework to explore how the welfare of humans, animals and the environment feed into the complex issue of roadkill, the thesis employs several avenues of approach. It uses:

a. literature reviews to analyse how WVC and roadkill, wildlife rescue and subsequent legislation evolved and to suggest the effect they may have on the wildlife volunteers rescuing injured or orphaned wildlife

b. an Australian online survey to investigate wildlife carers' experiences, knowledge and attitudes

c. a novel experimental design using a modified before, after, control and impact (BACI) design, and four methods of statistical analysis to investigate the efficacy of roadside roadkill mitigation devices

d. a novel mobile application that enables citizen scientists to provide reliable data on Australian roadkill. It also acknowledges the limitations of examining an issue as complex as roadkill and suggests where future research needs to be directed.

Chapter 2: Roadkill rescue

2.1. Preamble

One aim of the research undertaken was to obtain an understanding of the extent of Australian roadkill. Surprisingly, a literature review revealed that there are no national data on roadkill. There are data on wildlife vehicle collisions (WVC), produced by the car insurance agencies, but these are restricted to damage to vehicles or humans caused in an accident with native animals, so lack veracity on the numbers of animals killed or injured [31, 32]. There are also studies that have been conducted within the individual states and territories, but only one of these evaluates the roadkill in the whole of a state, that of Tasmania [13]. However, an estimate of Australian roadkill numbers was produced by combining data from these state-based studies, information from companies manufacturing replacement milk substitute quantifying the amount supplied and, therefore, number of mammals that had been rescued and were being raised, and data from the RSPCA and wildlife carer networks on animals that had been rescued as orphans or were injured. Once this approximation was produced it was possible to understand the size of the operation required to manage the injured and orphaned animals resulting from WVC.

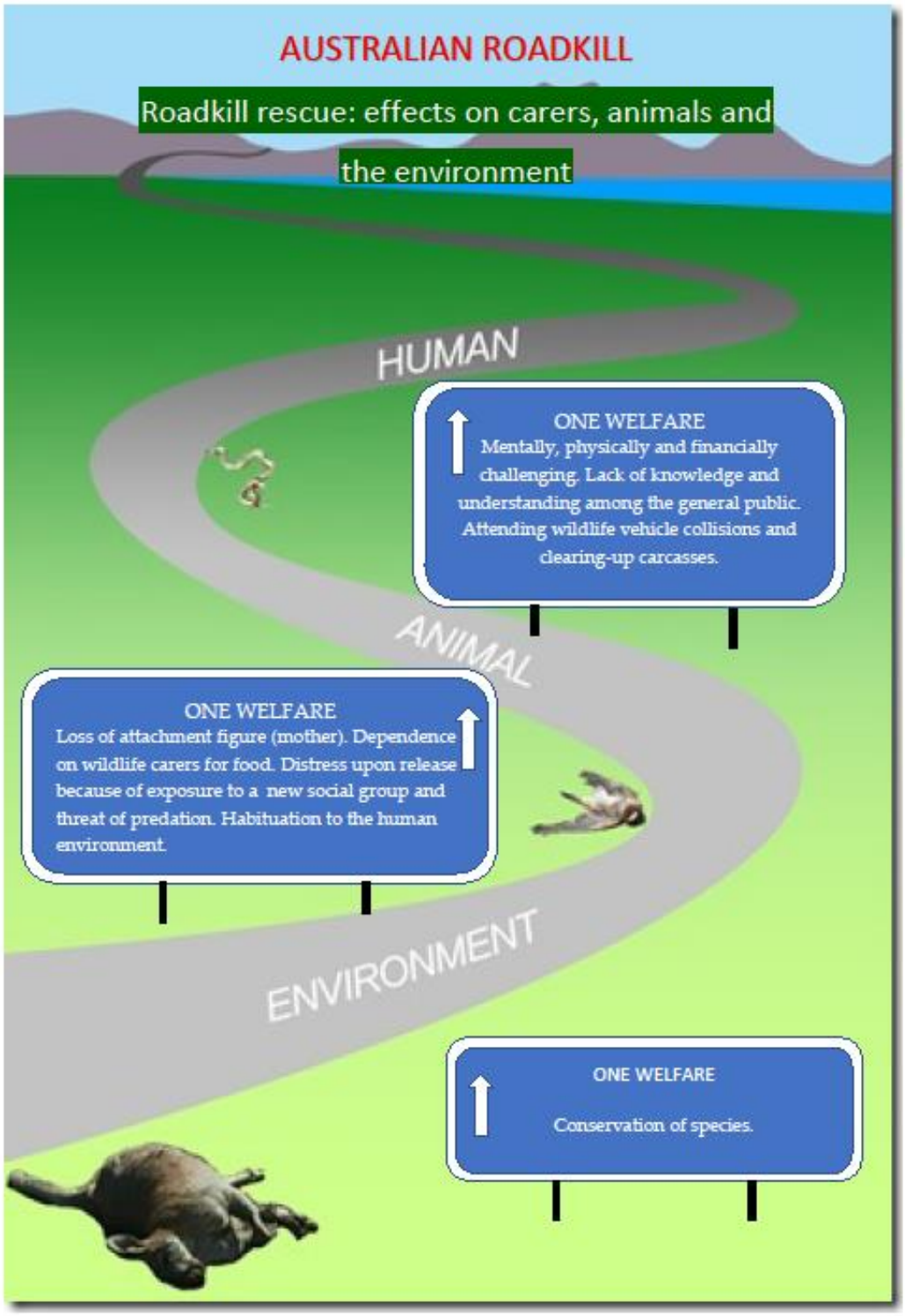
There are over 20,000 wildlife carers, mainly volunteers, who work to mitigate the environmental loss of Australian wildlife as a result of WVC and to alleviate the suffering of injured animals. They rescue, rehabilitate and release to the wild injured and orphaned wildlife resulting from WVC, as well as from anthropogenic and naturally occurring events. This work is undertaken at their own expense both in time and financial input and, in most states, they are even required to pay for their own training. It is interesting to note that the only state in which a comprehensive study of roadkill numbers has been conducted, Tasmania, is the only state where the

wildlife carers are managed by a department funded by the state, the wildlife management branch of the Department of Primary Industries, Parks, Water and Environment. They work in conjunction with the Tasmanian Rehabilitation Council, a not-for-profit organisation, and wildlife carers receive training without having to pay a fee – a model worth examining in other states and territories.

There is a knowledge gap in some aspects of the work of wildlife carers. Information is lacking on the number of roadkill rescues and rescues from other events, such as bush fires, attacks by companion animals, hunting and culling, and on how this work of rescue, rehabilitation and release affects the welfare of the wildlife carers, particularly their mental wellbeing. Another area where information is missing is how all the legislation, regulations and codes of practice that affect wildlife carers are implemented on a state-by-state basis, without any coherent national policy.

Chapter 2 presents a review that explores the rescue of injured and orphaned wildlife, the possible effect that this has on the wellbeing of the carers and whether their welfare is being managed appropriately. It quantifies the physical, financial and mental stressors that carers may face. It hypothesises about the different types of grief they may experience and how these could impact their work and mental health. The review has been published in *Wildlife Research*, the Commonwealth Scientific and Industrial Organisation journal.

Figure A2. One Welfare. Roadkill rescue



2.2. Published paper: A review of roadkill rescue: who cares for the mental, physical and financial welfare of Australian wildlife carers?

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Review

A review of roadkill rescue: who cares for the mental, physical and financial welfare of Australian wildlife carers?

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Abstract. The non-human animal deaths and injuries that result from collisions with motor vehicles are known colloquially as roadkill, and often lead to individuals from various taxa being orphaned. The complexities of multiple spatial and temporal variables in the available data on Australian roadkill and the scale of orphaning and injury make statistical analysis difficult. However, data that offer proxy measures of the roadkill problem suggest a conservative estimate of 4 million Australian mammalian roadkill per year. Also, Australian native mammals are mainly marsupial, so female casualties can have surviving young in their pouches, producing an estimated 560 000 orphans per year. A conservative estimate is that up to 50 000 of these are rescued, rehabilitated and released by volunteer wildlife carers. These roadkill-associated orphans are in addition to those produced by other anthropogenic and natural events and the injured adult animals in the care of volunteers. In accepting total responsibility for rescued animals, wildlife carers face many demands. Their knowledge base can require days of initial instruction with the need for continual updates, and their physical abilities and personal health can be tested by sleepless nights, demanding manual tasks and zoonoses. This review article explores the impact of this commitment and conservatively estimates carers' financial input to raise one joey at approximately $2000 a year, and their time input at 1000 h, equating to $31 000 per year, applying a dollar value of $31 per hour. It categorises relevant types of grief associated with hand-rearing orphans and rehabilitating injured animals, and suggests that wildlife carers most likely experience many types of grief but are also susceptible to burn-out through compassion fatigue. A perceived lack of understanding, empathy and appreciation for their work by government can add to the stressors they face. Volunteering is declining in Australia at 1% per year, social capital is eroding and the human population is aging, while the number of injured and orphaned animals is increasing. Wildlife carers are a strategic national asset, and they need to be acknowledged and supported if their health and the public service they provide is not to be compromised.

Additional keywords: carer burnout, carer stressors, orphaned wildlife, roadkill numbers, wildlife carer. Received 13 July 2017, accepted 21 February 2018, published online 4 May 2018

Introduction

The advent of motorised transport has not only altered environments but also presents a unique primary threat to wildlife. Human dependence on vehicular transport has had considerable influence on faunal populations across the world (Trombulak and Frissell 2000; Davenport and Davenport 2006; Coffin 2007). For example, in Europe, roadkill is the main cause of the decline of the critically endangered European mink (*Mustela lutreola*) (Palazón *et al*. 2012). Cars are among the biggest threats to the highly endangered Florida panther (*Puma concolor coryi*), with 10–15 panthers killed by vehicles each year (Fergus 1991; Waymer 2014) and, from the International Union for Conservation of Nature Red List, in less than a year, one tiger (*Panthera tigris*) and two leopards (*Panthera pardus*) were killed by speeding vehicles on the state highway at Sariska Tiger Reserve, Alwar, Rajasthan India (Baskaran and Boominathan 2010). In Australia, roadkill can be a tipping point and represent a threshold beyond which there are serious threats to species as the result of disease, habitat destruction or anthropogenic events. Examples are the endangered Tasmanian devil (*Sarcopholus harrisii*) (Jones 2000; Hobday and Minstrell 2008), endangered koalas (*Phascolarctos cinereus*) (Lunney 2013; McAlpine *et al*. 2015) and threatened wombats (*Vombatus ursinus*) (Roger *et al*. 2007).

Dead and injured animals are constantly found along Australian roads. The density of roadkill can reach as high as one dead animal every 3 km per day. As disturbing as this number is, the survey from which this statistic is drawn (Hobday and Minstrell 2008) included only those seen dead on the road or verge from a vehicle and not in the adjacent vegetation. There may be many physical, environmental and cognitive factors influencing the behaviour of both animals and human drivers involved in such incidents, and these contribute to the fundamental issue of drivers and wildlife being unable to adequately predict an imminent collision and act to avoid it. These lethal interactions involve most taxa and produce what has become known as roadkill. Non-lethal interactions often injure animals and may lead to a slow death unless victims can be rescued or euthanased. Managing the rescue and rehabilitation of animals injured by vehicles, as well as by other anthropogenic events, is a huge undertaking. In the United States, members of the National Wildlife Rehabilitators Association (NWRA) treat hundreds of thousands of wildlife animals annually and more than 75% of the animals cared for have been directly affected in some manner by human activities (NWRA 2017). In the United Kingdom, data from 2013 indicate that the Royal Society for the Prevention of Cruelty to Animals (RSPCA) UK, treated more than 15 000 sick, injured or orphaned wildlife animals (RSPCA 2017), while in Australia, 25 568 native wildlife animals were presented to the RSPCA Australia from July 2015 to June 2016 (RSPCA Australia 2016).

Generally, when pregnant or nursing females are killed by road traffic or by other anthropogenic means (including cat and dog attack, gunshot, habitat challenges and deliberate acts of cruelty), their offspring die with them, either by fatal injury at the time, or subsequent starvation in the nearby environment or maternal den. However, in Australia and parts of South America, deaths of lactating females present a unique problem. Australia has 357 indigenous mammal species, of which 159 are marsupials (Van Dyck and Strahan 2008; Chapman 2009). When marsupial females are killed, the young in the pouch often survive the mother and may be sufficiently developed to be hand-reared. There may also be joeys that have left the pouch but are still dependent on the mother for feeding and so stay near her. A decision must be made as to what to do about these orphans, if they are found while still alive. The options available are non-intervention, euthanasia or rescue and rehabilitation.

Some ecological theorists interpret animal rights theory as a 'hands-off' approach and decry human interference in the processes of the natural world (Bekoff and Hettinger 1994; Hettinger 1994; Francione 2008; Rolston 2012). This approach has been described as the 'laissez-faire intuition'

(Palmer 2010), but its critics argue that if it is accepted that wild animals have an inalienable right to life, then non-intervention and letting nature take its course is neither morally nor ethically acceptable (Callicott 1980; Cohen and Regan 2001). This could be seen as particularly valid when humans have caused the harm by roadkill, agricultural practices (culling) or by keeping uncontrolled domestic pets (predation). Those involved in wildlife rehabilitation argue that there is a moral obligation to give anthropogenic casualties a second chance wherever possible (McKeever 1979; Thomas 1993; Hutto 1995). Rescuing joeys from the pouch of a dead or injured mother could be seen as positive intervention that has the potential to remediate a human- caused harm.

Euthanasia of animals, perhaps especially where it concerns native species, is a sensitive and emotive topic and can raise vigorous public debate. The recent global outrage on social media when Harambe, a male western lowland gorilla (*Gorilla gorilla gorilla*) at Cincinnati Zoo, was shot and killed because a child's life seemed in danger (McCabe 2016) demonstrates the sensitivity of using euthanasia, even as a considered reaction to a human life-threatening event. RSPCA Australia policy on orphaned wildlife contends that only where rehabilitation and release is unlikely to be successful, should the animal be humanely euthanased (RSPCA Australia 2008). It is also questionable whether veterinarians, registered with the Australian Veterinary Association (AVA), could undertake euthanasia of a viable, healthy joey or injured animal that could be rehabilitated, without breaching the AVA's code of professional conduct (Australian Veterinary Association 2018: p.1, line 1) which states 'The community and your clients are entitled to expect that you will always consider the health, welfare and respectful treatment of the animal'.

If non-intervention and euthanasia are discounted, then rescue-and-rehabilitation is the remaining option. The current Australian environmental, ethical and legal frameworks for these orphaned animals involve their being hand-reared by volunteer wildlife carers, treated as wild animals and eventually returned to the wild, or held in captivity in perpetuity. This can be seen as a public service, as the financial cost for these frameworks is borne largely by the wildlife carers themselves.

There are no national data on roadkill numbers, orphaned or injured animals, the number of registered and unregistered wildlife carers, the period wildlife carers spend as carers, the rate of recruitment and resignations of wildlife carers; nor is there any monitoring of the survival rate and welfare of animals that are released. The national data collected by vehicle companies or RAC are animal-related crash data rather than road-kill statistics.

A few state-based studies hint at the scale of the problem presented by anthropogenic marsupial deaths and injuries (Guy and Banks 2010). Unfortunately, they tend to target differing species, animal densities, types of road, States and Territories, years, traffic volumes, traffic speeds and seasonal patterns. With inconsistent treatment of so many variables in different datasets, and roadkill fatalities being points in both time and space, the complex process of statistical analysis is very challenging, if not impossible.

For the current review, it was decided that conservative estimates would be sufficient to demonstrate the size of the roadkill problem, and so evaluate the merits of data that may act as possible proxy measures of the scale of roadkill-related injuries and the numbers of consequent deaths and orphans. There is little published information on the public service provided by wildlife carers to Australia in raising and rehabilitating orphan animals and rehabilitating injured animals. This leaves a large knowledge gap when we try to understand the size, complexity and sustainability of the present system of managing wildlife casualties.

This review uses the data available in the primary literature on the frequency of roadkill in certain Australian states and provides an estimate of the numbers of marsupial roadkill and how many orphans may require care as a result of roadkill. These figures, combined with an estimate of the financial costs of raising orphaned pouch-young and the number of active carers across the country, provide for the first time an approximation of the financial burden of raising orphaned pouch-young Australia-wide. The review goes on to describe the other facets of the possible impacts on wildlife carers, including the potential physical demands and psychological costs associated with the task.

Methods

A systematic search was performed of the Australian research literature from 1973 until 2017. Data were obtained using the search terms 'Australian wildlife roadkill', 'road crashes involving wildlife' and 'animal road mortality' in the University of Sydney library cross-search engine and Google Scholar. From this sweep, 14 studies of Australian mammalian roadkill were identified and extracted (Table 1). Particular emphasis was placed on findings from studies that included all or most mammal species extant in the study area, covered at least 50 km of roadway, and with data collected over at least a year. A calculation was made for the roadkill per km per day using data on the reported number of animals killed in these studies, the period over which the study took place and length of road under observation. The lowest rates obtained for roadkill per km were then extrapolated upwards, using the total road length in Australia, to arrive at a conservative estimate of the number of Australian mammalian roadkill per year. This figure needed further adjustment because the roadkill data in previous studies were recorded from vehicles travelling along highways. Surveys conducted on foot through the road verges and adjacent bush areas revealed that a further 30% of the number of roadkill observed from vehicles remain unrecorded by these studies (Hobday and Minstrell 2008; Knowler 2015; N. Mooney, wildlife biologist, pers. comm. 2017).

No national data are collected on the number of animals orphaned as a result of anthropogenic events. Several sources were consulted to obtain an estimated number, including the RSPCA Australia, not-for-profit wildlife carer organisations (Wildlife Rescue Service New South Wales, Wildlife Victoria, Native Animal Network of South Australia, Wildlife Preservation Society Queensland, Western Australian Wildlife Rehabilitation Council Inc., Tasmanian Wildlife Rehabilitation Council, Australian Capital Territory Wildlife, Wildcare Inc. Northern Territory), individual wildlife carers, manufacturersof replacement milk powder for rearing orphaned joeys (Wombaroo, Biolac, Passwell) and a roadkill survey (Department of State Growth 2017). Information was received on the number of joeys rescued, the number of joeys released, the number of active wildlife carers in Australia (Table 2), the quantity of milk powder produced (which was received as business-in-confidence information) (Table 3), the quantity required to raise joeys of various species (Table 4) and the ratio of rescued joeys to number of adult roadkill deaths (Department of State Growth 2017). Separate estimates were made using each dataset. A final estimate was obtained by taking an average of the results.

Representative organisations from each State were each contacted to obtain approximate data on the number of registered wildlife carers in Australia (Table 2). Specifically, the president or secretary of each registered rehabilitation centre or rehabilitation network organisation was asked for membership numbers. For those states (Western Australia and South Australia) with no centralised carer database, carers were contacted on a group or individual basis.

Details of the qualifications required of wildlife carers were sourced from individual carers,

carer organisations, food manufacturers, equipment suppliers, websites and personal experience from being a wildlife carer over a 10-year period. These data were collated to estimate the workload and financial costs involved in being a wildlife carer (Table 5).

The risks for wildlife carers suffering psychological stress while caring for injured, traumatised or suffering animals were investigated by conducting a search of peer-reviewed international literature (from 1989 to 2016), using the search terms 'grief', 'grief and loss', 'grief therapy', 'grief counselling' and 'definition of grief' in the University of Sydney library's cross-search engine and Google Scholar. Papers that discussed at least eight of the most commonly listed types of event that could cause human grief were examined. The types of grief reported in these papers were then tabulated against definitions for 18 recognised types of grief (Table 6). Finally, using these definitions, the different types of grief that could be experienced by a wildlife carer were predicted and comparisons made with the other types of grieving events and their resultant types of grief, to demonstrate how wildlife carers could experience 15 of the 18 defined types of grief. Compassion fatigue or secondary post-traumatic stress disorder may well affect wildlife carers, so a further search was conducted using the same search engines and the search terms 'compassion fatigue', 'traumatic events' and 'caregiver burnout'. These papers were examined to explore how compassion fatigue may affect wildlife carers.

Results

Calculating conservative estimates of marsupial roadkill

Using data from studies on roadkill conducted on a limited scale within individual Australian states (Table 1), specifically the column for average roadkill per km per day, the estimates for marsupial roadkill range from 0.0041 per km per day (footnote A), see Department of State Growth 2017; Table 1, on a remote Tasmanian road with little traffic (Department of State Growth 2017) to 0.074 per km per day (footnote C), see Hobday and Minstrell 2008; Table 1, on Tasmanian urban and rural highways with moderate-to-heavy traffic (Hobday and Minstrell 2008). In 2012, the total network length of both sealed (bitumen) and unsealed (gravel) roads in Australia was 900 082 km (D), (Bureau of Infrastructure, Transport and Regional Economics 2014). Local rural roads make up 60% of Australian road infrastructure, with 14% being local urban roads, 23% rural highways and arterial roads, and 3% urban highways and arterial roads. Therefore, with 60% of Australian roadways similar to (footnote A), a remote Tasmanian road with little traffic and (footnote B), a NSW rural road, then a figure of 0.011 per km per day (E), (midway between (footnote A) 0.0041 and (footnote B), 0.018) is favoured to produce a conservative estimate of marsupial roadkill. Using 0.011 per km per day, the estimated yearly figure for total Australian marsupial roadkill is 3 600 000 (i.e. E D 365). However, these data are for roadkill reported as present on the road or verges, counted from a vehicle, and do not include animals that have managed to move out of sight before dying of their injuries. To obtain these data, several roads were surveyed in Tasmania between 2002 and 2008 using a vehicular drive-through; they were then resurveyed until no more previously unseen roadkill were recorded. Then the verges were searched on foot. (N. Mooney, wildlife biologist, pers. comm. 2017). The number of new roadkill was estimated at a further 30% of the number of roadkill recorded from vehicle observation. Being conservative again by using only 15% (half the 30% factor) to further extrapolate the previous figure of 3 600 000 marsupial roadkill, the figure 4 140 000 is obtained. It is therefore concluded that a conservative estimate of marsupial roadkill in Australia exceeds 4 million per year.

Calculating an estimate of rescued joey orphans

The percentage of female marsupial roadkill that carry pouch- young is largely unrecorded. A roadkill monitoring report for the Tarkine Drive upgrade in Tasmania (Department of State Growth 2017) indicates the presence of viable joeys in ~14% of roadkill marsupials endemic to this area: the Tasmanian devil (*Sarcophilus harrisii*); Tasmanian pademelon (*Thylogale billardierii*); Bennetts wallaby (*Macropus rufogriseus*); common wombat (*Vombatus ursinus*); and the brushtail possum (*Trichosurus vulpecula*). However, no female marsupials carrying multiple joeys were recorded in the roadkill, despite this being expected in several species, including Tasmanian devils and quolls (*Dasyurus*). This outcome seems unlikely and may reflect insufficient pouch-checking by those monitoring the roadkill. If this were the case, such an omission would have reduced the reported percentage considerably. Proposing a national figure from just one small dataset is not without risk, but applying the Tarkine Drive percentage (14%) to the conservative 4-million estimate of roadkill (see above), gives an estimate of 560 000 pouch-young that may survive their mothers' deaths. The rural nature of most Australian roads (with 60% being remote and unpaved), the 30% of roadkill unseen from the road (see above) and the inherent lack of pouch-checking leads us to surmise that 85–95% of these young die before rescue. Thus, we estimate that between 28 000 and 84000 roadkill-related orphans might be rescued.

Milk replacer and number of orphans it would feed

After being approached by phone, the three major Australian manufacturers of milk replacer supplied data by email or verbally on the full understanding that it was 'commercial-in-confidence' information. Collated data show that ~127 tonnes of milk- replacement products are produced to hand-feed and rear joeys to the stage of weaning each year. Table 3 presents the maximum amount of milk replacer needed to rear a joey of each of the commonly hand-reared marsupial species from its minimum viable weight at rescue to a weight ready for weaning. Wallabies and kangaroos, and brushtail and ringtail possums are fed the same formula milk powder. The ratio of wallaby roadkill to kangaroo roadkill is ~10 : 1 and that of brushtail to ringtail possums 100 : 1 (Lee *et al*. 2004; Leeuwenburg *et al* 2004; Taylor and Goldingay 2004; Ramp *et al*. 2005; Hobday and Minstrell 2008). Applying these ratios, and assuming that joeys are rescued in the same ratio, ~97 000 kg of milk replacer could be fed to wallaby joeys and 10 000 kg to kangaroo joeys per year (10 : 1). Similarly, 6930 kg of milk replacer could be fed to brushtail possums and 70 kg to ringtail possums per year (100 : 1). When rescued, many joeys weighing more than the minimum viable weight have already received milk from their mothers, so require less time, and therefore less milk than estimated here, to rear them to weaning. Approximately 5–10% of joeys die early during the rearing process (milk manufacturer 2017, pers. comm; Tribe and Brown 2000; Department of Primary Industries Water and the Environment Tasmania 2017 pers. comm.), meaning that, relative to completed rearing, smaller volumes of milk may be consumed as a result of these deaths and can therefore be discounted. The cost of the powder and protracted 'best before' date of 18 months on milk replacer mean that wastage or stockpiling by carers is <1% and can also be discounted.

Using these data, and assuming minimal wastage, the milk powder produced is estimated to be sufficient to hand-rear a minimum of 40 000 joeys a year. The number of joeys that are reared using products other than milk powder substitute, e.g. goats' milk is discounted.

Table 1. Australian marsupial wildlife roadkill showing variables involved in producing an estimate of average roadkill per km per day Note that Osawa (1989), Driessen *et al*. (1996), Mallick *et al*. (1998), and Ramp and Ben-Ami (2006) reported on only one species; Coulson (1982) reported on only two species. For some papers, the set of species indicated for each study is a minimum set rather than the full set. MBP, mountain brushtail possum (*Trichosurus caninus*); B, bettong (*Bettongia*); NBB, northern brown bandicoot (*Isodon macrourus*); BFF, black flying fox (*Pteropus alecto*); P, pademelon (*Thylogale*); BTP, brushtail possum (*Trichosurus vulpecula*); RK, red kangaroo (*Macropus rufus*); BW, Bennett's wallaby (*Macropus rufogriseus rufogriseus*); RNW, red-necked wallaby (*Macropus rufogriseus*); CW, common wallaroo (*Macropus robustus*); RTP, ring-tail possum (*Pseudocheirus perigrinus*); EBB, eastern barred bandicoot (*Perameles gunnii*); SBB, southern brown bandicoot (*Isoodon obesulus*); EGK, eastern grey kangaroo (*Macropus giganteus*); STQ, spotted tail quoll (*Dasyurus maculatus*); EQ, eastern quoll (*Dasyurus viverrinus*); SW, swamp wallaby (*Wallabia bicolor*); GHFF, grey-headed flying fox (*Pteropus poliocephalus*); TDV, Tasmanian devil (*Sarcophilus harrisii*); K, koala (*Phascolarctos cinereus*); W, common wombat (Vombatus ursinus); LNB, long-nosed bandicoot (*Perameles nasuta*); WGK, western grey kangaroo (*Macropus fuliginosus*)

Author/originator	Location/date	Sampling rate/ Number of days	Length of road km	Traffic density vehicles per day	Speed limit km h^{-1}	Environment/ type of road	Species	Roadkill number	Average roadkill per km per day
Vestjens 1973	ACT NSW 1970–1972	Monthly 24	301.0	n/a	n/a	Rural/urban Bitumen/gravel	BTP, RTP, EGK, RK	72	0.010
Coulson 1982	Vic 1975–1980	Periodic 124	20.0	~1000	100	Rural bitumen	EGK, SW	32	0.013
Osawa 1989	Qld 1981–1982	Monthly 365	23.0	n/a	100	Rural–urban transition bitumen	SW	127	0.015
Driessen *et al*. 1996	Tas 1992–1996	Quarterly 1 149	1 699.0	n/a	50–100	Rural bitumen	EBB	2 350	0.0012
Mallick *et al*. 1998	Tas 1992–1996	Quarterly 180	96.0	n/a	80	Rural bitumen	EBB	256	0.015
Klöcker *et al*. 2006	NSW 2002	Every other day 168	21.2	~50	100	Rural bitumen	WGK, EGK, RK, CW	125	0.035
Taylor and Goldingay 2004	NSW 2000–2001	Weekly 120	100.3	5000–20 000	60–100	Rural bitumen	NBB, LNB,BTP, K,RTP, SW, BFF, GHFF	211	0.018[B]
Ramp *et al*. 2005	NSW 1998–2003	Twice daily 2 190	40.0	~500	100	Rural bitumen	EGK, SW, RNW, W	2 916	0.033
Giffney *et al*. 2009;	NSW 2004–2006	Bi-weekly 104	80.0	n/a	50–100	Urban bitumen	BTP, RTP	591	0.07
Ramp and Ben-Ami 2006	NSW 2003	Daily 143	22.0	~3000	60–80	Rural–urban transition bitumen	SW	14	0.0045
Bond and Jones 2008	Qld 2004	Twice weekly 122	1.3	n/a	70	Rural bitumen	RNW, SW, RTP, NBB	6	0.038
Hammond 2008	Tas 2007–2008	Every 4 days 91	21.7	n/a	100	Rural bitumen	BW, B, BTP, P, SBB, STQ	251	0.13
Hobday and Minstrell 2008	Tas 2001–2004	Periodic 616	99.2	n/a	50–110	Rural and urban bitumen and gravel	BTP, P, BW, TDV, W, EBB, RTP, SBB, STQ, B, EQ, LNB	4 533	0.074[C]
Department of State Growth 2017	Tas 2013–2016	Weekly 143	97.2	~70	60–100	Rural bitumen and gravel	P, TDV, BW, W, BTP	57	0.0041[A]

[A]57 (roadkill) divided by 97.2 (km of road) divided by 144 (days reported) = 0.0041 roadkill per km per day. [B]211 (roadkill) divided by 100.3 (km of road) divided by 120 (days reported) = 0.018 roadkill per km per day. [C]4533 (roadkill) divided by 99.2 (km of road) divided by 616 (days reported) = 0.074 roadkill per km per day.

Calculating the number of wildlife carers

There are more than 200 registered wildlife-carer organisations in Australia with ~15 560 registered members (Table 2). However, a wildlife organisation may have many branches, e.g. NSW WIRES (NSW Wildlife Information, Rescue and Education Service), which is the nation's largest carer organisation, has 31 branches, and South Australia NAN (Native Animal Network Inc.) has 14 branches. Nevertheless, each multi-branch organisation was counted in our calculation only once per State or Territory. As mentioned previously, there are a minimum of 5000 unregistered carers so, in total, the minimum number of wildlife carers is 20 560.

WIRES has ~2500 registered members, of whom ~1500–1600 (60–64%) are actively caring for animals. In 2016, these members released to the wild 14 004 bandicoots, possums, gliders, macropods, antechinus, wombats, quolls, koalas and pademelons. Unfortunately, no data are available on how many of these were hand-reared orphans and how many were injured and rehabilitated rescued animals. It has been suggested that, on average, carers hand-rear 3–4 animals per year (WIRES 2017, pers. comm.).

If we accept the above information for roadkill orphans and the milk powder manufactured to rear them as being enough to rear a minimum of 40 000 joeys per year, the number of wildlife carers as 20 560 (of which 60–65% or ~12 300–13 200 are active) and animals reared and released as 3–4 per carer per year, an approximate conservative estimate for the total number of joeys rescued, rehabilitated and released would be 50 000 a year.

Impact on carers

As well as the skills, commitment and time required of wildlife carers, there are two major costs that they experience: financial and emotional.

Financial costs

As a condition imposed under all State and Territory regulations, every wildlife carer must agree to be responsible for the total financial costs of rearing orphan joeys and rehabilitating injured adults. A specific case of rearing a red kangaroo, *Macropus rufus* (the largest terrestrial mammal native to Australia), is provided here (Table 5) to illustrate maximal possible costs involved in raising just one orphan joey. It is worth noting that only food costs are reduced significantly when smaller species are reared. Many of the other costs, including infrastructure, medical expenses, education, training courses and transport (Table 5) remain almost constant.

Although many veterinarians work pro bono or charge wildlife carers very little, other costs do often include veterinary treatment, the need to communicate with others, transport for an animal, travel to meetings and training courses and general shopping, and involve information technology and transport costs. Attending a basic course, such as the Rescue and Immediate Care Course (RICC) run by WIRES NSW, costs $175 (which also covers the first-year membership fee). This expense is non-negotiable because the RICC is a prerequisite to becoming a registered carer. Public and personal liability insurance is another essential cost, although this may be covered by membership of a wildlife carer organisation. Events such as a carer being injured when rescuing an animal, or having a released animal injure a member of the public or cause a vehicle accident, all call for insurance cover. The carer may not be responsible, but the legal costs can be considerable in defending any action brought against a wildlife carer.

Housing and pre-release pens cost up to $4000–$5000 but can be used for a series of joeys. Similarly, stationery and books, training courses, registration and insurance can be assigned to the rearing of multiple joeys.

Table 2. Wildlife carer organisations by State or Territory

State or Territory	Number of organisations	Number of carers
New South Wales	34	5000
Victoria	50	3300
Queensland	59	2200
Western Australia	25	3150
South Australia	20	1340
Tasmania	13	300
Australian Capital Territory	2	60
Northern Territory	4	210
Total		15 560

Table 3. Total milk replacer manufactured by Wombaroo, Biolac and Passwell

Species	Production per year (kg)	Allocation (kg)
Kangaroo/wallaby	107 000	K'roo 10 000 Wallaby 97 000
Wombat	8000	8000
Possum	7000	Brushtail 6930 Ringtail 70
Koala	5000	5000

Table 4. Estimates of milk usage by each of the commonly hand-reared marsupial species, cost of milk and animals reared

Species	Dry weight of replacement fed per orphan (kg)	Cost (AUD)	Approximate minimum number of animals that could be reared per year
Brushtail possum (*Trichosurus vulpecula*)	0.6	40	11 500
Common wombat (*Vombatus ursinus*)	10.0	260	800
Eastern grey kangaroo *Macropus giganteus*)	15.0	520	660
Koala (*Phascolarctos cinereus*)	5.0	130	1000
Red-necked wallaby (*Macropus rufogriseus*)	7.5	260	12 900
Ringtail possum (*Pseudocheirus perigrinus*)	0.3	20	230

Table 5. Cost of rearing a red kangaroo joey to point of release in Australian dollars

Description	Quantity	Cost per item (AUD)	Total (AUD)
Infrastructure and consumables			
Cage/box	1	50.00	50
Hanging pouch frame	1	30.00	30
Pouches	5	15.00	75
Heat pad	2	69.00	138
Feed bottles	5	6.00	30
Teats	40	1.30	52
Mixing bowl, jug and whisk	1	25.00	25
Playpen	1	119.00	119
Cloth towels	4	2.00	8
Paper towel rolls	12	2.08	25
Food bowl and water bowls	2	10.00	20
Weighing scales	1	140.00	140
Food			
Milk substitute	1		520
Macropod pellets	1		96
Proprietary/prophylactic medications			
Skin lotions toltrazuril, peptosyl, non-antibiotic scour treatment, avermectin			50
Education			
Stationery and books			200
Training courses, registration, insurance			250
Vehicle costs and vet checks			172
Total			$2000

Table 6. Types of events that cause grief and the types of grief associated with them

Types of grief	Death of a spouse (Waskowic and Chartier 2003)	Death of a parent (Umberson 2003)	Death of a child (Keesee *et al.* 2008; Lin and Lasker 1996)	Marriage break up (Bacon 2001)	Parents with disappeared children (Hollander 2016)	Palliative care to the end of life (Stada 2013; Matzo and Sherman 2014)	Children leaving home (Raup and Myers 1989)	Death of a pet (Adams 1996; Rochester 2011)	Rearing an animal for release
Abbreviated				X				X	X
Absent	X		X	X	X	X		X	X
Ambiguous					X				X
Anticipatory	X	X					X	X	X
Chronic		X	X		X	X			X
Collective	X								
Cumulative			X					X	X
Delayed		X				X		X	X
Disenfranchised	X			X	X		X	X	X
Distorted			X					X	X
Exaggerated			X					X	
Inhibited		X				X		X	X
Masked		X							X
Prolonged		X	X		X			X	X
Secondary				X					
Traumatic	**X**				X	X		X	X
Unanticipated	X	X	X			X		X	X
Unresolved					X		X		X

Psychological effects

A search of the literature revealed eight of the most general type of experiences that can cause grief, and these are correlated with 18 types of grief that can be experienced in human lifestyle and personal relationships (Table 6). The last column in Table 6 illustrates the number of different types of grief that could be experienced by wildlife carers while rearing an orphaned or rescued animal. For example, having the animal die could be compared with the death of a pet or child; releasing the animal to the wild and not knowing if it survives could be compared with children leaving home or having pets or children go missing.

Discussion

Australian native animals are injured and orphaned at rates that require more than 200 registered carer networks and over 15 560 registered volunteer wildlife carers to meet the challenge of caring for injured animals (mammals, birds, fish and reptiles) and hand-rearing rescued orphans. There are also many other unregistered people involved; Tasmanian authorities do not require wildlife carers to be registered if looking after certain species, such as possums and wallabies, and South Australia does not require registration to obtain a permit to keep injured or orphaned native wildlife. Western Australia has no registration system, although people can join registered organisations. So no data exist on the number of individual wildlife carers acting outside these carer-network registers. This gives some indication of the size of the problem presented by orphaned and injured animals in Australia. Volunteers dedicate not only their time, physical effort and emotional input to this endeavour, but also contribute a significant amount of their own money. To put these financial and time contributions into a national perspective, at a level of one carer raising just one joey a year with an input of 1000 h and $2000 as discussed previously, the contributions from the national volunteer force amount to 20 000 000 h and $40 000 000. In NSW last year the average number of animals reared and released by each wildlife carer was 9.3. Approximately 14 000 marsupial joeys were rehabilitated and released by 1500 active carers in NSW in 2016, which equates to an average of 9.3 animals per carer per year, i.e. 14 000 divided by 1500. (WIRES 2017, pers. comm.).

Using this approximation, the national figures would be 186 000 000 h and $370 000 000 of financial outlay. Applying a dollar value of $31 h^{-1} (Sunners 2015) to the 186,000 000 h put in by carers brings the value of this work to $5 766 000 000. When this is added to the financial outlay of $370,000 000, the total amounts to ~6 billion dollars of volunteer in-kind and financial input.

When carers undertake to rear and rehabilitate native orphan and injured animals, they assume significant responsibilities. Although they are volunteers, they need a broad knowledge base in animal care, behaviour, legislation and policy to successfully rehabilitate a wild animal. Professional wildlife parks, e.g. Healesville Sanctuary, Bonorong Wildlife Sanctuary and Currumbin Wildlife Sanctuary rear very few orphaned animals themselves. They take them in for veterinary assessment and then place them with wildlife carers. (Healesville Sanctuary, 2017 pers. comm; Bonorong Wildlife Sanctuary 2017; Currumbin Wildlife Sanctuary 2017). Wildlife carers must undertake training before obtaining any animal and must subsequently ensure that their skill-base is kept updated on rehabilitation techniques, legislation and policies. (Tribe and Brown 2000; Gage 2002; Stocker 2005; Tribe *et al*. 2005). The financial demands of caring for wildlife usually reflect the need to supply all food, medicines, laundered pouches, sterilised bottles and teats, as well as safe and secure housing. The physical effort involved in rearing an orphaned joey or rehabilitating an injured individual, particularly those of larger species, may have a deleterious effect on the health of wildlife carers, especially those who are experiencing sleep deprivation from adhering to a 1- to 3-hourly feeding schedule day and night (Gay *et al*. 2004; Klumpers *et al*. 2015). There are over 60 diseases in Australia that can be transmitted from animals to humans, and carers are exposed to faeces, respiratory secretions and blood on a daily basis (Fowler 2007). Carers

need to be constantly aware of the occupational risks they encounter as a result of being exposed to wildlife. (Garland-Lewis *et al*. 2017).

Carers need to rely on personal resilience to deal with traumatic events, such as an animal dying, having an animal in care attacked by a predator or requiring an animal to be euthanased because it is unsuitable for release. These coping strategies can be likened to related needs in veterinarians (Rollo 2015) and volunteers who care for foster children, the disabled or elderly (Pickin *et al*. 2011; Li *et al*. 2014; Mishra *et al*. 2016).

During the time wildlife carers have responsibility for an animal, there is the potential for many events that can compromise their mental wellbeing. These can relate to the animal dying, becoming ill or suffering an injury and having to enter veterinary care, or even accidentally escaping. The carer may become unwell, have a personal crisis or a demanding family event, or be attacked by an animal in their care. They may be told to toughen up by friends or other carers if they are showing emotion towards an animal dying or leaving. Additional to these stressors, there is the knowledge that the animal must be released with no certainty of survival. After release, the animal may be shot or returned to care after having been injured. Volunteer respondents to a recent survey (PwC 2016: p. vii) reported that 'in their experience, both complacency and ignorance of the enormous value volunteers contribute results in a lack of regard of volunteers as strategic assets'. Similarly, an earlier study of wildlife carers (Guy and Banks 2010), revealed that carers perceived government attitudes towards wildlife rehabilitation as more negative than positive. Wildlife carers do not even have the comfort of knowing that their efforts are serving any useful purpose, other than the personal satisfaction of caring. Even though monitoring of released animals is recommended by the RSPCA Australia (RSPCA 2017), there is an almost total lack of post-release monitoring, supervision or data collection (Guy and Banks 2010). However, when reared individuals are of a threatened or endangered species, they are usually identifiable by microchip, tracking collar, tag or distinct coat markings. This enables some post-release monitoring and the contribution made to wildlife conservation can be estimated. Well-trained volunteer carers are expert in being available to undertake this rearing and release work and receive the reward of knowing this contribution is valued. Another valuable contribution wildlife carers can make is as environmental educators (Croll 2011). They have the ability and expertise to work through environment education centres to provide talks to schools, clubs and community groups. This adds to the self-worth felt by carers as well as educating others about the environmental effects of roadkill on native animals.

The stressors mentioned previously may expose wildlife carers to the risk of compassion fatigue and considerable grief of different kinds. When relationships with animals are believed to be inferior to human relationships this can be problematic at times of a loss (Hafen *et al*. 2017) The question of whether certain kinds of grief can be applied equally to the loss of an animal to which a person has bonded merits consideration. Support for this prospect lies in attachment theory and in how wildlife carers view the animals for which they care. Furthermore, there are theoretical arguments and empirical evidence that indicate attachment theory offers the means to examine human–animal bonds, and that humans can form attachment bonds with animals that are similar

to exclusively human attachment bonds. By way of examples, Beck and Madresh (2008) and Zilcha-Mano (2017) developed questionnaires based on a validated human relationship questionnaire (RQ) (Bartholomew and Horowitz 1991) and a revised version of the experiences-in- close-relationships (ECR-R) questionnaire (Fraley *et al*. 2011) to examine, first, human relationships with animals as distinct from those with human partners and, second, the attachment quality of human–animal attachment. Zilcha-Mano (2017) concluded that human–animal bonds meet the four prerequisites of attachment bond, namely proximity seeking, safe haven, secure base and separation distress, and that animals can therefore be viewed as attachment figures. Kurdek (2009) compared the way owners used their dogs as a safe haven as opposed to other key attachment figures, such as spouses, parents and friends. From these studies, it was concluded that humans can form attachment bonds with animals in ways that are consistent with the literature on attachment theory.

Because attachment brings with it the risk of loss, and therefore grief, the links among the three have been demonstrated in theories of grief and mourning (Weiss 1988; Parkes *et al*. 1993; Shaver and Cassidy 1999). Archer (1999) argues that grief is a natural reaction to losses of many kinds, even the death of an animal. He brings together experimental psychology, ethology and evolutionary psychology to demonstrate this. This definition of grief, as a reaction to a loss, is supported by more recent authors (Morris 2013; Rainer 2013; Winokuer *et al*. 2016). Worden (2002) has argued that the strength of one's attachment to another, the security of the attachment and the ambivalence in the relationship are all factors that will affect the grief reaction to the loss of that other.

It may seem perverse to suggest that wildlife carers could suffer from more types of grief than someone experiencing the death of a spouse. However, this is possible not least because there are many different experiences of loss that can occur during the time a wildlife carer is rescuing, rehabilitating and releasing an animal, whereas the death of a spouse is a single event. Feelings of loss are very personal, and only the affected individuals know the strength of relevant attachment bonds and hence the size of a loss. However, it can be seen from Table 6 that four of the major types of grief, anticipatory, disenfranchised, traumatic and unanticipated are proposed for both the death of a spouse and the rearing of an animal for release. It is beyond the scope of the current review to attribute degree of loss and grieving experienced by a wildlife carer. This is an area for further research. It would require the use of a validated grief instrument, suitably modified to apply to those who care for rescued wildlife, to be administered by way of an individual questionnaire survey or interview. This would then need to be evaluated to quantify the degree of grief experienced by carers resulting from the different events that occur during the rescue, rehabilitation and release of orphaned wildlife.

The death of an animal in care could be a trigger for several forms of grief, particularly if sudden, or if the carer is inexperienced. Abbreviated grief (Appendix 1a) is a short- lived but normal form of grief. The grieving process is shortened because the role of the deceased animal is immediately filled by another joey, and because there has often been little or no attachment to the deceased animal (Averill 1968; Irving *et al*. 1999). However, chronic grief (Appendix 1e) is a severe grief reaction that may have features in common with most early stages of grief, but is one that does not abate and can last for an extended period and fail to reach a conclusion. Carers can experience

extreme distress over the death of an animal and make little or no progress in coping with normal living (Lasker and Toedter 1991; Middleton *et al*. 1996; Bonanno and Kaltman 2001). This form of grief can be maintained by feelings of insecurity and insecure attachment to the deceased animal (Middleton *et al*. 1996) For an inexperienced carer, especially one who is psychologically unprepared, experiencing the death of an animal for the first time is likely to result in prolonged grief (Appendix 1n). The carer could be incapacitated by grief and daily function might be impaired on a long-term basis. The carer might spend much time contemplating the death, longing for reunion, and be unable to adjust to life without the animal (Barnes *et al*. 2012; Kersting *et al*. 2013). In a similar way, the sudden death of an animal can trigger unanticipated grief (Appendix 1q). This form of grief is characterised by great difficulty in accepting the loss and is accompanied by overwhelming feelings that could result in the carer becoming less able than normal to function in various areas of their daily life (Parkes 1986). Although affected carers could intellectually acknowledge the death, they would often have great difficulty in accepting the loss, due to it being sudden and unexpected. In this situation, grief symptoms will tend to persist much longer than normal grief reactions (Waskowic and Chartier 2003). This could damage the affected carer's adaptive capabilities and require a complicated recovery. If a carer experiences a second loss while still grieving the first (Pivnick 2011), then the two losses become cumulative (Appendix 1g), compounding the grief (Marino 1998). Delayed grief (Appendix 1h) may occur when a carer, consciously or subconsciously, avoids the reality and pain of a loss and suppresses feelings of grief outside the normal immediate timescale after an animal's death. The grief may later be triggered by an event related to the original loss, such as finding the pouches in which a joey was reared or a bottle from which it was fed (Bonanno *et al*. 2002; Claxton and Reynolds 2012).

Events external to the actual caring for the animal can also trigger grief, which is said to be disenfranchised (Appendix 1i), when feedback from a support group or friends (however inadvertently) makes carers feel that their loss, and/or their grief, is invalidated and insignificant (Doka 1989). Consequently, such grief cannot be acknowledged openly and has a depth that is not socially recognised. For example, displaying certain characteristics, such as crying uncontrollably over the death of an animal and placing flowers on its grave, could be regarded as not socially recognised and so has a silencing effect on the carer (Doka 1989). This grief could be triggered by remarks by others about the need for carers to toughen up, be less emotionally weak or even to care less for an animal (Raup and Myers 1989). Distorted grief (Appendix 1j) might be triggered when a carer must return an animal to authorities in the knowledge that it is extremely unlikely to survive if subjected to immediate placement into the wild (so-called hard release). Extreme, intense or atypical reactions to this event could cause carers to show unusual behaviour and self-destructive actions (Middleton *et al*. 1993). Anger and hostility towards oneself or others are common in instances of distorted grief (Remondet and Hansson 1987).

Another form of grief typically lacks clarity and is termed ambiguous (Appendix 1c). It can lead to carers holding different views about what has been lost (Lagoni *et al*. 1994). Individuals and those around them may question whether a real loss has occurred or whether the loss is of a scale sufficient to justify strong emotional responses, such as may arise with disenfranchised deaths

(Boss 1999; Hollander 2016). When an animal is released to the wild and is unable to be tracked, carers are unlikely to know what happens to this creature they have been committed to for up to 2 years. A sense of grief, termed unresolved grief (Appendix 1r), could develop and may be punctuated with hope that the animal will survive and be identified at some time in the future (Boss 1999; Barton Ross and Baron- Sorensen 2007). There may also be an element of self-doubt or guilt in believing that it is best for the animal to be returned to the wild. The resolution of grief could be elusive if carers cling to the hope that information will come to light that their animal has survived (Middleton *et al*. 1993; Boss 1999).

There are other grieving phenomena that can be experienced internally by a carer. Absent (Appendix 1b) and inhibited grief (Appendix 1l) can stem from denial or inhibition. Absent grief may be demonstrated by the carer behaving as though nothing has happened. The affected carer typically shows no feelings of grief and become detached from reality as if the event never occurred (Middleton *et al*. 1996; Bowlby 1998; Solomon and Gupta 2014). This is similar to inhibited grief where, again, the carer shows no outward signs of grief, but where symptoms can last for an extended period. Physical problems, such as lack of energy, headaches, gastrointestinal symptoms and chest pain, might develop if carers deny themselves the opportunity to experience the pain of grief directly (Middleton *et al*. 1998). It is possible for grief reactions to impair normal functioning, but a carer might fail to associate these signs with a loss. Symptoms are often masked as either physical symptoms or other maladaptive behaviours (Bonanno and Kaltman 2001), so this is termed masked grief (Appendix 1m).

Other forms of grief can occur over many months or years. Anticipatory grief (Appendix 1d) can happen when a carer believes some kind of loss will eventuate. The loss is not necessarily caused by anticipating a death or injury to an animal that is not healing and thus likely to die. By way of an example, carers may feel this grief before an animal is due to be released, particularly if they do not know what will happen to it. This grieving progress could continue for an extended period and so be emotionally draining because the latency before the loss is unpredictable (Rando 1986; Middleton *et al*. 1993; Simon 2008; Holley and Mast 2009). Normal grief responses experienced in combination with traumatic distress (e.g. due to some frightening, horrifying, unexpected, violent or traumatic event) can trigger traumatic grief (Appendix 1p). An example could be when animals placed in an enclosure to prepare them for release to the wild are killed by predators or are shot. The distress caused could be extreme enough to impair daily functioning and trigger reactions such as avoiding people, places or activities that remind the carer of the animal, or experiencing unusual levels of sleep disturbance, loss of interest in work, social caretaking or recreational activities to a maladaptive degree (Prigerson *et al*. 1999; Levinson and Prigerson 2000; Lobb *et al*. 2006; Bonanno and Mancini 2008; Kersting *et al*. 2013).

Although the mental state of wildlife carers can be adversely affected when the day-to-day caring for an animal comes to an end, and the animal is released, there can also be relief and happiness at the success of getting the animal to this stage and ready for release. The situation can be compared with the time when children leave home and the effect this has on their parents. The void left by the departure can cause loneliness and grief. The effect of the adjustment needed and

satisfaction with life can be significant (Raup and Myers 1989; Mitchell and Lovegreen 2009; Bouchard and McNair 2016). Accordingly, it is wise to prepare emotionally for the departure and wildlife carers can adopt strategies similar to those suggested to human carers to reduce the possibility of grief and be able to experience a positive outcome (Hebert *et al*. 2006; Holm *et al*. 2015).

Other factors that may cause distress to carers

Charles Figley's original general description of compassion fatigue (Figley 1995) has been applied to many professions, including healthcare professionals (Meadors and Lamson 2008), psychotherapists (Figley 2002), social workers (Bourassa 2012), foster parents (Blanchette 2011) and family caregivers (Perry *et al*. 2010), demonstrating how widespread this phenomenon can be. Compassion fatigue in the animal care community is described as 'exhaustion due to compassion stress, [and] the demands of being empathic and helpful to those who are suffering' (Figley and Roop 2006: p.12). The result can be disruptive, depressive and irritating, and can lead to occupational burn out. This is a situation where wildlife carers become physically and emotionally exhausted, usually after prolonged stress, frustration and sleeplessness, and become unable to function (Rank *et al*. 2009; Moore *et al*. 2014; Galazka 2017).

Even though they are volunteers, wildlife carers are subject to government authority and policies. (Australian Capital Territory Parliamentary Council 2014; New South Wales Government 2010; Queensland Government 2006; Western Australian Government 1970; South Australian Government 1972; Tasmanian Government 2002; Victorian Government 1975; Northern Territory Government 2014). Many of them find this a problem. When regulations are so prescriptive that monitoring of compliance is virtually impossible, or the expertise to make assessments of behavioural suitability for return to the wild is unavailable, those who do try to comply with regulations feel let down when others are not being required to comply (Bardach and Kagan 1982; Nie 2008). This can lead to conflict both between individual carers and within organisations (Jacobs 2017).

All wildlife carers expend time, money and effort in rearing orphans or rehabilitating injured animals. Depending upon state regulations, they are mandated to release the animals back to the location where they were found, once they have been reared. For example, Section 12.2.1.4. of the NSW Code of Practice for Injured, Sick and Orphaned Protected Fauna (New South Wales Government 2010: p.20) states 'If there is no information about where fauna was found, it must not be released.' Also, Section 7.1.1.2 p.8. states: 'Fauna must be euthanased when there is no suitable release location'.

These release sites may be close to farmland or a settlement where native animals are regarded as pests. In NSW in 2016, 107 687 native animals and birds were killed by property owners using an 's121' licence (intended to control native animals if they are causing damage or economic hardship on a property). Additionally, the NSW Office of Environment and Heritage issued permits for the killing of 170 290 eastern grey kangaroos, 308 swamp wallabies and 136 common wombats (NSW Parks and Wildlife Service 2017, pers. comm.). In the same period, 14 004 marsupial species were returned to the wild by wildlife carers who were members of the NSW

WIRES (WIRES, 2017, pers. comm.). By way of comparison, in Victoria in 2011, the figure for mammals returned to the wild was 6 442 (Department of Environment, Land, Water and Planning, 2017, pers. comm.). Regardless of interstate differences in numbers, the paradox here is that animals are being rescued, raised for release and returned to the site of the traumatic event that brought them into care at the same time as similar species are being killed under licence. This can be challenging for volunteers; whose primary objective is the welfare of animals.

Wildlife rehabilitation is defined as 'the managed process whereby a displaced, sick, injured or orphaned wild animal regains the health and skills it requires to function normally and live self-sufficiently' (Molony *et al*. 2006: p. 530). There are notable success stories of hand-reared or captive-bred animals being successfully reintroduced to the wild, often featuring wombats (Saran *et al*. 2011) and Tasmanian devils (Rogers *et al*. 2016). However, many reintroductions are unsuccessful or meet limited success. For example, between 1972 and 1988 six separate attempts to reintroduce threatened wallabies (Marsupialia: Macropodoidea) in areas of Western Australia and New South Wales all ended in failure (Short *et al*. 1992; Priddel and Wheeler 2004; Short 2010; Moseby *et al*. 2011). After a comprehensive evaluation of 380 translocations in Australia, involving 102 species (beginning with koala to Phillip and French Islands in Victoria in the 1880s to 125 translocations since 2000), a survival rate of 54% was recorded (Short 2010). There is no requirement under any state legislation for animals to be reliably identifiable (by means of microchipping, tagging or branding) either during care or before release. In contrast, one of the main reasons that wildlife carers undertake the rehabilitation of injured and orphaned animals is for conservation purposes (Guy and Banks 2010). Whether this aim is being satisfactorily fulfilled is impossible to ascertain without reliable data from post-release monitoring. A system of individually identifying animals from the moment they enter care would enable better governance during the rehabilitation process and informed decisions about future release activities. A simple microchip would enable any road kill or culled animal to be scanned and identified as a hand-raised orphan or not. The cost of approximately $20 per animal is one that could be met by state or federal authorities so as not to increase the financial burden on carers.

The problem presented by roadkill and other anthropogenic and natural events that result in orphaned and injured animals is a national one. A priority recommendation for future research would be two national roadkill survey days separated by 3 months with citizen scientists using a suitable smartphone or tablet app to record roadkill, so that temporal and spatial data are confined to these days. A national approach that includes a consensus of best practices for training wildlife carers, raising and rehabilitating orphaned and injured animals, monitoring release outcomes and financial and counselling support would increase the professional standing of carers and, through them, outcomes for the animals for whom they care.

The need for volunteers in contexts beyond wildlife care is increasing, with 86% of organisations that use volunteers stating the need for more personnel (PwC 2016). In 2014, the Australian Bureau of Statistics recorded the first decline in both men and women volunteering since records began. The social capital of Australia has declined from its 1967 situation, when 33% of Australians were members of organisations such as Rotary clubs, Lions clubs and the Scouts, to just 18% by 2004

(Caneva 2015; Cowlishaw *et al*. 2010).

Animal welfare ranks fourth among sectors in which volunteers would be most interested to volunteer (PwC 2016). The number of wild animals presented to the RSPCA Australia has increased from 8310 in 1998–99 to 25 568 in 2015–16. Clearly, as the number of wildlife requiring treatment increases, so does the need for wildlife carers. In NSW, WIRES reports that its membership numbers are presently 2333 and that these numbers fluctuate between 2100 and 2500 as new volunteers are recruited and trained (WIRES, 2017, pers. comm.). This illustrates a considerable turnover in volunteer numbers and the requirement for constant management of recruitment. Should recruitment become unsustainable then pressure on the remaining experienced carers will continue to build as the workload increases.

One way to ease the workload for wildlife carers would be to reduce the number of animals requiring rescue and rehabilitation.

There are several methods by which this could be achieved, one being the adoption of a non-intervention or euthanasia policy for all or some injured or orphaned animals but, as discussed earlier with the case of Harambe, the gorilla at Cincinnati zoo, this would be unlikely to meet with general approval from the public. From an Australian perspective, possibly the first public reaction on a conservation issue related to animal culling occurred in Queensland in August 1927. The government passed legislation that allowed the killing of Koalas and 600 000 were killed in the following month. The public backlash led to the killing being banned. Roe (2017). There are many recent examples in Australia where animal advocates and activists have managed to focus public attention on the issue of animal welfare and gain political consensus. This sensitivity is highlighted in the following statement of the Department of Territory and Municipal Services Australian Capitol Territory (2017: 142): 'It is inevitable that kangaroo management in the nation's capital will be affected by 'animal rights' campaigns against any management actions that involve the killing of kangaroos'. Similarly, a plan to cull koalas on Kangaroo Island in 2015 was cancelled by the South Australian government after a public backlash. The AKF's stance on koala relocation and culling (Australian Koala Foundation 2017: p.1) states it 'will never condone the culling of koalas or any other native animal'. This position is likely to be supported by the nation's 15 600 wildlife carers (Table 2).

Another option to ease the workload for wildlife carers is to recruit more of them, or reduce the number leaving. Similarly, a reduction in roadkill (and therefore the need to rescue orphans) would have the effect of reducing the burden on wildlife carers. This is an area where a national research approach on roadkill mitigation measures could be highly beneficial.

Conclusion

The physical, financial and mental contributions made by wildlife carers are considerable. The current review reveals that wildlife rehabilitation in Australia is a major operation undertaken by volunteers acting individually or as part of an organisation. It is managed on a state-by-state basis, each with its own set of rules and regulations, and by more than 200 organisations and by individuals acting alone, so its operation is fragmented. Future research opportunities should

focus on a national system of licensing and training carers and on an overall plan for the way animals are prepared for release on a species-by-species basis. A system of individually identifying animals from the moment they enter care would enable better governance during the rehabilitation process and informed decisions to be made about future release activities. As roadkill increases, volunteering declines and financial costs increase, those volunteers still acting as wildlife carers are likely to experience increasing workloads, and with that, increasing financial, mental and physical stressors. The wildlife carers who manage Australia's injured and orphaned native animals are a national asset that requires strategic nurturing with empathy, understanding, financial and psychological support if it is to remain viable and sustainable. The skills learned in this process often contribute to the success of captive breeding and may be the salvation of many animal species threatened with extinction in the future.

Conflicts of interest

The authors declare no conflicts of interest.

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Appendix 1. General definitions of grief

There are no standard definitions for types of grief. The following are a consensus drawn from several authors (Archer 1999; Worden 2002; Bryant and Peck 2009; Wilson 2013; Boss and Yeats 2014).

(a) *Abbreviated*. A short-lived form of grief. The grieving process often seems shorter than for normal grief because the role of what is lost is immediately filled by someone or something else.
(b) *Absent*. This is characterised by the bereaved acting as though nothing has happened for a period up to approximately a week.
(c) *Ambiguous*. The losses are hard to define and therefore also hard to identify. Ambiguity makes it hard to grieve the loss fully because it seems so inconsequential and insignificant.
(d) *Anticipatory*. The reaction to an expected loss and grief begins once it is accepted and understood that a loss will occur.
(e) *Chronic*. These are strong grief reactions that do not subside and last over a long period. The person is continually experiencing extreme distress over the loss with no progress towards feeling better or improving functioning.
(f) *Collective*. Grief reactions that are felt by a collective group such as a community, society, village, or nation.
(g) *Cumulative*. When a second loss is experienced while the affected individual is still grieving a previous loss.
(h) *Delayed*. Lack of time to mourn when a loss occurs means grieving is delayed and symptoms and reactions are not experienced until an extended period has elapsed.
(i) *Disenfranchised*. This is a term describing grief that is not acknowledged by society. Even widely recognised forms of grief can become disenfranchised when well-meaning friends and family attempt to set a time limit on a person's right to grieve.
(j) *Distorted*. This is characterised by intense, extreme, or atypical reactions to a loss
(k) *Exaggerated*. This type occurs when there is a cumulative effect of losses.
(l) *Inhibited*. This is a form of unresolved grief where the bereaved person displays no outward signs of normal mourning for an extended period.
(m) *Masked*. These are reactions that impair normal functioning even though the individual is unable to recognise these symptoms and behaviours are related to the loss.
(n) *Prolonged*. Reactions that are prolonged and intense. The griever is incapacitated by grief and daily function is impaired on a long-term basis.
(o) *Secondary*. This occurs when a loss affects many areas of life, creating multiple losses stemming from the primary loss.
(p) *Traumatic*. Normal grief responses experienced in combination with traumatic distress suffered as a result of a frightening, horrifying, unexpected, violent and/or traumatic event.
(q) *Unanticipated*. These reactions result from an entirely unexpected sudden loss.
(r) *Unresolved*. This grief lasts longer than usual and does not go away.

Chapter 3: Wildlife legislation, regulation and codes of practice

3.1. Preamble

The previous chapter highlighted that the welfare of wildlife carers could be adversely impacted in many ways, one of which was the way that legislation could affect their wellbeing, that of the animals for which they care, and the environment.

The World Animal Protection index established a grading system of 50 countries with regard to their commitment to animal welfare in legislation, regulation and policy [33]. Austria, Switzerland, New Zealand and the UK are the only countries to be rated an A in this index, where A is the highest grade and G the lowest. Reasons for a grade A include regulations that animals be held in equal value to humankind, acknowledging that animals are sentient and that animal suffering, infliction of pain or harm, exposure to humiliation or anxiety, or activities that can be recognised as degrading are to be avoided. Interestingly, these are four countries that have a centralised system of government. This enables national laws on animal welfare to be enacted by one authority.

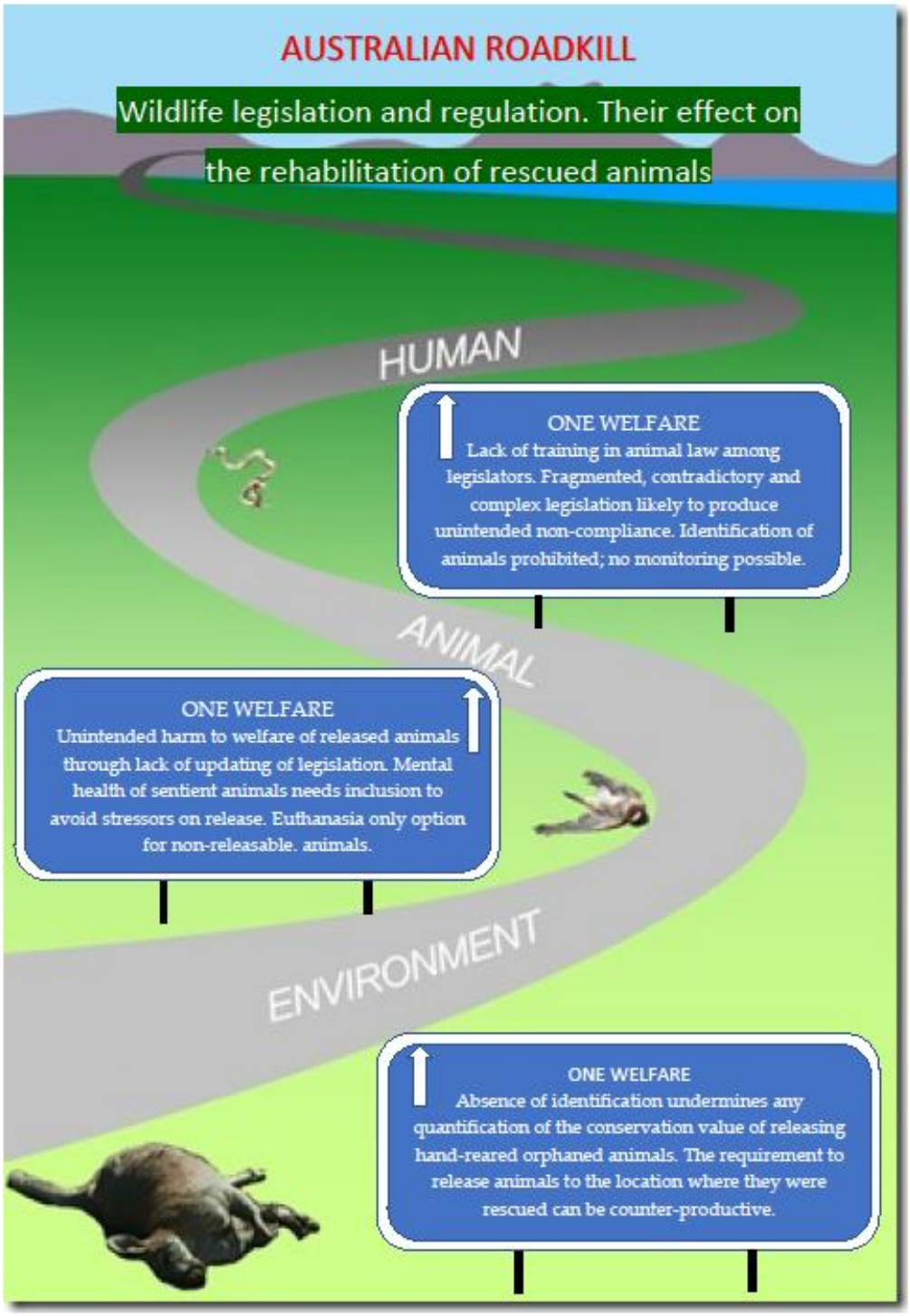
By recognising that animals be valued as equal to humans and as sentient beings with mental needs, the three elements of human, animal and environment contained in the One Welfare ideal are being met. In some countries, for example Brazil, graded a C [34] and Germany, graded a B [35], animal welfare is included in the constitution. Two countries with a state and federal system of government, the US and Australia, rate poorly as a D and moderately as a C respectively, and have constitutions that do not include animal welfare.

The concept that human existence, native animals and the environment are inexorably linked, i.e. the One Welfare concept, does not seem to have featured in the preparation of the Australian constitution. This is particularly perplexing when there was a model available of a civilisation, based on these principles, that had survived in Australia for over 60,000 years, that of indigenous Australians, the Aboriginal and Torres Strait Islander peoples. While human needs and, to some extent, those of the environment are included, the constitution does not mention native animals or animal welfare. Responsibility for native animals or animal welfare was retained by the States when the Australian constitution was enacted. Without the requirement of Federal input, the States and Territories have gone their independent ways. Furthermore, 16 of the 39 State and Territory acts that provide the regulatory framework for animal welfare were drafted prior to 2008, before the first candidates in animal law graduated from the University of New South Wales. This was the first law school in

Australia to teach animal law, and introduced a course as a post-graduate elective in 2005. Unfortunately, prior to this, Australia's lack of legal expertise in animal welfare may have contributed to the fragmented, complex, contradictory and inconsistent system of regulatory management that has evolved.

To understand more fully how Australian legislation on animal welfare evolved, a literature review was conducted. The review included the 39 state and territory legislations, regulations, codes of practice and guidelines that govern animal welfare. The mantra on which the legislation seems to be based is that the best place for rehabilitated wildlife is in the wild. The common human intuitive reaction is to place rehabilitated animals back into the wild. This approach may not consider what the animal feels, and has been challenged by others [36-38]. It is also challenged here, based on the recognition of animal sentience by many authorities and the premise that the mental welfare of animals is equally important to their physical welfare.

Figure A3. One Welfare. Wildlife legislation and regulation


AUSTRALIAN ROADKILL
Wildlife legislation and regulation. Their effect on the rehabilitation of rescued animals
HUMAN
ONE WELFARE
Lack of training in animal law among legislators. Fragmented, contradictory and complex legislation likely to produce unintended non-compliance. Identification of animals prohibited; no monitoring possible.
ANIMAL
ONE WELFARE
Unintended harm to welfare of released animals through lack of updating of legislation. Mental health of sentient animals needs inclusion to avoid stressors on release. Euthanasia only option for non-releasable. animals.
ENVIRONMENT
ONE WELFARE
Absence of identification undermines any quantification of the conservation value of releasing hand-reared orphaned animals. The requirement to release animals to the location where they were rescued can be counter-productive.

3.2. Published paper: A review of Australian animal welfare legislation, regulation, codes of practice, and policy, and their influence on stakeholders caring for wildlife and the animals for whom they care

The problems created by the legislation, regulation and codes of practice and policies are complex for the volunteers rescuing and rehabilitating native animals. The following review was published in the open access journal *Animals.*

Review

A review of Australian animal welfare legislation, regulation, codes of practice, and policy, and their influence on stakeholders caring for wildlife and the animals for whom they care

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Simple Summary:
The Australian constitution does not mention native animals. Responsibility for animal welfare is largely retained by the states and territories via a fragmented, complex, contradictory, inconsistent system of regulatory management. The problem this creates for volunteers undertaking the rescue and rehabilitation of native animals is complex. Capturing and rehabilitating wild animals goes against regulations. In most jurisdictions, it is illegal to microchip, band, or mark an animal, making it almost impossible to monitor their survival. A minimum of 50,000 rehabilitated native animals are released back to the wild each year, with few checks afterwards to see how well or if they are surviving. Whilst it can be appropriate to rehabilitate and release injured native animals back to the wild, there may be moral, ethical, and practical reasons for not releasing hand-reared orphan native animals. With no reliable method of identification, no instructions on how to get animals ready for release or see if they are suitable, and little post-release checking, the practice of placing hand-reared native animals into the wild, and the regulatory framework enabling it, should be reviewed.

Abstract:
The Australian constitution makes no mention of native animals. Responsibility for animal welfare is largely retained by the states and territories via a fragmented, complex, contradictory, inconsistent system of regulatory management. Given that most jurisdictions have expressly made the possession of wildlife unlawful, the action of taking and possessing an animal, to rehabilitate it, defies the regulatory process. In most jurisdictions, it is illegal to microchip, band, or mark an animal, meaning that no reliable method is available to monitor an animal. Each year, a minimum of 50,000 rehabilitated native animals are released back to the wild, with little post-release monitoring. Where required, the assessments of behavioural and health requirements to confirm suitability for release may be undertaken by people with either negligible or questionable qualifications. Whilst it can be

appropriate to rehabilitate and release injured native animals back to the wild, there may be moral, ethical, and practical reasons for not releasing hand-reared orphan native animals. This article examines the evolution, and explains the consequences, of decentralised regulation on wildlife carers and rehabilitating animals. It recommends that the practice of placing hand-reared native animals into the wild, and the regulatory framework that provides for it, should be reviewed.

Keywords: wildlife; native animals; wildlife care; legislation; mental well-being; physical well-being

1. Introduction

Australian animal law is similar to the legislation of other nations in that, on the one hand, there are laws that allow extermination, culling, and harming animals while, on the other hand, there are those that protect animals from being killed, harvested, or harmed. A plethora of legislation, regulation, codes of practice, and guidelines are intended to guide the volunteer wildlife carers who rescue and rehabilitate injured and orphaned native wildlife (Table 1). These carers must navigate laws and policies that differ in complexity, aim, and implementation, both between and within Australian states and territories. The well-being of rescued wildlife is a central objective for wildlife carers, but navigating these laws successfully is an additional challenge to that of rehabilitating the injured and orphaned animals.

Table 1. Australian state and territory animal legislation, regulation, and codes of practice/policy.

State/Territory	Legislation	Regulations	Codes of Practice/Policy
Australian Capital Territory (ACT)	Animal Welfare Act 1992 (ACT) [1] Nature Conservation Act 2014 (ACT) [5]	Nature Conservation Regulation 2015 (ACT) [2]	Code of Practice for the Welfare of Captive Birds 1995 (ACT) [3] Reptile Policy 2018 (ACT) [4] Code of Practice for the Welfare of Amphibians in Captivity 2004 (ACT) [6]
New South Wales (NSW)	Prevention of Cruelty to Animals Act 1979 (NSW) [7] Biodiversity Conservation Act 2016 (NSW) [10]	Prevention of Cruelty to Animals Regulation 2012 (NSW) [8]	Code of practice for injured, sick and orphaned protected fauna 2011 (NSW) [9] The Rehabilitation of protected fauna policy 2010 [11]
Northern Territory (NT)	Animal Welfare Act 2017 (NT) [12] Territory Parks and Wildlife Conservation Act 2014 (NT) [15] Animal Protection Bill 2018 (NT) Pending [16]	Animal Welfare Regulations 2013 (NT) [13]	Guide for Caring for Wildlife 2018 (NT) [14]
Queensland (QLD)	Nature Conservation Act 1992 (QLD) [17] Animal Care and Protection Act 2001 (Qld) [20]	Animal Care and Protection Regulation 2012 (Qld) [18]	Code of Practice Care of Sick, Injured or Orphaned Protected Animals in Queensland 2013 (Qld) [19]
South Australia (SA)	National Parks and Wildlife Act 1972 (SA) [21] Animal welfare Act 1985 (SA) [24]	National Parks and Wildlife (Wildlife) Regulations 2016 (SA) [22]	General Guidelines for the Management of Protected Wildlife in Captivity in South Australia 2010 (SA) [23] Guidelines on Taking from the Wild (SA) [25]

Tasmania (TAS)	Nature Conservation Act 2002 (TAS) [26] Threatened Species Protection Act 1995 (TAS) [29] Animal welfare Act 1993 (TAS) [30]	Wildlife Regulations 1999 (TAS) [27]	General Requirements for the Care and Rehabilitation of Injured and Orphaned Wildlife in Tasmania, 2008 (TAS) [28]
Victoria (VIC)	Wildlife Act 1975 (VIC) [31] Prevention of Cruelty to Animals Act 1986 (VIC) [34]	Wildlife Regulations 2002 (VIC) [32] Prevention of Cruelty to Animals Regulations 2008 (VIC) [35]	Code of Practice for the Welfare of Wildlife during Rehabilitation 2017 (VIC) [33]
Western Australia (WA)	Biodiversity Conservation Act 2016 (WA) [36] Animal Welfare Act 2002 (WA) [39]	Wildlife Conservation Regulations 1970 (WA) [37]	Standards for wildlife rehabilitation in Western Australia 2014 (WA) [38]

Native animals play a critical role in the Australian environment and Australia's identity. A kangaroo (*Macropus rufus*) and emu (*Dromaius novaehollandiae*) feature on the Australian coat-of-arms, while various state and territory coats-of-arms also bear images of kangaroos, in the company of black swans (*Cygnus atratus*), brolgas (*Grus rubicunda*), shrikes (*Laniidae*), wedge-tailed eagles (*Aquila audax*), and Tasmanian tigers (*Thylacinus cynocephalus*). However, the appearance of animals in Australian heraldry emblems seems to have offered no guarantee of the welfare of featured species being promoted by legislators. Indeed, two Tasmanian tigers feature on the Tasmanian (TAS) coat-of-arms, which was granted a royal warrant from King George V, and the current Tasmanian Government logo, issued under the *Trademark Act 1995* (Commonwealth) [40], features a Tasmanian tiger peering through long grass. This logo is now used as an identifying device and a visual communication of the Tasmanian brand [41]. Paradoxically, it was legislation passed by the Tasmanian parliament in 1888, placing a £1 bounty on Tasmanian tigers, that contributed to their extinction. In contrast, some introduced species appear in the heraldry of Queensland (QLD). The QLD Government introduced legislation in 1997 confirming the appearance of the red deer (*Cervus elaphus*) as a feature on the state's coat-of-arms, despite this being an introduced species. Some 11 years later, the same legislative body declared the red deer as, "invasive animals that are listed as restricted matter", permitting them to be shot in environmentally sensitive areas [42]. Although others [43–46] have deliberated about what constitutes an introduced or native species, and some cite the dingo as a blurring distinction [47,48], the *dingo* is recognised as a *native animal* under the laws of all Australian jurisdictions. These recognise a native species as one that was present in Australia before the year 1400 [49] (Section 528). By way of contrast, red deer did not exist in Australia until early white settlers introduced them from Europe in the 19th century, confirming them as a non- native species.

These two instances show that the states use legislation both to identify themselves by aligning with species-specific emblems and also, when it suits, to euthanase members of the very same species.

Australian states are sovereign powers. The states formed the Commonwealth and delegated to it specific responsibilities, such as defence, income tax, and external treaties. There are no national laws mandating minimum animal welfare requirements. The responsibility for the management of wildlife and other animals remains with the states and territories. However, by way of an exception, there are legally binding national codes of practice for animal welfare in research, i.e., the Australian code for the care and use of animals for scientific purposes [50]. The Commonwealth's role is limited in this domain because the Australian constitution mentions neither animal welfare nor animals, other than fisheries, in any detail [51].

The Australian philosopher Peter Singer gave direction to the treatment of animals in 1989, with his book *Animal Liberation* [52]. Animal law is an independent and unique discipline and, over the last 10–20 years, has emerged as a focal domain among Australian legal scholars, reflecting societal interest

in animal rights, welfare, and protection. Increasingly, law journals are publishing articles on animal law, and several textbooks dedicated to the topic have been written by Australian academics [53–57]. In parallel with this, community expectations and public awareness of animal welfare issues have been heightened by the use of social media and the broadcast of covert video recordings by animal protection organisations in Australia. These trends are likely to continue to grow, despite the so-called ag-gag laws that have been proposed to shut down covert surveillance activities [58].

World Animal Protection is an international non-profit animal welfare organization that recently published a classification of 50 countries around the world, according to their commitments to protect animals and improve animal welfare in policy and legislation [59]. Largely due to its lack of a national approach to animal welfare, Australia rated overall as a 'C', in contrast to New Zealand, who was rated a superlative 'A' on a scale from 'A' to 'G' (with 'A' representing the highest scoring and 'G' the most room for improvement). Australia also scored poorly due to its failure to either support the Universal Declaration on Animal Welfare or to legislate Office International des Epizooties (OIE, the World Organisation for Animal Health) welfare standards, as well as perceived deficits in OIE participation and a dearth of education programs.

Despite Australia's current failures on the animal welfare front, it is worth noting that a cohesive national approach to animal welfare policy has been attempted in the recent past. The Australian Animal Welfare Strategy (AAWS) was developed to outline directions for future improvements in the welfare of animals and to provide national and international communities with an appreciation of animal welfare arrangements in Australia. The Primary Industries Ministerial Council, a part of the Council of Australian Governments process and the appropriate forum for attention to achieve a national approach and harmonisation of laws, endorsed the AAWS in May 2004 and the first National Implementation Plan for the strategy in May 2006 [60]. From 2010 to 2013, the Commonwealth contributed funding to the AAWS. The AAWS aimed to help build partnerships, improve coordination, reduce duplication of effort, and deliver a more effective and consistent national approach to improving the welfare of animals [61]. However, as part of its 2014–2015 Budget measures, the incoming Liberal–National Party coalition ceased funding of the AAWS.

An issue that might benefit from a national approach is the injuring and killing of native animals on Australian roads by wildlife vehicle collisions (WVCs), colloquially called roadkill. The annual Australian mammalian roadkill has been conservatively estimated at 4,000,000 animals [62], Subsequent to collisions with vehicles, the numbers of animals that suffer a slow death in the bush is uncountable. WVCs and other events, both anthropogenic and natural, produce a massive burden of injured and orphaned animals, the care of which is often left to volunteers. Voluntary wildlife carers must be competent in understanding the federal and state legislation, regulations, codes of practice, general guides, and guidelines so that they operate within the law (Table 1). Non-compliance has consequences that can prove financially costly as well as emotionally draining. Similarly, training is required so that carers can manage the complex tasks involved in rehabilitating wildlife and continue their professional development with ongoing education.

The noun 'rehabilitation' comes from the Latin prefix 're', meaning again and the verb 'habitare', meaning to live, hence wildlife rehabilitation is generally defined as: "providing professional care to sick, injured, and orphaned wild animals so ultimately they can be returned to their natural habitat or state" [11] (p. 4 Objectives), [63,64] However, other definitions, both in legislation and in literature change the words "natural habitat or state", to be more specifically, "the wild". For example, Tasmanian legislation mentions, "with the goal of releasing them back to the wild" [28] (p. 3 Definitions, bullet point 2), and Australian and USA literature mentions, "to appropriate habitats in the wild" [65,66]. The semantics of using the terms "natural state or habitat" and "the wild" in rehabilitation reveal much more than a pedantic difference. In changing "natural habitat or state" to become "the wild", laws that mandate the wild for the release of hand-reared animals leave no flexibility to accommodate those animals whose well-being might be optimised by remaining in captivity. It is suggested that a difference should be acknowledged between rescue animals that are injured (having been raised in the

wild as their natural habitat or state) and those that are hand-reared (for whom captivity has become their natural habitat or state).

A history of the establishment of regulation in Australia is presented to explain how the complex regulatory system evolved, how its implementation has produced practices that may be questionable in their efficacy in maximising the well-being of rescued native animals and the people who care for them, and how seemingly minor changes of nomenclature in laws can have far-reaching consequences. Previous reports have considered the complexities of wildlife care in Australia [67–71], but none have considered human and animal welfare in the light of the disjointed legal frameworks that apply across the states and territories. This article provides a review and critique of the current systems of regulating the rescue and management of injured and orphaned native animals in Australia. It explains the consequences of the current decentralised system of regulation for wildlife carers and the animals for which they care. A consideration of regulation from both the USA and the UK is included to either illustrate perceived deficiencies in approaches that exist in Australia or to explore alternatives that might improve the current system. Sections of this review investigate the regulatory processes of:

(a) Rescue and taking of injured and orphaned wildlife;

(b) Licensing, permitting, and training requirements for wildlife carers to keep wildlife;

(c) Facility requirements, standards of care, monitoring and identifying of animals;

(d) Funding and monitoring of carers/carer networks;

(e) Releasing animals;

(f) Pre-release methodology and post-release monitoring.

2. Discussion

2.1. Rescue and Taking of Injured and Orphaned Wildlife

A system has evolved in Australia involving government, not-for-profit organisations, and volunteer groups managing the rescue and rehabilitation of injured and orphaned animals. Generally, vehicular casualties are reported to local authorities, for example, police, National Parks and Wildlife Service, state infrastructure, and environment departments or to designated organisations via wildlife rescue helplines, for example, Royal Society for the Prevention of Cruelty to Animals Australia (RSPCAA), Wildlife Information, Rescue and Education Service (WIRES), and Wildcare Australia Inc. Information about the casualties is then passed to volunteer wildlife carers and rehabilitators, who take on the animals.

Regulations relevant to wildlife vehicle collisions (WVCs) vary from a legal requirement to stop and assist and to report the incident in the Australian Capital Territory (ACT) and Victoria (VIC), to no requirement to stop or report in TAS and Western Australia (WA), where any action depends on the morality of the individual driver. In the ACT, for example, it is an offence to kill a native animal and a person commits an offence if they "engage in conduct" that causes the death of a native animal. One might consider that "engaging in conduct" could include driving a vehicle. However, a subsequent clause states that the above does not apply if the death of the animal is caused by an accidental collision with a motor vehicle [5] (Chapter 6, Part 6.1. Division a 6.1.2, 130 {1}]).

Legal provisions also require the alleviation of an animal's suffering to be an obligation under the ACT Animal Welfare Act 1992 [1], "a person commits an offence if they injure a live member of a vertebrate species and do not take reasonable steps to alleviate any pain suffered by the animal". Furthermore, if there is no person in charge of the animal, the person in attendance must report the injury within 72 h. Examples given of animals that may have no person in charge include kangaroos (*Macropus spp*), foxes (*Vulpes vulpes*), and galahs (*Eolophus roseicapilla*), which may effectively be seen as meaning all wildlife. Thus, a vehicle driver who accidentally kills such an animal is exempt but is required to assist and report if the animal is injured. Under legislation proposed in the ACT [72], the

requirement to report an accident will change to a requirement to tell a relevant person within 2 h where the animal was injured. An example is given of "circumstances, if a car hits a mammal—the size of the animal, witness accounts that the driver swerved or braked to avoid the animal or stopped after hitting the animal".

In all states except WA, it is illegal to take or possess wildlife, dead or alive, whole or parts of, unless exempted by reason of having a permit to do so (Table 2). This disbars the general public from doing any more than rendering assistance to an injured WVC victim. Clearly, they must not remove the animal or its offspring. With the exception of WA, all states and territories require licensing of people wishing to possess wildlife for the purpose of rehabilitation. In TAS, no permit or license is required for taking and keeping Bennett's wallabies (*Macropus rufogriseus*), pademelons (*Thylogale*), and common brushtail possums (*Trichosurus vulpecula*) (Table 2). In the Northern Territory (NT), ACT, and South Australia (SA) it is also possible to obtain licences to keep, sell, import, and export certain native wildlife on a permanent basis, and in the NT, rescued wildlife becomes the property of the licence owner. In QLD, an animal taken in for rehabilitation under licence ceases to be the property of the state, until released (Table 2). Given that most states have expressly made the possession of wildlife unlawful, the action of taking and possessing an animal, to rehabilitate it, defies the regulatory categories. However, if statutorily provided, a permit or license serves as an exemption. This allows governments to regulate wildlife care and rehabilitation via a licensing system and thus ensure that those who seek to become wildlife carers are adequately qualified, have the necessary finance and facilities to do so, and can operate in a lawful manner.

Table 2. Taking of Injured and Orphaned Wildlife.

State/Territory	Assisting with Dead or Injured Wildlife	Requirement for Subsequent Action	Possess an Animal
ACT	Animal Welfare Act 1992 (ACT) [1] (Pt 2 10 {1}) Nature Conservation Regulation 2015 (ACT). [2] (Section 8 s 269) Must assist.	Animal Welfare Act 1992 (ACT) [1] (Pt 2 10 {2}) Yes, within 72 h	Nature Conservation Act 2014 (ACT) [5] (Chapter 6 Pt 6 Div. 6.1.2. Sect 132) Nature Conservation Act 2014 (ACT) [5] (Chapter 11 Pt 11.1) Licences to keep, sell, import into, and export out of the ACT are required for all animals (dead or alive, whole or parts of) except those scheduled as exempt species under the Act.
NSW	Biodiversity Conservation Act 2016 (NSW) [10] (No 63. Part 2.1) Only if licensed.	Code of Practice for Sick and Orphaned Protected Wildlife (NSW) [9] (4.1.2) Assessment within 24 h required. Biodiversity Conservation Regulation 2017 [Answers.com., #731] [73] Notification of possession within 3 days.	Biodiversity Conservation Act 2016. [10] (Part 2. 2.5 {1} {2}) A person who possesses or attempts to possess a protected animal is guilty of an offence.
NT	Territory Parks and Wildlife Conservation Act 2014 (NT) [15] (Division 8 {1}) A person must not take or interfere with protected wildlife unless authorised to do so as in the rescue of an animal.	Not required.	Territory Parks and Wildlife Conservation Act 2014 (NT) [15] (Division 6. 55 (c) & 62) Yes, can keep protected wildlife, and can bring protected wildlife into, release or take out of the Territory. Permit required. Wildlife becomes the property of the holder of the permit. 1237 permits were issued in 2017 to keep native protected wildlife.
QLD	Nature Conservation (Wildlife Management) Regulation 2006 (Qld) [74] (Section 59 {4}) Rescuer of animal must surrender the animal to a licensed rehabilitator or conservation officer.	Nature Conservation (Wildlife Management) Regulation 2006 (Qld) [74] (Section 59 {4}) Within 72 h.	Nature Conservation Act 1992 (Qld) [17] (Part 5 Division 3. 83) Yes. An animal ceases to be the property of the State if the animal is taken under a licence or permit.
SA	Animal Welfare Act 1985. (SA) [24] (Pt3. 15A) Duty of person in charge of a vehicle in case of accidents involving animals. Must inform the owner or an inspector of the accident occurring.	Animal Welfare Act 1985 (SA) [24] (Pt3. 15A) Within 24 h	National Parks and Wildlife Act 1972 (SA) [21] (Sect. 51 Taking of protected animals {1}) Subject to this part, a person must not take a protected animal or the eggs of a protected animal.
TAS	No requirement to stop or report.	General Requirements for the Care and Rehabilitation of Injured and Orphaned Wildlife in Tasmania 2012 (TAS) [28] Report at the earliest opportunity on the first day of business after receiving the animal.	Wildlife Regulations 1999. (TAS) [27] (Part 2. Section 16) Except as may be authorised by a permit, a person must not take, buy, sell or have possession of any form of specially protected wildlife or the products of such wildlife.
VIC	Road Safety Act 1986 (Vic) [75] (Section 61) If an accident occurs whereby any person is injured or any property (including any animal) is damaged or destroyed, the driver of the motor vehicle—(a) must immediately stop the motor_vehicle; and (b) must immediately render such assistance as he or she can.	Code of Practice for the Welfare of Wildlife during Rehabilitation 2017 (Vic) [33] Animals must be assessed accurately and without delay.	Wildlife Act 1975 (Vic) [31] (Part 111A 28 A {1} and {F})The Secretary may give written authorisation to a person to take wildlife for the purposes of enabling the care, treatment or rehabilitation of sick, injured or orphaned wildlife.
WA	Reporting a Traffic Crash 2018 [76] No requirement to stop, assist or report.	None	Anyone can rehabilitate wildlife without a permit.

2.2. Licensing, Permitting, and Training Requirements for Wildlife Carers to Keep Wildlife

Australia started to regulate the activity of wildlife rehabilitation in the 1990s on a state or territory basis, but no national guidance is yet available. All states and territories, except WA, mandate that to undertake the care and rehabilitation of protected wildlife, a licence or permit is required. Protected

wildlife is generally defined as native birds, reptiles, amphibians, and mammals, with some exceptions, such as dingoes in NSW [10] and common brushtail possums, Bennett's wallabies, and Tasmanian pademelons in TAS [27]. NSW legislation attempts to ensure that no rescued wildlife suffers at the hands of untrained personnel by making it compulsory to undertake professional training and comply with the standards in the code of practice. This is a condition for obtaining a license, issued under Section 120 of the National Parks and Wildlife Act 1974. NSW, to practice as a wildlife carer [77]. More advanced training is required within three years, if a licence is to be renewed. Most other states and territories have less structured training requirements, some stating that proof of experience in wildlife care in a permit application is sufficient. The worst-case scenario is in TAS, where any untrained person can take in a rescued common brushtail possum, Bennett's wallaby, or pademelon, without the requirement for a licence or notification to authorities.

Licences or permits may be issued to an individual or to an organisation, such as a wildlife shelter or a wildlife care group. By regulating the issuing of a licence or permit, authorities can require and monitor a minimum level of professionalism and preparedness of candidate carers. This level of skill and knowledge can serve as a benchmark in deciding if individuals are ready to take on the complex task of caring for rescued wildlife.

The Federal/State system of government in the USA is similar to that in Australia, with the regulation of wildlife rehabilitation and care being delegated to individual states. However, in 1982, it was found desirable and possible to create the American National Wildlife Rehabilitators Association (NWRA). This addressed the need to gather and disseminate information on the care of rescued animals and to establish universal national standards. This association is now funded mainly by tax deductible donations and also by some government grants [78]. In the United Kingdom (UK), the centralised system of government facilitates a similarly unified approach through the British Wildlife Rehabilitation Council, which was formed in 1987 and operates as a charitable organisation. Australia still has no equivalent organisation or council, even though its six states and two territories must be more manageable than 50 USA states and 100 UK counties. An Australian Wildlife Rehabilitation Conference has been a biennial event since 2008 but receives no government funding.

In the USA and the UK, having nationally recognised organisations has led to government recognition of the contribution and value of wildlife carers. These organisations have moved to ensure that a minimum standard of preparedness, knowledge, and standard of care is provided to wildlife undergoing rehabilitation. The regulations produced not only provide protection for the animals in care but also for the people engaged in rehabilitation activities and the general public. Their guiding principle is that by regulating who can be a wildlife carer, wildlife and the general public can be protected from unsatisfactory and inappropriate practices, albeit often by well-meaning but unskilled people.

In Australia, states apply varying levels of scrutiny when vetting candidate carers. For example, in NSW, WIRES, the largest Australian wildlife carer network, requires a prospective carer must have attended their in-house Rescue and Immediate Care Course. This course comprises an online theory component and a one-day practical workshop, and costs AUD$125, which candidates have to fund themselves. The course provides information about the responsibilities of a rehabilitator, and basic rescue and immediate care techniques. Upon successful completion of the course, a written authority is issued allowing the applicant to operate under the terms of the group's Office of Environmental Health licence. To participate in WIRES training and volunteer with WIRES, the prospective carer must be over 18 years old. Furthermore, for any given species the prospective carers hope to take in, undertaking advanced training in the care of that species is required, along with facilities approval, subject to inspection by an experienced rehabilitator. Specialist courses are available for avian species in general and for raptors, small mammals, macropods, wombats, large mammals, reptiles, flying foxes, microbats, possums, and glider species [79].

In the mid-range of this notional scale of regulation implementation, are QLD, SA, TAS, NT, and the ACT, which administer a written questionnaire as part of a candidate's application to care for wildlife. For example, QLD issues a form that requires details of the length of wildlife caring experience,

facilities and training obtained, the available support networks, and the anticipated conservation benefits that will be achieved from caring. If wanting to care for specialist species or genera, candidates must supply the details of two referees, who have to demonstrate their own knowledge and experience and can verify the skills of the applicant [80]. Specialist species and genera include cassowary (*Casuarius casuarius*), emu (*Dromaius novaehollandiae*), echidna (*Tachyglossus aculeatus*), koala (*Phascolarctos cinereus*), platypus (*Ornithorhynchus anatinus*), hawks, eagles, kites and falcons (*Falconiformes*), owls (*Strigiformes*), diving-petrels, storm-petrels (*Procellariiformes*), penguins (*Sphenisciformes*), tropical birds, frigatebirds, cormorants, shags, darters, gannets, boobies and pelicans (*Pelecaniformes*), whales and dolphins (*Cetacea*), bats (*Chiroptera*), dugongs (*Sirenia*), sea turtles (*Cheloniidae*), and snakes (*Serpentes*). In SA, applicants are required to provide detailed information to substantiate their knowledge and experience in the husbandry of the species to be kept [81]. In the ACT, a candidate must become a financial member of a licensed wildlife organisation and then complete basic training to care for wildlife. Under the carers' code of conduct and code of ethics issued by ACT Wildlife, carers are expected to complete ongoing advanced training, as required, relevant to the species for which they hope to care [82].

At the less demanding end of the scale is WA, where potential wildlife carers are merely encouraged to "explore and understand the principles underlying the standards for wildlife rehabilitation in Western Australia" [38] but, since no permit is required to rehabilitate wildlife, this is not mandatory. Licences are issued for periods from one to three years, depending on which state or territory issues them.

In recent years, many Australian states, territories, and wildlife organisations have tried to raise the professional status of carers through educational courses, workshops, and conferences. Although some wildlife carers see this as burdensome, many accept that the complexity of wildlife caring and rehabilitation demands very high standards if the welfare of the animals is not to be compromised. The American state of North Dakota sets such a high standard in this regard that it requires all wildlife rehabilitators to be licensed veterinarians. This has proved prohibitive to the extent that, in 2008, North Dakota had only one remaining wildlife rehabilitator [83]. Washington State requires that permitted wildlife rehabilitator applicants must be a licensed veterinarian or, if not a veterinarian, a wildlife rehabilitator with six months experience in wildlife rehabilitation, with a veterinarian acting as a sponsor to provide guidance [84]. Most other American states require passing an examination as the sole method of demonstrating competency to obtain a rehabilitation licence [85]. By contrast, in the UK, the Royal Society for the Prevention of Cruelty to Animals (RSPCA) has expressed concern that a licence is not currently required to rehabilitate wild animals and accordingly supports the introduction of a system that requires all wildlife rehabilitation centres and rehabilitators to be licensed [86]. The British Wildlife Rehabilitation Council only "encourages rehabilitators to learn about the ecology and behaviour of wildlife and from each other" and publishes a guide for rehabilitators, a similar system to that operating in WA. That said, the need for training is implied in UK legislation. "A person commits an offence if he does not take such steps as are reasonable in all circumstances to ensure the needs of an animal for which they are responsible are met to the extent required by good practice" [87]. An initiative to produce a consensus in Australia on national competency standards for wildlife care is yet to emerge. There is considerable variation in requirements among states and territories (Table 3).

Table 3. Australian licensing, permitting, and training requirements for wildlife carers to keep wildlife.

State/Territory	Permit/License Required	Demonstration of Competence	Permit Renewal
ACT	Nature Conservation Act 2014 (ACT) Required. Must be a member of a wildlife organisation and reside in the ACT [5] (Chapter 6 Division 6.1.4) Licences will not be issued for the hand-rearing of	Minimum requirement is to have attended an orientation course and basic bird care course. Carers are expected to complete ongoing advanced	Yearly

	young kangaroos or their release in the ACT [88]	training, as required, relevant to the species they care for [82]	
NSW	Biodiversity Conservation Act 2016 (NSW) Required [10] (No 63 Division 3. 2.17. {f})	Code of Practice for Injured, Sick, and Orphaned Protected Fauna (NSW). New fauna rehabilitators must undertake an introductory training course. Fauna rehabilitators must attend an advanced training course every three years [9] (13.1.1.)	As specified in the licence
NT	Territory Parks and Wildlife Conservation Act 2014 (NT) Permit required [15] (Division 6. 55 {1})	Territory Parks and Wildlife Conservation Act 2014 (NT). Must have previous experience, qualifications, membership to a local wildlife care group, an experienced mentor, facilities, and resources. An application for a permit may be refused if these requirements are not met [15]	Bi-annually
QLD	Nature Conservation Act 1992. (Qld) Rehabilitation permit required [17] (Pt 5. 88 {2})	Nature Conservation Act 1992. (QLD) Individuals must be experienced in wildlife care/rehabilitation or obtain endorsement by wildlife care group under their license. Specialist species require two referee reports from people of professional standing in the relevant wildlife management field [17] (Pt 5. 88 {2})	Nature Conservation Administration Regulation 2017 (Qld) Tri-annually or for the life of a protected animal kept under the permit [80] (Section 21 p. 17 e)
SA	National Parks and Wildlife Act 1972 (SA) Permit required. The Minister may grant to any person a permit to take and hold protected animals or the eggs of protected animals, if satisfied that it is desirable to grant the permit [21] (Section 53)	General Guidelines for the Management of Protected Wildlife in Captivity in South Australia 2010 (SA). Permit will only be issued if the applicant can demonstrate that they have the necessary skills, experience, and resources and resides in SA [23] (Section 4.1.)	Annually
TAS	Wildlife Regulations 1999 (TAS) Yes. A licence is required except for brushtail possum, Bennett's wallaby, and Tasmanian pademelon [27] (Part 2.5.)	Wildlife (General) Regulations 1999 (TAS). Permit to rehabilitate wildlife only given after vetting process via registration form	Variable. Given with regard to the time an animal is expected to be in care
VIC	Wildlife Act 1975 (Vic) A wildlife shelter permit is required for the purposes of wildlife rehabilitation [31] (Section 28)	Code of Practice for the Welfare of Wildlife during Rehabilitation 2017 (Vic) Wildlife rehabilitators need to demonstrate that they have acquired appropriate	Choice of one year or three years

		training or the required knowledge [33]	
WA	Wildlife Conservation Regulations 1970 (WA) Keeping of fauna in captivity. No permit or licence required. A person may temporarily keep in captivity or confinement fauna that is sick, diseased, or injured or that is abandoned juvenile fauna, for the purpose of caring for it until it recovers or becomes capable of fending for itself [37] (Part 4 28A {1})	Not required	Not applicable

2.2.1. Inter-State/Territory Differences and Anomalies in Regulations

The main differences and regulation anomalies among states and territories lie in the areas of rescuing, identifying, monitoring post-release, training, and assessment criteria for the ability to release to the wild. For example, in the ACT, the eastern grey kangaroo is designated a controlled animal so, in 2016, annual permits were issued for 11,130 to be culled for commercial reasons, and 1989 adults and 800 pouch young to be culled for conservation reasons [88]. It would seem contradictory, on the one hand, to be killing members of a species and, on the other, rescuing and raising them. Hence regulation exists, in the ACT, which confirms, "licences will not be issued for the hand-rearing of young kangaroos or their release in the ACT" [88](Section 4.3.1 {f}). Volunteer wildlife carers attending a wildlife rescue in the ACT should know that if they find a pouch young eastern grey kangaroo neonate (joey) they will be expected to euthanase it because it is classed as a non-releasable animal, and "non-releasable animals have a right to euthanasia"[89].

A practical solution to this apparent contradiction has been engineered, whereby any rescued eastern grey joeys are taken from the ACT across the NSW border to be raised by carers at Wildcare, Queanbeyan, with the proviso that they cannot be returned to the ACT for release. However, this solution raises other considerations for NSW wildlife carers, because "a person who imports wildlife into New South Wales is guilty of an offence" and "a person who, without authority, liberates a captured protected animal in a place other than the place of its capture is guilty of an offence" [73]. So, an exemption licence is required to allow this border crossing. Because, after rehabilitation, the animal cannot be released back to its place of capture, i.e., in the ACT, the NSW law requires it be euthanased or referred to the Department of Environment, Climate Change, and Water [11]. The ensuing effect on the animal's welfare as a result of time delays, travel, and changing environments could be considerable, just as the effects on the moral and mental well-being of wildlife carers could be significant.

The identification of animals is critical for the analysis of projected needs and welfare outcomes when seeking to monitor the progress of native fauna through the rescue, rehabilitation, and release procedure. None of the states or territories require a rescued animal to be individually identifiable. Indeed, the opposite is the case in VIC, WA, and QLD, where marking an animal is discouraged and those who do so are guilty of an offence. The Victorian Wildlife Act, 1975 (VIC), part v11 section 51 states: "any person who marks protected wildlife by means of a ring, band, dye, or other means whatsoever without the authority in writing of the secretary shall be guilty of an offence against this act" [31]. The Western Australian Standards for Wildlife Rehabilitation 2015 (WA) Chapter 2 (Regulation 28A) states: "if fauna are marked in any way the rehabilitator is no longer 'caring for sick or injured fauna' but is conducting research" [38]. Also, in section 57 of the Wildlife Conservation Regulations (WA) 1970: "a current licence is required from the department to mark fauna for identifying purposes post release" [37]. In QLD, the Nature Conservation Act 1992 (Qld) section 15.2.10,

states that, "tagging, banding, or other marking, including microchip or passive integrated transponder (PIT) implanting must only be performed as part of a Department of Environment and Science approved program" [17]. NSW takes a somewhat different approach and actively encourages the marking of rehabilitated wildlife, albeit not until the time of release of the animal. The NSW Code of Practice for injured, sick, and orphaned protected fauna 2011 (NSW) Section 12.3.1.4. states: "fauna rehabilitators should arrange for fauna to be tagged, banded, and micro-chipped or marked as appropriate for individual identification prior to release" [9]. TAS and NT are the only jurisdictions in which marking or labelling a rescued animal may be required by the relevant authority upon acquisition by the rehabilitator. In TAS, the Nature Conservation Act 2002 (TAS) Part 2 section 17 states: "the Secretary may direct the holder of a licence to mark or tag wildlife in the holder's possession" [26]. In the NT, the Territory Parks and Wildlife Conservation Act 2014 (NT) Pt 1V Division 6. 57. (1) (b) v, states: "authority may require the labelling or applying of markings to the animal" [15].

There are lengthy descriptions in all the state and territory regulations about how and when inspectors can examine rescued animals and their records, and how all the different permits operate. So, it is surprising to note that none of the states or territories require any rescued animal to be individually identifiable. This abiding lack of identification negates the ability of any third party's inspection of a carer's practice to be reliable. Conceivably, for any record being kept, an animal that escapes, dies, or is even given away can easily be substituted with another. An animal that is inappropriately returned to the wild and then causes damage or injury cannot be traced back to the person who released it. This renders unsustainable any investigation of the offence of releasing an animal without a permit or any claim for damages. These seem to represent compelling reasons for identification being required by regulation.

An illustrative case of human wildlife-related injury from TAS is one of a wombat (*Vombatus ursinus*) attacking people and causing injuries that resulted in their needing hospital treatment [90]. The suspicion was that the wombat had been hand-reared and was not able to cope after being released to the wild. The wombat was captured and euthanased by the Tasmanian Parks and Wildlife service. Had it been identifiable by a microchip or tag, the outcome could have been different for the animal, as well as for the person who may have released it [90]. Had its history been known, the wombat may have been suitable for captive housing in a wildlife park and a life in captivity, and the carer identified and appraised in rearing-and-release protocols.

Similarly, if rehabilitated wildlife that was returned to the wild were identified by microchip, all roadkill and other casualties could be scanned to reveal if they were originally injured animals or ones that had been hand-reared. This would provide a valuable source of data for research into survival rates.

2.2.2. Intra-State/Territory Differences and Anomalies in Regulations

Beyond the apparent inter-state/territory contradictions in legislations described above, there are similar apparent contradictions and anomalies in intra-state legislation. One example concerns the following aspect of QLD legislation: "tagging, banding, or other marking, including microchip or passive integrated transponder (PIT) implanting must only be performed as part of a Department of Environment and Science approved program" [91] (Section 15.2.10). This provision seems to be contradicted by two others, namely that "a register must be kept by each wildlife rehabilitator for all protected animals rescued or cared for including identifying number or name" [19] (Section 16.2.1. 2.), and "to gauge the effectiveness of various rehabilitation and release techniques, post-release sightings of known rehabilitated wildlife should be recorded and kept" [19] (Section 16.3.2.).

A further example of contradictions in intra-state legislation appears in Victorian legislation, stipulating that, "The release site should be suitable habitat in the general facility from which the animal was originally collected. For instance, if the animal were found injured on a highway, an area of bushland adjacent to the highway would be a suitable release site" [33] (Case assessment, bullet point 15), but requiring that "the factors that lead to the original injury or condition must not pose an unacceptable risk to the animal upon release' [33] (Case assessment, bullet point 18). The suggested

example of using an area of bushland adjacent to the highway to release the animal where it was injured would surely present an unacceptable risk of a WVC.

2.3. Standards of Care, Facility Requirements, and Identifying of Animals

All Australian states have issued either a code of practice, general guidelines, requirements, or standards for the rehabilitation of wildlife. For the territories, the NT has individual guides for caring for animals and the ACT is in the process of developing a code of practice. A draft code of practice for the Welfare of Native Wildlife—Rescue, Rehabilitation, and Release is currently with the ACT Animal Welfare Advisory Committee [92].

In general, the purpose of these codes is to benchmark standards to provide the minimum acceptable animal welfare and conservation outcomes. Notably, all states and territories stipulate that all costs involved in providing wildlife rehabilitation are to be met by the wildlife carers (Table 4).

Table 4. Standards of care, facility requirements, and identifying of animals in the states and territories of Australia.

State/Territory	Code of Practice Issued	Provision of Facilities	Animal Identification
ACT	No (In progress)	Must provide all equipment and some food costs [82] (para. Equipment)	Rings for identification allowed on birds, otherwise not required
NSW	Yes	The Department of Environment, Climate Change, and Water will not provide recompense for expenses incurred by rehabilitators [11]	Yes. Fauna rehabilitators encouraged to mark animals [73] (section 2.36)
NT	No code, but individual guides available, e.g., about caring for wildlife, caring for macropods, caring for raptors [14]	Wildlife carer's responsibilities. A wildlife carer's work is voluntary and costs for food, bedding, cages, equipment, and vets must be met by the carer [14]	Can be required to do so [16]
QLD	Yes [19,91]	All costs met by wildlife carers [19]	No, except if authorised [19]
SA	Yes [23]	All costs associated with the rescue, transport, and rehabilitation of protected wildlife are to be met by the individual carers [23]	No requirement
TAS	Yes [28]	Funding for feed, housing, veterinary care, and emergency situations is the responsibility of the individual carer [28]	Can be instructed to mark the animal [26]
VIC	Yes [33]	All costs met by wildlife carers but grants are available to help with infrastructure and training costs [93]	Must not mark animals without a licence [31]
WA	Yes [38]	No government funding.	No requirement

2.4. Funding and Monitoring of Carers/Carer Networks

All states and territories require at least some record-keeping by wildlife carers, but the complexity of the records required varies. Four states—NSW, QLD, TAS, and WA—specify very comprehensive standards and guidelines for what records are required from wildlife carers in their codes of practice. VIC and SA require that records, pertinent to the taking and release of the animal, are maintained, but wildlife rehabilitators are only "encouraged to keep their own additional details regarding the care, treatment, and release of the animals" in VIC [94]whereas in SA, a record book is provided with a requirement for "records to be provided in accordance with the prescribed process on the permit" [23]. The NT requires a wildlife report and an application for release form to be completed by wildlife carers, but requires no details of the care provided during rehabilitation [14]. The ACT requires carers to keep records and submit returns annually[95].

NSW requires that "records must be submitted to the Wildlife Licensing and Management Unit of the Office of Environment and Heritage in an approved electronic format on an annual basis" [9] (Section 14. 1.1. para 7. 201). VIC lists key record-keeping obligations, including the need to "submit a completed return form by no later than 14 April detailing wildlife in your possession", and also supplies a record-book in which "all wildlife transactions must be recorded" [94]. However, this does not include details of the care or health status of the animals under rehabilitation. In Tasmania, wildlife carers are strongly encouraged to keep records for all animals in rehabilitation, and there is a requirement under the Wildlife (General) Regulations 2010 (TAS) section 30 for record-keeping and for providing a return of the records for inspection, when requested [27]. SA requires returns to be made annually by 30 June [22] (Schedule 5). Other states require similar records to be kept in a manner that can be readily examined, analysed, and clearly understood and made available to officials on request. Little state or territory funding is supplied to wildlife carers to meet the costs of providing these records (Table 5). All states legislate to enable officials to require wildlife carers to submit to inspection of their records, premises, or facilities they provide for rescued rehabilitating animals (Table 5).

Table 5. Funding and monitoring of wildlife carers and carer networks in Australia.

State/Territory	Financial Support Provided by State/Territory Government	Inspection of Premises and Records	Compliance and Legal Issues
ACT	No	Enforcement powers given [96] (Part 7 Divisions 7.3 and 7.4)	Obtaining Court orders and corporate penalties [96] (Part 7 Division 7.11)
NSW	Some funding given to licensed groups.	At least once every three years [11]	A person who contravenes a condition of a biodiversity conservation license is guilty of an offence [10] (No 63. 2.14 {4})
NT	Funding of AUD$50,000 per year divided between three organisations [97]	Inspection every two years [15]	The role of Conservation Officers is to implement and enforce compliance [15]
QLD	None by State. Some grants available from local councils, e.g., Brisbane [98]	Inspections may be carried out as part of a new permit application assessment, information or complaints received from the public or randomly selected audits on permit holders [18]	The Department of Environment and Science is responsible for the assessment and licensing of individuals and organisations [99]

SA	None.	Wardens can undertake random inspections [22]	Failure to comply, penalties, regulations and codes of practice enforced by wardens of the Department of Environment and Natural Resources [21]
TAS	Management of wildlife carers provided by the State Department of Primary Industries Parks, Water and Environment. No other funding given.	Random checks may occur [28]	An authorised officer may inspect facilities and records [28]
VIC	Wildlife rehabilitator grants of up to AUD$2000 per applicant available. 2018/2019 A total AUD$170,000 has been allocated [93]	Inspections are often conducted to monitor general compliance trends among authorisation holders [100]	Authorised Officers have the power to enter, inspect, or search any property and any buildings or structures other than a dwelling, as well as vehicles or boats, with or without notice [31]
WA	No funding given	No inspection requirement.	Not monitored.

2.5. Pre- and Post-Release Preparations, Behaviour Modification, Protocols to Be Followed, and Monitoring of Rehabilitated Wildlife Being Returned to the Wild

Depending on the species, the rescuing, rearing, and releasing of native animals to the wild can take up to two years. Before releasing an animal, an assessment can evaluate whether an animal has developed maladaptive behaviour. It can also evaluate whether an animal has learned behaviours appropriate to it surviving in the wild, so that its well-being will not be compromised. Post-release monitoring can validate the accuracy of the assessment.

2.5.1. Pre-Release Behavioural Advice

Advice is given to wildlife carers in all jurisdictions to avoid desensitising animals undergoing rehabilitation, i.e., treating them as a pet or member of the family and allowing exposure to humans (Table 6). It describes methods that may prevent this happening. Also, advice is given on how to prevent animals becoming desensitised to those stimuli that may require them to show a flight or fight response to survive, when released. Further advice is supplied about allowing animals to learn behaviours to equip them for life in the wild. Summarised from the codes of practice and regulations, Table 6 provides the type of advice relating to pre-release behaviour methodology on a state/territory basis. None of the codes or guidelines mention the provision of behavioural conditioning prior to release, via a behaviour modification programme, to sensitise and condition an animal to relevant environmental hazards, such as road traffic, predators, and humans. The ACT has no code of practice to offer advice, although one is under preparation [92].

Table 6. List of advice given to Australian wildlife carers for pre-release behavior conditions. When not identical, similarities in advice have been placed into a single item.

Element of Advice	NT	NSW	QLD	SA	TAS	VIC	WA
Fauna undergoing rehabilitation must be prevented from coming into contact with domestic pets/animals	✓	✓	✓		✓	✓	✓

Housing must be designed and/or positioned so that fauna cannot see domestic pets or incompatible species	✓	✓	✓		✓		✓
Housing must be positioned so that fauna is not exposed to strong vibrations, noxious smells, or loud noises		✓	✓		✓	✓	✓
Failure to recognise pet species as predators will preclude rehabilitated wildlife from being released into the wild. Animals are not suitable for release unless they display instinctual fear and avoidance towards humans and domestic pets	✓	✓		✓	✓	✓	✓
Hand-reared gregarious species must be exposed to members of the same species or family		✓	✓		✓	✓	✓
During pre-lease, exposure to humans should be greatly reduced	✓	✓			✓	✓	✓
Species that manipulate their physical environment, e.g., dig burrows or build nests, should be given an opportunity to do so		✓	✓	✓	✓		
Enclosures must allow for the display of natural behaviours and sufficient room to avoid 'stress' behaviours. They should be large enough for the animal to learn or relearn behaviours, and if occupied by several animals of the same species must be large enough to allow for normal patterns of group behaviour	✓	✓	✓	✓	✓	✓	✓
Prior to release food must be offered in a way that encourages natural feeding behaviour. Good feeding management is essential for maximum development of natural behaviour and survival techniques		✓	✓	✓	✓	✓	✓
Housing should be provided in such a way as to enable training for survival in the wild						✓	

2.5.2. Protocols for Releasing or Not Releasing Rehabilitated Native Fauna to the Wild

All states and territories declare the intention that all rehabilitated wildlife is returned to the wild, (Table 7). NSW, SA, and NT require a permit to be obtained to release an animal. TAS requires notification of the Department of Primary Industries, Parks, Water, and Environment prior to release. QLD, NSW, and VIC stipulate release but only if wildlife is assessed as being physically and behaviourally fit (Table 7). VIC stipulates release within 24 h of an animal being ready for release. WA states that release must be as soon as practicable after the animal recovers or becomes capable of fending for itself. The ACT requires rehabilitated animals to be returned to the wild but stipulates no time limit on holding suitable animals prior to release.

Table 7. Published Australian protocols for releasing or not releasing rehabilitated native fauna to the wild.

State/Territory	Release to the Wild Required	Assessment of Suitability for Release Required	Action on Non-Releasable Wildlife	Allowed to Be Kept Permanently in Captivity
ACT	Yes [96] (Section 18 Pt 2 Section 3)	No	Non-releasable animals which are inappropriate for education, foster-	Yes, under conditions [2]

			parenting, or captive breeding have a right to euthanasia [89]	
NSW	A biodiversity conservation licence under Division 3 required [10] (No 63 Part 2.6)	Yes. To be carried out by a wildlife veterinarian or an experienced rehabilitator [11]	Animal must be euthanased or referred to the Department of Environment, Climate Change and Water [11]	Yes, given specified conditions can be met [11]
NT	Must release to wild unless a permit to keep is issued [14]	No	Euthanase [14]	Yes. Permit required [14]
QLD	Only if wildlife is assessed as physically and behaviourally fit to be released [91]	Yes. To be carried out by a wildlife veterinarian or an experienced rehabilitator [91]	Euthanase or refer to authority [19]	Yes. Permit to keep wildlife can be issued under specified purposes or a recreational wildlife licence to keep a prescribed protected animal for recreational purposes (i.e., as a pet) [80]
SA	National Parks and Wildlife Act 1972 (SA) Permit to release required [21]	No. The release of long-term captive animals is rarely justified on conservation or animal welfare grounds [101]	Rescued protected wildlife which cannot be released or retained with a good expectation of quality of life must be euthanased	Permits are available to take protected wildlife from the wild, to keep wildlife as pets and to buy and sell wildlife [21]
TAS	The Secretary must be notified prior to the intended release of any animal referred to on a permit [28]	Details of the process for the release of protected wildlife and the common wombat should be provided to conservation branch prior to release [28]	No direction given	Permits required for partly or wholly protected species Brushtail possums, Bennett's wallabies and pademelons can be kept without a permit [27]
VIC	Victorian Wildlife Shelter and Foster Carer Authorisation Guide Condition 21. Yes, must release wildlife within 24 h of it being ready for release (condition 21)	All animals to be released must be inspected by a veterinarian or experienced wildlife rehabilitator [33]	Wildlife that cannot be released must be euthanased [100] (condition 22)	No. Note: only legal to keep certain classes of wild pigs as pets [31,32]
WA	Yes. A person who keeps fauna under	Self-regulated. Advice on release	Must give the animal to	Not allowed [36]

sub regulation (1) must release it	evaluation given [38]	appropriate authority	

2.5.3. Suitability of Rehabilitated Animals for Release

Assessment of the suitability for a rehabilitated animal to be released varies considerably among the states and territories. NSW and QLD require an assessment by a wildlife veterinarian or an experienced rehabilitator, while VIC requires an assessment by a veterinarian or experienced wildlife rehabilitator. No definition is given to explain the difference between a veterinarian and a wildlife veterinarian or what constitutes an experienced wildlife rehabilitator. TAS requires a declaration of the process that will be used for the release of protected wildlife to be submitted to and approved by the conservation branch of the Department of Primary Industries, Parks, Water, and Environment. WA has adopted a self-regulated approach, with advice on the suitability of rehabilitated animals for release being given to wildlife carers. SA does not require an inspection but, significantly, states that: "the release of long-term captive animals is rarely justified on conservation or animal welfare grounds" [101]. The NT and ACT require no assessment of the suitability of rehabilitated animals for release. None of the states discuss or give protocols necessary for an objective behavioural assessment of such suitability, how it has to be undertaken, or where the necessary assessment facilities should be located, nor is there any mention of how assessors should be remunerated for their work.

Assessment of the suitability of rehabilitated animals to return to the wild can involve the assessors making life-or-death decisions for the animals, in some states. Both the animals' health and behavioural suitability for release need assessment (Table 8). In a 2012 survey of current rehabilitation practices in Australia, Guy and Banks [102] observed that there were no consistent criteria for the suitability of an animal for release, although as early as 1989, Kleiman proposed five behavioural attributes as a minimum required for animals to survive in the wild after rehabilitation [103]. These were the ability to avoid predators, acquire and process food, interact with conspecifics, construct and discover shelter sites or nest sites, and to navigate through its natural habitat. Additionally, Kleiman added that to "achieve a wild state, an animal must show a fear and avoidance of humans". All states and the NT stipulate that some criteria, similar to those behavioural attributes in the above list, must be met, but the details differ considerably among jurisdictions [9,14,19,23,28,38,104], as summarised in (Table 8). Evidence of the need to adopt behavioural attributes is suggested from the analysis of survival rates of Tasmanian devils (*Sarcophilus harrisii*) recently released to the wild during a rewilding project [105]. The longer the period in captivity, the higher the mortality when released, particularly from WVCs. This suggests that a lifetime of habituation towards humans and vehicles, as a result of daily exposure in captive facilities, compromised the Tasmanian devils' vehicle avoidance skills. In their analysis of survival rates, Grueber et al. concluded: "Our results imply that long-term captive breeding programs may produce animals that are naïve to the risks of the post-release environment. Growing evidence suggests that behavioural and genetic changes mediated by a captive-rearing environment may negatively impact the suitability of captive animals for release" [105].

Table 8. Standards of health and behaviour required of rehabilitated wildlife in various jurisdictions prior to release in Australia.

	NSW	NT	QLD	SA	TAS	VIC	WA
Health							
1. Recovered from injury and/or disease	✓	✓	✓	✓	✓	✓	✓
2. Not known to be sterile/unable to reproduce			✓				
3. Weight and condition are within an appropriate range	✓	✓	✓		✓		✓
4. Appropriate fitness levels	✓	✓		✓	✓	✓	✓

5. Not a biosecurity risk				✓			
6. Pelage, plumage, scales or skin is adequate for survival	✓	✓					✓
7. Acclimated to prevailing climatic conditions	✓		✓		✓	✓	
8. Must be independent of its natural mother					✓		
9. Salt tolerant, for marine species	✓						
10. Of sufficiently mature age for independent survival							✓
11. Weaned off all unnatural feedstuffs					✓		
12. Must be released before sexual maturity					✓		
Behaviour							
13. Can recognise, catch and consume appropriate, naturally available food	✓	✓	✓	✓	✓	✓	✓
14. Can recognise and successfully avoid predators and domestic carnivores	✓		✓		✓	✓	✓
15. Not attracted to humans or sights, sounds, or smells that are specific to captivity	✓		✓				
16. Show a fight or flight response similar to that shown by wild conspecifics		✓				✓	✓
17. Can navigate effectively through its natural environment	✓		✓		✓	✓	✓
18. Can recognise and interact normally with other members of the same species	✓	✓	✓		✓	✓	
19. Mark its territory, if applicable						✓	✓
20. Find or construct shelter						✓	✓

The complexity and detail of the above standards mean that it is unlikely that, regardless of their experience, lay rehabilitators would be qualified to assess the health requirements in Table 9. If these are to be met, it should be mandated that all animals be examined by a veterinarian, qualified in wildlife medicine, before release. Undertaking behavioural assessments that could lead to either euthanasia or the inappropriate release of an animal requires considerable expertise and facilities and the need for a thorough knowledge of the species' wild behaviour [106]. The Australian regulations concerning the release of rehabilitated wildlife seem to assume that a veterinarian or experienced wildlife carer has these competences. This appears to be a flawed assumption when one considers that the training time allocated to animal behaviour and veterinary behavioural medicine during the training period to become qualified as a veterinarian is constrained.

Table 9. Pre-release and post-release behaviour methodology and monitoring of native wildlife in care, as proposed in the states and territories of Australia.

State/Territory	Pre-Release Behaviour	Programmed Behaviour Modification Before Release	Post-Release Monitoring
ACT	Code of ethics. Releasable native fauna should be maintained in a wild condition and released as soon as appropriate [89]	None specified	No
NSW	Comprehensive advice given [9] (8.2.1.5. &10.1.1.4.)	None specified	Encouraged [107] (12.3.1.1&4)
NT	Advice on housing and avoiding interaction with pets and humans [14]	None specified	None

QLD	Comprehensive advice given [91]	None suggested	Few studies undertaken
SA	Advice given not to humanise or allow the animal to be imprinted [23]	None specified	None
TAS	Comprehensive advice given [28]	None specified	None
VIC	Comprehensive advice given [104]	Housing should be provided in such a way as to enable training for survival in the wild [104]	None specified
WA	Advice on pre-release conditioning given [38] (Chapter 1. (7) Stage 3)	None specified	Suggested [38] (Chapter 1 {9})

It is one matter to set behaviour standards, but the practicality of complying with these assessment standards could be time-consuming and difficult to implement, and, for some species, require large infrastructure and technical facilities. Items 15 and 16 in the behaviour assessment (Table 9) state: "an animal must not be attracted to humans or sights, sounds, or smells that are specific to captivity" and "must show a fight or flight response". Legislated animal behaviour traits cannot be objectively assessed unless appropriate facilities are provided. Reintroduction biology is an emerging field of science[103]. However, the release of thousands of hand-reared native animals back to the wild through the wildlife carer networks [62] provides research opportunities for reintroduction research that includes experimental studies.

2.5.4. Action on Non-Releasable, Rehabilitated Wildlife

If, during the rescue, rehabilitation, or assessment prior to release, an animal is deemed to be non-releasable, the protocols to be followed vary from mandatory euthanasia to being retained in captivity. In VIC, "non-releasable" equates to euthanasia. The Wildlife Shelter and Foster Carer Authorisation Code 2018 states, "you must euthanise wildlife that cannot be released" [100] (condition 22). The other jurisdictions demonstrate more flexibility. For example, NSW states the Department of Environment, Climate Change, and Water will assess whether a rehabilitator or others could hold un-releasable animals permanently in captivity, e.g., if the animal will serve as a companion in a social group, or will be involved in a recognised education program, or will be involved in scientific research [11]. If the above is not possible, the animal must be euthanased. In SA, it is possible to obtain a permit to keep rehabilitated wildlife and to buy and sell wildlife [21] (Schedule 2, Section 3). However, if a reasonable expectation of a high quality-of-life is not available then the animal must be euthanased. Neither the timescale for assessing satisfactory expectations nor the methodology to obtain them are specified. The ACT allows the keeping of a non-releasable animal under licence as long as it was not taken from the wild, which would be an offence under the Nature Conservation Act [5]. The NT permits the keeping of wildlife as pets under licence, but advise that if the animal is unsuitable for release and has a poor chance of survival in the wild then it should be humanely euthanased [14] (option 1.2.). QLD has a mechanism whereby a non-releasable animal can be referred for placement through the Queensland Species Management Plan and, if not, must be euthanased [91] (Section 12.3.2.). WA regulates that wildlife carers cannot keep non-releasable animals and must surrender them to a wildlife officer or, with permission from the Minister for Agriculture and Food, give them to a person who is licensed to keep them. It states that, if it is unlikely that the animal cannot be fully rehabilitated, euthanasia should always be considered the preferred option [38] (Chapter 5 paragraph 3). In the ACT, Jones comments that, "we strongly discourage the keeping of any non-releasable wildlife by carers and have thus far been successful in finding new homes for these creatures in wildlife parks, etc." [95].

2.5.5. Release Methodology

Post-release behavioural advice is aimed at using the most likely method to secure a successful release. However, post-release monitoring is minimal (Table 9). Most jurisdictions recommend either a hard release or soft release and that, either way, the animals should be returned to the location where they were found, if possible. A hard release is generally defined as one where the animal is released without any support, whereas a soft release is where an animal is provided with temporary post-release support, which may include supplementary feeding, shelter provision, or protection from predators. There is consensus that the sooner an animal is returned to the wild the better. NSW is the only jurisdiction to actively encourage post-release monitoring. "Fauna rehabilitators should arrange for fauna to be tagged, banded, and microchipped or marked for individual identification prior to release. Fauna rehabilitation groups and zoological parks are encouraged to participate in post-release monitoring programs to determine survivorship" [9] (Section 12.3.1.4.). In WA, the Standards for Wildlife Rehabilitation in Western Australia state to "monitor post-release if possible" [38] (Chapter 1, {9}).

2.6. To Release or Not Release Rehabilitated Injured and Orphaned Native Animals to the Wild

In general, all Australian jurisdictions require that rehabilitated animals be returned to the wild. No distinction is made between those who are injured and those who have been orphaned and will require hand-rearing, and thus will spend a considerable time in rehabilitation. The consequences of this lack of distinction is examined.

2.6.1. Returning Rehabilitated Injured Animals to the Wild

Adult and juvenile native animals raised in the wild have all their innate and learned behaviour instincts intact when they are injured and rescued. Unless they remain in captivity for a prolonged period or are subjected to inappropriate housing and handling, these behaviour patterns will persist and become operative once released. As long as release protocols are followed (Table 9), the animal will have an opportunity to survive and a life equally worth living as that which would have been expected prior to the injury. From a conservation perspective, there is also another advantage to returning a rehabilitated, previously injured, animal to the wild. Learning theory would suggest that an animal subjected to a traumatic event, such as being struck by a vehicle, and able to pair this with a salient stimulus, such as the sound or headlights of a vehicle, can remember this after only one incident [108]. The animal learns to avoid the aversive stimulus [109]. Similarly, it was shown that avoidance behaviour can be extremely persistent [110]. Also, there is evidence of aversive sign-tracking systems, in which animals' tendencies to withdraw are determined by Pavlovian contingencies, which render them reliable signals for the occurrence of an aversive stimulus, such as a WVC [111].

The significance of this is that an animal who has been injured by a road vehicle could remember the sound, sight, or even odour that pre-empted the impact with the vehicle. When returned to the wild and encountering salient visual, olfactory, or auditory signals, the animal is likely to show a flight avoidance response and hence boost its chances of survival. A female could pass this behaviour to its offspring through association with avoidance responses and enhance their chances of survival. Innate behaviour is the result of the interaction of the species with its environment during evolution, and success at staying alive leads to coded information in the genome [112]. In a gregarious species, such as a kangaroo, an individual survivor might well become the animal that passes this behaviour to the rest of its mob. When a critical number of animals have survived such encounters with vehicles and have learned avoidance, then over multiple generations the avoidance behaviour may eventually become an innate behaviour similar to the way that, after being hunted during the last 50,000 years, kangaroos became innately wary of humans.

2.6.2. Releasing Hand-Reared Orphan Animals to the Wild

Requiring that hand-reared orphans must be returned to the wild may be difficult to defend on conservation, ethical, moral, and practical grounds. Indeed, the effectiveness of releasing hand-reared

animals to the wild for conservation reasons is questioned by many authorities both in Australia and worldwide. VIC and SA state that rehabilitation has limited benefits for biodiversity conservation [23,33], and that releasing long-term captive animals is rarely justified on conservation or animal welfare grounds [16]. In an example specific to the ACT, it is stated that, "there is no justification for the hand-rearing and release on conservation grounds, as the eastern grey kangaroo is an abundant species" [2]. In the UK, orphans who have been hand-reared are released back into the wild[113]. However, Kelly et al. state, "such releases receive very little attention from conservation biologists as they are seen to have little conservation value"[114]. This may be because the most numerous species that are hand-reared in the UK, such as badgers (*Taxidea taxus*) and fox (*Vulpes vulpes*) cubs, deer (*Cervidae cervinae*), squirrels (*Sciuridae*), and hares (*Lepus Europaeus*) are well represented in the wild [113] and the reintroduction of hand-reared orphans into the wild has limited success [115–118].

The consequences of returning hand-reared animals to the wild are not confined to their own generation. For example, there may be behavioural aspects to the release of hand-reared orphan animals that may actually compromise conservation by associative learning and the subsequent social transmission of conditioned responses. This runs counter to the positive effects of rehabilitating and releasing injured animals mentioned previously. For example, consider a female marsupial roadkill victim that did not react to a motor vehicle threat and get out of the way quickly enough. This might be because of a diminished fear or startle response to a stimulus or an innate slower reaction time. If an orphan is rescued from this female's pouch, hand-reared, and released to the wild, the high-risk behaviour of the mother may have been vertically transmitted to her offspring, thereby making the released orphan itself more likely to become a roadkill victim. It could be argued that releasing hand-reared orphans to the wild over several generations inadvertently selects for roadkill.

2.7. The Mental and Physical Well-Being of Rehabilitated Native Animals Released to the Wild

Contemporary animal welfare practice has evolved from just the physical provisions included in the Five Freedoms [119] to the need to consider a life worth living [120–122]. It has been argued that a new moral framework is needed that connects the treatment of animals more directly to fundamental principles of liberal-democratic justice and human rights [123]. These encompass the mental as well as the physical well-being of animals. Mental well-being encompasses the mental and emotional state of an animal [124–126]. This concept requires that animals should feel well mentally and should not be subjected to excessive negative emotions, such as fear, stress, pain, and hunger. In addition to avoiding negative emotions, animals should be able to experience positive emotions in the forms of pleasure or contentment if they are to have a life worth living, for example, by being able to perform important, normal behaviours, such as rest, play, or social interactions with conspecifics. Ethologists accept that animals have feelings [127–130] and thus it can be expected that hand-reared animals released to the wild could suffer.

2.7.1. Post-Release Behavioural Effects of Habituation or Desensitisation to Humans Prior to Their Release, on Hand-Reared Rehabilitated Animals

Australian marsupial pouch young (joeys) account for most of the orphans that require hand-rearing and release, but there are also birds (*Aves*) and bats (*Chiroptera*). The ability to rear chicks successfully to the time of release and survival in the wild has been demonstrated in several conservation projects, such as saving the Californian condor (*Gymnogyps californianus*) [131], and the whooping crane (*Grus Americana*) [132]. By avoiding human contact, wearing costumes that resembled adult birds and feeding using 'puppet heads' or artificial bills, any possibility of imprinting, habituation, or desensitisation to humans was successfully avoided. Similarly, with hand-reared bats (*Chiroptera*), it is possible to rear orphans without imprinting them on humans. Specifically, as demonstrated with pipistrelle bats (*Pipistrellus* spp.), as long as there is extensive pre-release conditioning of flight training in a large flight cage, release to the wild is not characterised by reduced survivorship [114]. Thus, there is no moral or ethical reason to believe the welfare of hand-reared birds or bats is compromised when released to the wild.

In contrast, when hand-rearing juvenile mammals (joeys) that require milk substitute feeding from a bottle it is difficult, if not impossible, to avoid habituating them to the olfactory, auditory, and visual stimuli of human presence and the human environment. A joey in a cloth pouch hanging from a hook in a kitchen does not experience the smells of a female's pouch, the sounds of the bush, or the natural movements of a mother. Indeed, the likelihood of the joey associating anthropogenic stimuli with food and comfort seems considerable. As it matures, the joey may well experience the interior of a vehicle when travelling to be examined by a veterinarian, or hear the sounds of a neighbour's dog barking, or other auditory and olfactory attributes of urban existence. All of these experiences may extend the joey's desensitisation to the human environment and to stimuli, that ordinarily would produce a fear/flight response when released, if the chance for survival is to be optimised. Taking all of this together, the possibility that hand-rearing rescued offspring of wild animals and releasing them to the wild through a hard, soft, or managed release operation leads to a life worth living and the ability to exhibit normal behaviour seems remote. Under these circumstances, the normal behaviour for such animals is the behaviour they have learned in captivity. Many wildlife carers may equate releasing an animal and seeing it disappear into its natural habitat with success but, without post-release monitoring, this may be an unfortunate convenient illusion. The mental state of the released animal may not be the happy state that carers may prefer to assume.

2.7.2. Assessment of Likely Mental and Emotional State of Rehabilitated Animals Released to the Wild

To comment on an animal's welfare, one must be able to assess objectively whether it is compromised by its mental and emotional state. Although there is the danger that the use of a definition based on human experience and feelings will increase the myopia that already exists when considering animal welfare, the definitions stated by Broom [133] and Hemsworth et al. [134] define animal welfare appropriately, as "the state of an individual animal as regards its attempts to cope with its environment" and as "a state within the animal, and a simplistic definition might be how the animal feels now". The common human intuitive reaction that placing rehabilitated animals into the wild is the natural approach to take may not consider what the animal feels. In defining 'naturalness', Yeates concludes that "the vague assumptions that naturalness is reliably associated with better well-being are unfounded" [121]. Australia is set to follow Europe, New Zealand, and Canada in recognising, for the first time in animal welfare legislation, that animals have sentience, i.e., the ability to have feelings such as pain and, pleasure and the ability to suffer. The ACT is leading other states and territories in introducing this concept in the Animal Welfare Legislation Amendment Bill (ACT) 2019 [72].

The basic question for hand-reared animals is what price they will have to pay to try to survive once released. Before releasing a hand-reared animal to the wild, it is appropriate to assess indicators of its likely mental, emotional, behavioural, and cognitive state after release. Socialisation and learning to live in an environment are vital for the survival of all animals. When striving to obtain and secure food, prey species are particularly exposed to agonistic behaviour from other animals motivated to defend themselves. If predator species have not learned how to react to auditory, olfactory, and visual stimuli of conspecifics, when confronted, their chances of survival are diminished. Similarly, with no cognitive map of the habitat into which they have been released to the wild, naïve orphans could go through a period of fear and stress, as their normal behaviour is no longer appropriate to surviving in their new environment.

Resource holding potential is a term describing the motivation and capacity an individual has to continue to fight, work, or endure [135,136]. It can be estimated as a function of the motivation to obtain a resource, what needs to be done to acquire the resource, and what it will cost obtaining and defending that resource. The number of days released animals remain in a fearful state before their resource holding potential is adequate, and the level of pain from hunger or thirst sufficient to motivate them to overcome their fear of a novel environment, are difficult to estimate.

2.7.3. Practical Issues of Undertaking Mental and Physical Assessments of Released Rehabilitated Animals

To understand whether the mental and physical well-being of a released animal are being compromised would require the tracking of all released animals over an extended period and a reliable system for identifying them. The financial cost of meeting such a tracking requirement would be considerable, with suitable tracking devices costing over $1000 each and, furthermore, it would be unlikely to obtain ethics committee approval [137]. In contrast, identifying each animal by means of a microchip, leg-band, or tag could cost as little as $20/animal, and so would be a practical proposition.

2.7.4. Alternatives to Releasing Rehabilitated Animals to the Wild

If it is considered that for conservation, ethical, moral, and practical reasons, rehabilitated, hand-reared wildlife should not be placed into the wild, then an alternative is required. Available alternatives include euthanasia or lifelong captivity. If it is possible to provide these animals with a life worth living in captivity, this may represent a workable solution. This idea already has a degree of acceptance. The National Parks and Wildlife Service Rehabilitation of Fauna Policy 2010 (NSW 2010) states: "The NPWS will consider on its merits any application from a zoo or fauna park licensed under the Exhibited Animals Protection Act 1986, to recruit protected fauna which has been hand-raised or is undergoing rehabilitation into the exhibition stock holdings of that park. Approval for the acquisition or retention of such an animal will be subject to the concurrent approval of the Registrar of the Exhibited Animals Protection Act."

With the exception of VIC and WA (Table 6), carers or institutions are permitted to keep wildlife under licence. The idea of keeping wildlife as pets has been explored by a feasibility study featuring conservation, welfare, and industry metrics [138]. This study advocated a conservation-through-sustainable-use strategy and placing a monetary value on native animals as companions to contribute to their protection in the wild. The study also says that replacing some non-native companion animals, such as cats, with natives could also have conservation benefits. A separate report by Hopwood also argued that the Government had erred by preventing people from keeping some species of Australian fauna as pets [139]. Enriched enclosures can be built to provide wildlife with an area appropriate to their behavioural needs. Examples of how wildlife can be kept in captivity in natural surroundings and enjoy a life worth living include Mulligans Flats in the ACT, Australian Reptile Park's 'Devil Ark' facilities at Barrington Tops in NSW, and the Devil Island Project's free-range enclosure facilities in TAS.

Many wildlife rehabilitators equate release with success, but very few post-release survival studies have been conducted to support or, for that matter, challenge this view. Those that have been conducted present a largely unsuccessful report, with the notable exception of wombats released onto private property [69]. The attempt to re-introduce Eastern Quolls (n = 20) on the Australian mainland in 2018 met with an over 70% death rate in three months due to vehicles, natural predators, and foxes [140]. A similar fate befell Tasmanian devils (n = 39) released on Tasmania's Forrester Peninsula, with a 25% mortality within the first weeks of release [141]. It is important to consider how wildlife carers might grieve or otherwise cope if they knew for certain that 70% of the animals they release die within three months [62].

3. Conclusions

This article has reviewed and critiqued the systems that currently regulate the care of Australian native animals that are rescued, rehabilitated, and released. It has found that the current systems are far from perfect, rely on many assumptions, are riddled with inconsistencies, and condone practices that may compromise the environment and the welfare of wildlife carers. The systems have evolved this way because of the influence of the state-by-state approach to policy development that prevails in Australia. Also, because the subject matter is complicated, it may have been left to policy-makers who may have been ill-equipped to craft appropriately reflective regulation. This is unsurprising, given that animal law as a discipline has been emerging only since 2010 in Australia and, in Australian tertiary education, currently exists as the inclusion of only one or two elective units in legal degrees. Many aspects of relevant policy rely on assumptions that are not based on scientific evidence. Vague

assumptions that naturalness in releasing animals to the wild is reliably associated with better well-being are unfounded [121]. This review and critique confirms that such assumptions are often wrong. It indicates the need for an evidence-driven approach to wildlife rehabilitation. Such an approach is problematic when legislation and regulatory systems are fragmented, contradictive, unenforceable or unenforced, and thwart the possibility of data collection.

A reformed national initiative could resolve many of the current difficulties that face care-givers/rehabilitators to the many and varied Australian native animals that would otherwise perish or require euthanasia. It is hoped that such a national initiative and review would result in the preparation of nationally consistent science-based, best practice guidelines for native animals in care and beyond. This could provide a significant boost to the well-being of both carers and native animals. The current regulatory controls are heavily weighted towards the rescue and rearing of the animals. However, the release of the animals to the wild raises concern for their post-release well-being. It is essential that all rescued animals that are to be released to the wild are reliably identifiable. Based on the existing regulations, expert assessment of animals' suitability for release is also required. Unless these criteria are met, releasing hand-reared wildlife to the wild should be discontinued, and other options, including allowing the keeping of some native animals and the use of large-scale facilities, such as national parks, islands, and fenced enclosures, explored.

Regulatory frameworks need to balance the needs of rescued wildlife, wildlife carers, and conservation. The public and wildlife carers need to be confident that regulation is consistent among jurisdictions and reflective of best practice for the rescued wildlife and the environment. The protection of Australian injured or orphaned native wildlife should be recognised as an important animal welfare issue.

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Chapter 4. Survey of Australian wildlife carers

4.1. Preamble. Understanding the triad of humans, wildlife and the environment in a rescue operation by seeking the views of the humans involved

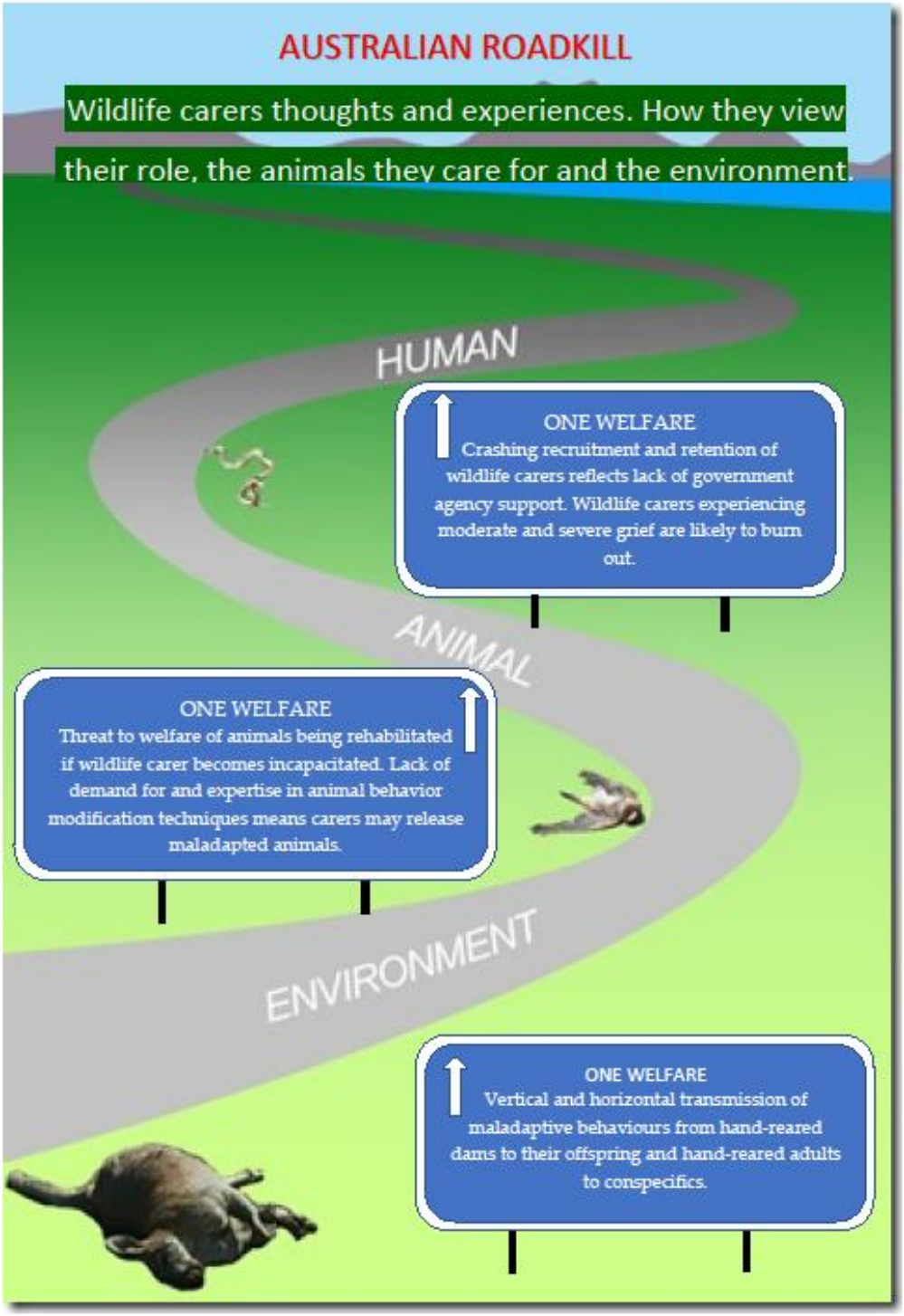
The conclusions from the previous two chapters suggested that more research was required on how roadkill rescue and wildlife rescue in general was being managed. Several gaps were exposed in the training wildlife carers receive in animal behaviour modification, in reliable animal identification systems, whether carers experience mental health issues and how they are managed and consulted by the state authorities. Information was needed from the wildlife carers to address these gaps in knowledge. The opinions of Australian wildlife carers to quantify and qualify the situation as it exists from their experienced viewpoint was used to test the hypotheses and conclusions of the previous studies. A survey of Australian wildlife carers was undertaken to address this knowledge gap.

A literature review in 2016 of global wildlife rescue, rehabilitation and release [39, 40] revealed little data on the number of people registered as wildlife carers, nor the effect that caring for injured and orphaned wildlife has on the physical, financial and mental health of the carers. Studies on Australian wildlife carers were limited in scope and demography [41-43]. The focus of these studies was mainly on the welfare of the rescued wildlife. None attempted to measure the financial and physical challenges, and time commitment associated with the emotional and mental health of those caring for the animals.

The previous two studies in Chapters 2 and 3 revealed how wildlife carers are mandated to return rehabilitated animals to the wild, yet are not allowed to identify the animals in a reliable manner such as with a microchip or passive integrated transponder. This may result in people who are motivated to help animals and the environment having to release animals, with which they may have formed an emotional bond, never knowing what happens to the animals. They do not know whether the animals experienced a life worth living or suffered in any way. Furthermore, if the prime motivation for becoming a wildlife carer is to help conserve the environment and legislation makes it impossible to know whether this is being achieved, the effect on the wildlife carers could be demotivation and mental health deterioration. This is a prime reason why human, animal and environmental considerations need to be integrated, as in the One Welfare concept. This is just one example of why an understanding of those who volunteer to help native wildlife is required, and yet is currently lacking. Likewise, a better understanding of the opinions, motivations and welfare of carers might enable actions to mitigate the continuing difficulty animal shelters and rescue organisations face in recruiting and retaining volunteers.

An international study conducted on the potential impacts on the mental health of wildlife carers revealed that their work was associated with suicidal ideations [44]. Chapter 4 addresses this issue within a quantitative and qualitative survey of Australian wildlife carers by including a novel diagnostic instrument to measure their grief. In combination with the other financial and physical stressors, the wildlife carers face the possibility of compassion fatigue and burnout.

Figure A4. One Welfare. Wildlife carers, their thoughts and experiences



4.2. Published paper: The demography and practice of Australians caring for native wildlife and the psychological, physical and financial effects of rescue, rehabilitation and release of wildlife on the welfare of carers.

The following article was published in the open access journal *Animals.*

The demography and practice of Australians caring for native wildlife and the psychological, physical and financial effects of rescue, rehabilitation and release of wildlife on the welfare of carers.

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Simple Summary: Little is known about the physical, financial and emotional effects on the 20,000 volunteers who rescue, rehabilitate and release injured and/or orphaned Australian wildlife. A survey for wildlife carers was instigated to address this knowledge gap. Collated survey responses from the wildlife carers suggested that their physical workload is on average 32 h per week but can be up to 100 h, their financial input is on average AUD5300 per year and over a lifetime of caring can go up to AUD800,000. The emotional input is such that 28% of the respondents were experiencing moderate to severe grief. Grief increases as more joeys die in care and as expenditure rises, and is age dependent. Burnout and compassion fatigue are likely outcomes. Over 65% of respondents felt that their welfare, and that of the animals for whom they care, is neglected and unappreciated by government agencies. Unless these deficiencies are corrected by financial and emotional support and workload is reduced for carers, animal and wildlife carer welfare will be compromised.

Abstract: The rescue, rehabilitation and release of injured and orphaned Australian wildlife is managed by over 20,000 carers, mostly voluntarily. These volunteers experience mental, physical and financial challenges that have not been researched adequately. This study collated the responses (n = 316) to a survey conducted among Australian wildlife carers who actively foster orphaned joeys for hand-raising and injured adult mammals for rehabilitation and release. It confirmed 86% of rehabilitators are female, 70% are over the age of 46 years and their prime motivation is an affinity with animals. The average time spent in the sector is 11.5 years, and the work week is 31.6 h, caring for 15 animals per year, with an average of 2.6 dying. The average financial commitment is AUD5300 annually and up to AUD800,000 over a lifetime. Regarding the grief experienced by carers, the lower the age, the longer the time spent, the greater the financial input and the more joeys that died, the more severe is the grief experienced. Moderate to severe grief is experienced by 28% of carers, which, coupled with other factors, could lead to burnout or compassion fatigue. Soon, wildlife carer welfare will likely be compromised unless financial and mental support is provided and their workload reduced.

Keywords: wildlife carers; One Welfare; Australian native wildlife; burnout; compassion fatigue; rehabilitation; grief; roadkill rescue

1. Introduction

The first law to prevent cruelty to animals in Australia was passed in Van Diemen's Land (now Tasmania) in 1837 by Governor Sir John Franklin. It was most probably based on British legislation enacted in 1835 (known as Pease's Act) [1]. By the 1870s, there were four animal protection societies in Australia and, over the next 80 years, the animal protection movement developed to rescue and rehabilitate injured and orphaned wildlife [2]. Currently, wildlife rescue and rehabilitation in Australia can be regarded as a major industry, with over 20,000 employees, albeit mostly volunteers, approximately 17,500 of whom are registered with State or Territory authorities [2,3]. Unfortunately, with no national political leadership or initiative to oversee this operation, it is a fragmented system that is managed on a state or territory basis. This fragmentation has resulted in considerable differences in the legislative and policy framework [4].

Any operation that relies on self-funding and the goodwill of volunteers is vulnerable to internal threats. In this instance, examples of such internal threats include financial pressures, inefficient management, unreliability of workforce numbers, as well as the widespread lack of operational health and safety monitoring of the workforce, scientific and practical protocols for wildlife rehabilitation and monitoring outcomes. External threats involve an increasing number of animals being presented for rescue and rehabilitation [5] and a decrease in the recruitment of new wildlife carers [6]. The increase in rescues reflects several anthropogenic factors, including habitat destruction, predation by introduced species and domestic pets, wildlife–vehicle collisions (WVC), farming practices and culling/hunting. Climate change increases the risk of bushfires, floods and hazardous weather events that influence the need for animal rescue. Habitat destruction has a major effect on the ecosystem, with inestimable numbers of mammals being displaced [7] and requiring rescue.

The threats from WVC and predation are exacerbated by the increasing number of vehicles on Australian roads and pet ownership. The 2015 Motor Vehicle Census reveals over 18 million registered motor vehicles in Australia, a total that had increased to 19.2 million by 2018 [8]. In 1994, in Australia, there were 3.1 million owned dogs (*Canis lupus familiaris*) and 2.5 million owned cats (*Felis catus*) [9]. By 2016, these numbers had increased to 4.8 million dogs and 3.9 million cats [10]. The role that the human–animal interaction and companion animals play in creating the need for wildlife rescue is confirmed by New South Wales's Wildlife Information Rescue and Education Service (WIRES), the largest Australian wildlife rehabilitation network, with 28 branches and over 2500 volunteers [11]. The service reported a 19% increase in the number of calls about injured animals in the previous year (2016–2017). For a twelve-month period in 2013–2014, RSPCA Queensland reported that around 8500 native animals and birds were presented at its Wacol wildlife hospital. By 2017–2018, that figure had swollen to well over 23,000.

Predation by introduced species such as the red fox (*Vulpes vulpe*s), feral and domestic cats and, to a lesser extent, domestic dogs, is a major threat [12]. The animals that, as a result, are injured or orphaned and rescued then face additional challenges in successful reintroduction. Wildlife carers are mandated to return the animals to the location from which they were rescued; so after rehabilitation, the continued presence of these predators in their habitat reduces the rescued animals' chances of survival [4].

Many injured and orphaned wildlife require euthanasia, but those assessed as suitable for rehabilitation require wildlife carers to take total responsibility for them. The duration of full rehabilitation and release can be lengthy, e.g., up to two years for some hand-reared marsupial neonates (joeys). What the animals learn during this time can be critical to the success of their eventual release. An animal that has been humanized, i.e., desensitized to sights, sounds and odors associated with humans, is unlikely to survive in the wild or have a life worth living, as it is likely to experience anxiety, fear, panic, frustration, anger, helplessness, loneliness, boredom and depression [13]. Codes of practice recognize this and state that 'releasing wildlife is the most difficult part of the rehabilitation process but should be considered the most important' and 'release options and procedures should be of the highest priority and taken into consideration at the time of acquisition of any wildlife' [14]. However, in 2000 [15], 2005 [16] and 2010 [7], it was reported that scientific studies on the pre-release treatment of animals

in animal welfare-based rehabilitation of Australian mammals simply did not exist. Eight years later they still do not exist. The lack of pre- and post-release, science-based protocols for rescued wildlife monitoring presents a worrying gap in knowledge.

Another major threat to the wildlife carer network is the finite number of trained volunteers to meet increasing demand. A recent report on wildlife volunteers in New South Wales highlighted 'some groups reporting fluctuations in annual membership of 25% and in some areas 60%' [2]. Although wildlife carers may be highly motivated volunteers, they require the same attention to occupational health and safety as commercial operators would be required to undertake in a well-functioning and reliable workforce.

Over the past 20 years, codes of practice designed to benefit the wellbeing of wildlife have been established in all Australian states to address wildlife carer training, regulation and the general structure of wildlife rehabilitation. However, two significant and persistent gaps in understanding and addressing the wellbeing of wildlife carers related to compassion fatigue (CF), described as 'a state of exhaustion and dysfunction—biologically, psychologically, and socially—as a result of prolonged exposure to compassion stress and all that it evokes' [17], and to burnout, defined as a 'state of emotional, physical, and mental exhaustion caused by excessive and prolonged stress.…[occurring]… when you feel overwhelmed, emotionally drained, and unable to meet constant demands' [18,19]. Both CF and burnout are recognized in other healthcare professionals such as intensive care unit nurses and doctors [20], military medical staff [21], audiologists [22] and mental health practitioners [23], but the possibility of their occurrence in the wildlife rehabilitation sector has not been explored in Australia. A recent survey of wildlife carers in New Zealand [24] concluded that there were significant differences in CF among New Zealand wildlife carers based on their age, gender, financial capacity and years of experience but that, overall, the incidence of CF was considered low, at an estimated 20%. However, the sample size of wildlife carers in that study, undertaken with attendees at the Wildlife Rehabilitators Network of New Zealand Conference held at Massey University in 2016, was small (n = 30), so the results should be viewed with caution. It is possible that the estimated CF percentage represents an under-estimate, especially if wildlife carers with the time, financial means and motivation to attend such a conference are not those likely to be suffering burnout or CF.

An international survey of wildlife carers (n = 534) was conducted to collect respondents' thoughts and feelings on mental health awareness [25]. Bullying, isolation, grief and financial burden were the four most significant themes explored and were proposed as reasons why carers might leave organizations or give up caring altogether. It concluded that deeper research is required to ensure that the impacts of positive and negative mental health consequences on wildlife carers are better understood.

The current study is the first to encompass wildlife carers from all states and territories of Australia. It involved a survey that was open to all Australian adults who rescued, rehabilitated and released marsupial mammals. The focus on mammals was implemented to target on the areas where hand-rearing, time in captivity, financial resources, animal behavior modification and wellbeing of carers combine to be of major significance. The aim was to assess what motivates people to become wildlife carers, their demographics, the number of animals and the species they rehabilitate, their opinion on the release methodology they use, release protocols, how well they feel supported in their role as wildlife carers, their overall financial and time contribution and the effect that wildlife rescue and rehabilitation has on their mental wellbeing as measured by the amount of grief experienced. The combined effect of these factors, coupled with lack of recognition or support and work overload, could indicate how exposed the wildlife carers are to CF or burnout.

Our overarching aim was to understand the complexities and size of the wildlife rescue and rehabilitation operation. The results were expected to reveal the threats to the future of such operations and to identify areas for potential improvement that could optimize the wellbeing of native animals and wildlife carers.

2. Materials and Methods

A quantitative and qualitative questionnaire was developed and set up online using the Research Electronic Data Capture (REDCap) system developed by a multi-institutional consortium initiated at Vanderbilt University USA in 2017. Data collection was customized for the current study by the research team with guidance from support staff at the University of Sydney. Ethics approval was obtained from the Human Research Ethics Committee at the University of Sydney under approval number: 2017/492.

The questionnaire was open to all adult Australian wildlife carers who rescued, rehabilitated and released marsupial mammals. A hyperlink containing a participant information statement, information about the project and the survey questionnaire was sent by email to wildlife carer networks across Australia inviting them to distribute it to appropriate contacts. It was also sent to individual wildlife carer license holders registered on websites. It was proposed that wildlife carers who did not have access to the internet could be invited by friends or colleagues to use their facilities to complete the questionnaire online. It was explained that all replies would be anonymous so that an individual's information would not be identifiable. However, respondents had the option to submit their email address via a separate email to the project supervisor if they wished to receive feedback on the survey. Participation in the survey was voluntary and participants could withdraw from the survey at any point by not submitting a response. Conversely, participation implied consent to the research.

The survey had four sections: demographics and motivations; contributions; knowledge and experience and mental health. Questions in the first three sections were mandatory, whereas the mental health section was optional. The questions on motivation were formulated using studies on the motivations of general volunteers and volunteers in wildlife rehabilitation [26,27]. A grief diagnostic instrument was used to evaluate the grief experienced by wildlife carers [28]. The category scores in the mental health section were based on the Diagnostic and Statistical Manual of Mental Disorders (DSM-1V) of the American Psychiatric Association [29] by the authors of the General Diagnostic Instrument [28]. A free-text general question invited the respondents to add any comments that they felt might clarify the role carers play in rescuing, rehabilitating and releasing native marsupial wildlife. If the respondents declined to answer the mental health section, they were directed to the final question, asking them if they wished to receive feedback on the results of the survey.

The questionnaire collected information on respondents' gender, age, ethnicity, birthplace, first language and place of abode. There were 13 options on what motivated the participants to become wildlife carers, and participants were asked to select as many as applied to them. The options were: I had an affinity with animals and a desire to help them; to help conserve the environment; to contribute to my community; I had the skills to rear animals; I related better with animals than humans; to learn something new; to handle animals; to challenge myself; I believed that I had a special gift in relating to animals; to socialize and meet people; to do something different; to fill a void in my life and to help me get a future job working with animals (see Table 4). Note that respondents could nominate more than one category of motivation, so the proportions across categories were not independent. Therefore, we used Markov Chain Monte Carlo (MCMC) sampling to model the set of multivariate binomial responses (i.e., response of "yes" or "no" for each of the 13 categories of motivation) using the MCMCglmm function in the MCMCglmm R-software library [30]. The null hypothesis that the proportion of males and females identifying a motivation category as a "yes" was the same, was tested by the male parameter estimate for each motivation category and determining that the MCMC 95% support interval did not include zero (i.e., reject null hypothesis at the $p < 0.05$ probability level).

Respondents were asked to report the species of animals that they had hand-reared, number of each species, the number of animals that died whilst in care and the financial and time input devoted to caring for the animals. They were also asked to give an approximation of the average hours per week they had spent doing volunteer work as a wildlife carer, during the year 1st July 2016 to 30th June 2017.

There were 16 questions on the length of service of wildlife carers, formal instruction or training received, methods used for the pre- and post-release of animals, identification of animals, and

respondents' opinion on housing, release methodology and improvements that could be made to optimize welfare for the animals and the financial and time input of carers (Table 1)

Table 1. Questions asked of wildlife carers.

	Questions Relevant to the year July 1st 2016 to June 30th 2017
1	In which year did you become a volunteer wildlife carer
2	Which of the following Australian native marsupial joeys; rescued as orphans, have you hand-reared
3	How did the joeys you hand-reared become orphans
4	Give an approximate figure for the total number of rescued joeys that died whilst you were caring for them
5	What teaching did the animals receive before they were released
6	What was the method used to release any animals
7	Were the animals identifiable
8	Did you receive any feedback indicating what happened to the animals after they were released?
9	Have you received any formal instruction or training in animal behavior modification techniques?
10	Which method would you prefer to use to release the animals you reared
11	Which method would you prefer to see used as a temporary home
12	What should happen to an animal that cannot be released?
13	Are the methods of release that are currently being employed optimal for the welfare of the animals
14	Approximately how many hours per week on average did you spend doing volunteer work as a wildlife carer?
15	What was your approximate personal financial contribution?
16	During the time you have been a wildlife carer give an approximate overall total for the personal money you have spent on caring for animals?

Respondents could select one of five methods that they used when releasing rehabilitated animals. Respondents were also asked to mention the methods of release they would prefer to use if given a choice; i.e., one not constrained by regulations or codes of practice. Respondents were asked to select the place where they would prefer to house a rehabilitated animal if it required temporary housing prior to release and about the fate of an animal if it was deemed unsuitable for release into the wild after rehabilitation.

Three questions explored how well wildlife carers thought they were supported in their role. Nine further questions concerned distressing events, regarded as losses, that may have occurred in the last year and participants were asked to select those they had experienced (Table 2).

Table 2. Distressing events that had impacted wildlife carers.

	Questions Concerning Distressing Events Wildlife Carers Had Experienced
1	Receiving adverse comments about the way you care for animals
2	Rescuing pouch-young or an injured joey from a dead female
3	The death or serious illness of an animal in your care
4	Releasing animals to the wild and not knowing whether they survived or had a life worth living
5	Had an animal taken by authorities, hard released to the wild and which died shortly afterwards
6	Financial hardship as a result of the cost of caring for animals
7	Health-related loss
8	Loss of freedom through being a carer
9	Other

A validated grief diagnostic instrument (GDI) [28] was used to assess the grief experienced by wildlife carers (Table 3).

Table 3. Losses experienced by the wildlife carers and the effect of these losses on their thoughts and behavior.

	Questions about the Effect of Experiencing the Losses Experienced by the Wildlife Carers
1	Have thoughts of the losses made it difficult for you to concentrate, remember things or make decisions
2	Have you experienced images of the losses surrounding the event
3	Have you found yourself longing for what has been or will be lost
4	Have reminders of the loss caused you to feel longing for what has been or will be lost
5	Have thoughts or reminders of the loss caused you to feel guilt
6	Have thoughts or reminders of what has been or will be lost caused you to feel sick or ill in any way
7	Have thoughts of the loss come into your mind whether you wish it or not
8	Have you felt distress by the reality of the loss
9	Have thoughts or reminders of the loss caused you to feel dread of the future
10	Have thoughts of your loss caused you to be more irritable with others
11	Overall, how much have thoughts and feelings about your loss or losses distressed you
12	Have other animals, people or familiar objects reminded you of the loss
13	Have thoughts or reminders of the loss caused your emotions to feel numb
14	Have you found yourself imagining that the loss did not or will not occur
15	Have reminders of the loss caused you to feel sadness
16	Have thoughts or reminders of the loss caused you to feel anger?

Participants were asked to score on a Likert scale of 1–4 with the options: a lot of the time; quite a bit of the time; a little bit of the time and never.

Several methods were used to undertake statistical analyses. Cronbach's alpha was used to measure the reliability of the GDI diagnostic instrument. To investigate potential risk factors that predict GDI scores, five predictor variables were considered; age of carer at the time of the survey, years of experience as a carer, number of joeys that died in their care in the previous year (2016–2017), average total hours per week spent caring in the previous year and the Napierian logarithmic transform of total financial cost of caring in previous year. All erroneously recorded values such as zero total hours spent and zero costs and total hours per week that were unrealistically high (i.e., greater than 100 hrs per week) were excluded. After this processing, data for 207 participants were useable for modeling. The logarithmic transform of total financial cost of caring in previous year was required to avoid excessive leverage of this variable on the GAM fit because of its extreme range in values (i.e., range of AUD150 to AUD89,000).

The age of carer categories was fitted in a Generalised Additive Model (GAM) [31] as a factor, while the other four continuous variables were each fitted as cubic smoothing spline terms with a basis dimension of five in each case. These five predictor variables were fitted jointly to GDI score, assuming Gaussian errors for the scores combined with the identity link function. Residual plots showed that the residuals from the fit had positive skew. However, transforming the scores using the square root function and refitting the GAM to this transformed response variable removed the skew in residuals and gave very close to Gaussian-distributed residuals as verified using a quantile–quantile (qq) plot. The predictor variables were combined additively in the GAM and interactions between them (such as fitting the smooths separately to each age category) were not investigated due to insufficient data. Combinations of the predictor variables were fitted, and the best model was selected based on its Akaike Information Criterion (AIC), which combines the lack of fit (i.e., minus twice the log-likelihood) with a penalty for number of parameters fitted. To demonstrate the effect of these variables on GDI score, for each variable in turn, the predicted mean GDI score (on the square root scale) was graphed against that variable while the other two variables were fixed at a single value.

When comparing the financial input and time spent in caring the fitted locally estimated scatterplot smoothing (LOESS) smooth (solid line) and the gamma generalized linear model (GLM) fit, used the identity link function (i.e., a linear model) with intercept constrained to zero. The comparison of the LOESS and GLM fit was close, so that a linear relationship was justified.

3. Results

Of the 316 responses received, 270 were complete. This represented 7.6% of the approximately 4150 wildlife carers in Australia eligible and able to access the survey, i.e., those caring for marsupials in the year 2016–2017 and with internet access. Optional questions, designed to measure general grief,

were declined by 36 participants. A further ten participants failed to complete one or more questions in the other section of the questionnaire. This meant that data from a total of 270 respondents were available for full analysis.

Participants represented all states and territories. Figure 1 shows the proportion of respondents by state, along with double standard error bars (i.e., the bars represent twice the standard error above the predicted mean; based on an assumed multinomial sampling distribution) and how the current survey data relate to the 2016 Australian census [32]. If the error bars do not overlap the census proportion, then the hypothesis that the survey is representative for that state can be rejected assuming an approximate normal distribution for the corresponding z-score test statistic ($p < 0.05$). On this basis, relative to the general population, the Australian Capital Territory, Northern Territory, New South Wales, Queensland and Tasmania were over-represented, Victoria and Western Australia were under-represented and South Australia was greatly underrepresented.

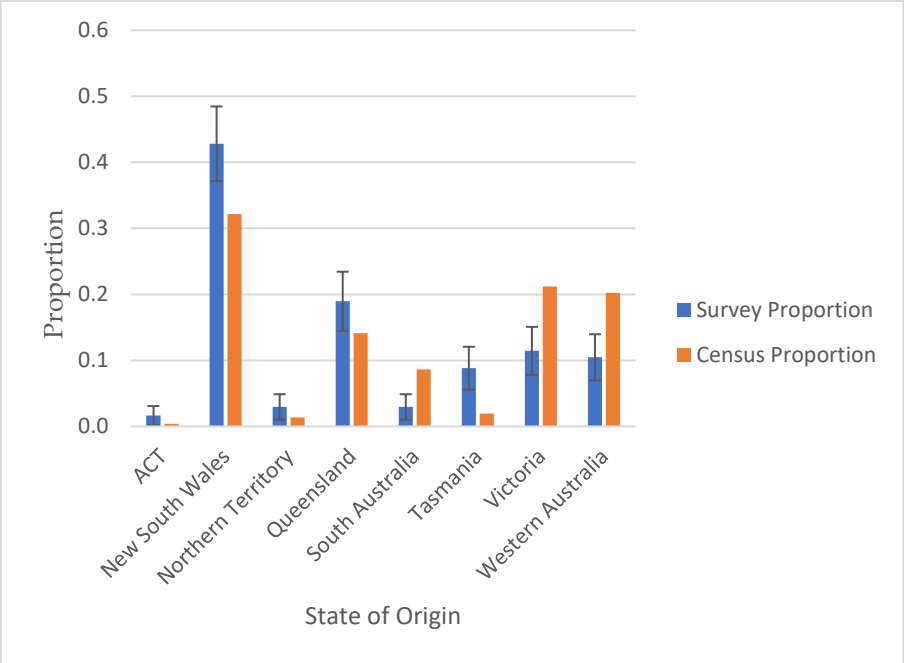


Figure 1. Representation of survey participants relative to general population, as reflected by the 2016 Australian census. Double standard error bars are shown, i.e., the bars represent twice the standard error above and below the predicted mean.

Among the respondents who completed the survey, 262 (85.6%) were female and 44 (14.4%) were male. The highest proportion of female participants was in the 46–60 age group and the highest proportion of males were over 60 years of age. Most respondents were over the age of 46 years ($n = 213$, 69.6%). Few respondents were under the age of 30 years ($n = 38$, 12.4%). Nine participants identified as aboriginal and/or Torres Strait islander ($n = 306$, 2.9%). Most respondents were born in Australia ($n = 228$, 74.8%) and had English as their first language ($n = 306$, 96.4%).

A significantly higher proportion of females nominated the motivation to help animals compared with males, and a significantly higher proportion of males nominated the motivation to conserve the environment compared with females. Other motivations where there were significant differences were securing a future job with animals, having the skills to rear animals, socializing and meeting people and filling a void in their lives. All of these were stronger motivations for females than for males (See Table 4).

Table 4. Motivations that prompted respondents into becoming a wildlife carer. The respondents could select as many of the motivational reasons as they felt applied to them.

Motivation	Number of Respondents			Percentage of Responses
	Female	Male	Total	
I had an affinity with animals and a desire to help them	209	29	238	88.5
To help conserve the environment	155	31	186	69.1
To contribute to my community	93	14	107	39.8
I had the skills to rear animals	93	11	104	38.7
I related better with animals than humans	70	9	79	29.4
To learn something new	62	7	69	25.7
To handle animals	63	10	73	27.1
To challenge myself	57	4	61	22.7
I believed that I had a special gift in relating to animals	56	8	64	23.8
To socialise and meet people	21	2	23	8.6
To do something different	23	3	26	9.7
To fill a void in my life	24	1	25	9.3
To help me get a future job with animals	12	1	13	4.8

The longest period that a participant was involved as a wildlife carer was 50 years (from 1968 until 2017), with four participants having been involved for over 45 years. The average duration of participation in the sector was approximately 11.4 years (n = 305). Approximately one third (32.8%) of the respondents had spent less than five years caring for wildlife (n = 100).

Respondents (n = 236) reported an approximate total time spent on training, mentoring and attending meetings, rescuing and/or rehabilitating fauna, record keeping, fundraising activities, travelling or community public relations activities in 2016–2017. The total of the average hours worked across respondents was 9994, at an average of 31.6 h.week^{-1}. wildlife carer^{-1}.

Respondents reported the total cost of animal food and associated equipment and infrastructure, travel and stationery, veterinary fees, training course fees, membership fees and insurance. For the year ending 30th June 2017, the total financial contribution by the respondents (n = 232) was AUD1,560,269 (mean of AUD5307 per carer; standard deviation of AUD10,574), with the maximum being AUD89,000.

Respondents gave an approximate overall total for the personal funds they have spent on caring for animals during the time they have been a wildlife carer. This included any income they lost if they needed to take time off work (e.g., to take a joey to a veterinary clinic or nurse a sick joey). The total expenditure by the respondents (n = 293) was AUD13,607,915 (mean AUD46,443; standard deviation AUD98,098). Figure 2 shows the relationship between years as a carer and total expenditure (on the base 10 logarithmic scale) by wildlife carers during the time they have been caring for wildlife. The model with the lowest AIC and therefore the best model according to this criterion included the above terms with the exception of average hours, so this last term was dropped from the model. The fitted GAM explained 16.7% of the null deviance (i.e., in the case of a Gaussian response variable the null deviance is the residual variance for the simple mean model). The GLM predicts that the average annual expenditure per carer is AUD3513 (SE = 340).

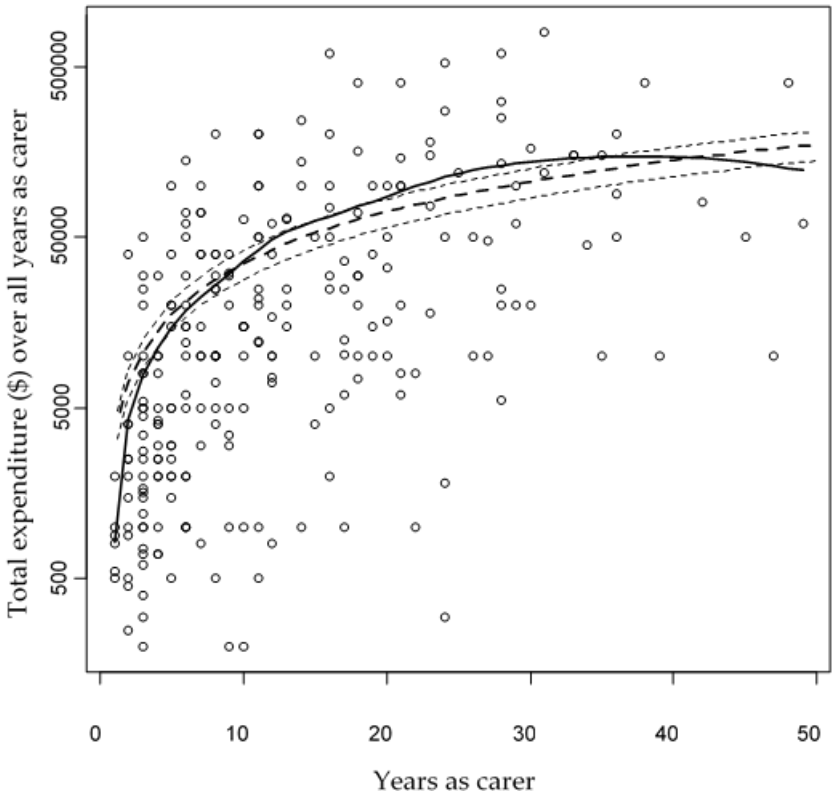


Figure 2. Total expenditure by wildlife carers during the time they have been caring for wildlife versus years as carer. The solid line is LOESS fit and dashed line is the generalized linear model (GLM) fit with double standard error bounds shown as thin dashed lines.

There are two major types of event that differentiate the rescue of smaller and larger marsupials. Having been attacked by domestic pet or feral animal is the main reason that *Antechinus*, bandicoots and gliders (i.e., smaller species) require rescue, while roadkill is the main reason for the rescue of kangaroos, koalas, pademelons, possums, wallabies, wallaroos and wombats (i.e., relatively larger animals). Proportionately, roadkill and attack by domestic or feral animal accounted for most of the injured or orphan animals taken in by carers (Figure 3).

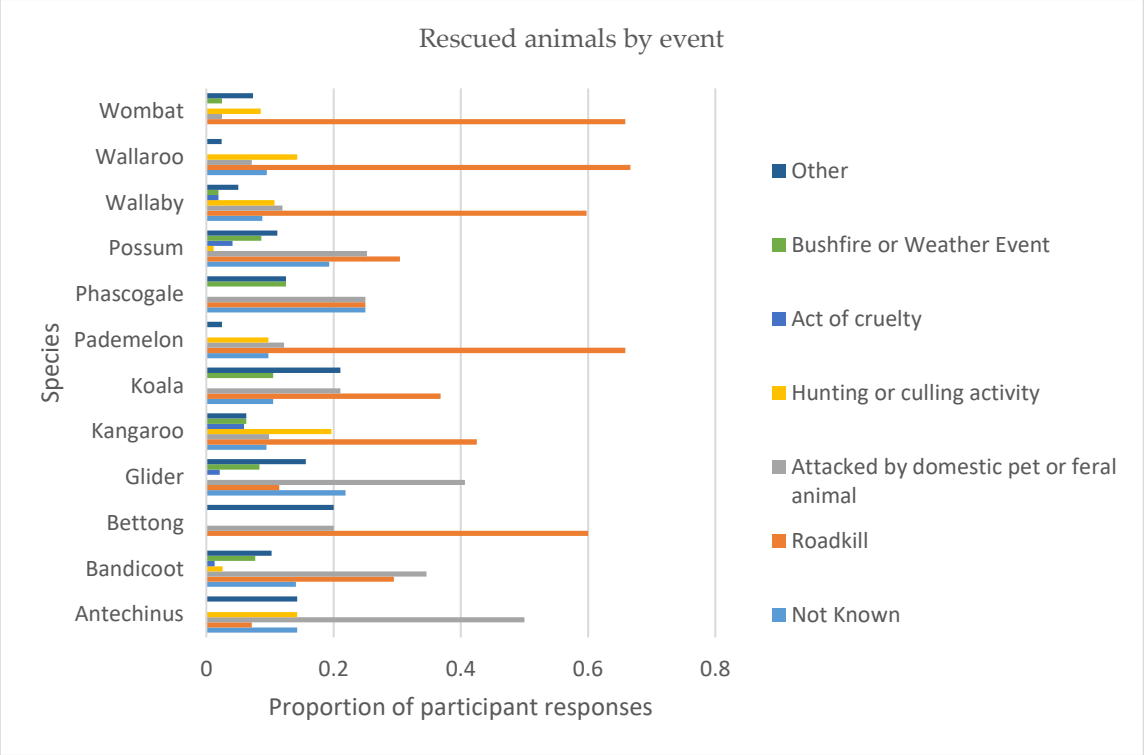


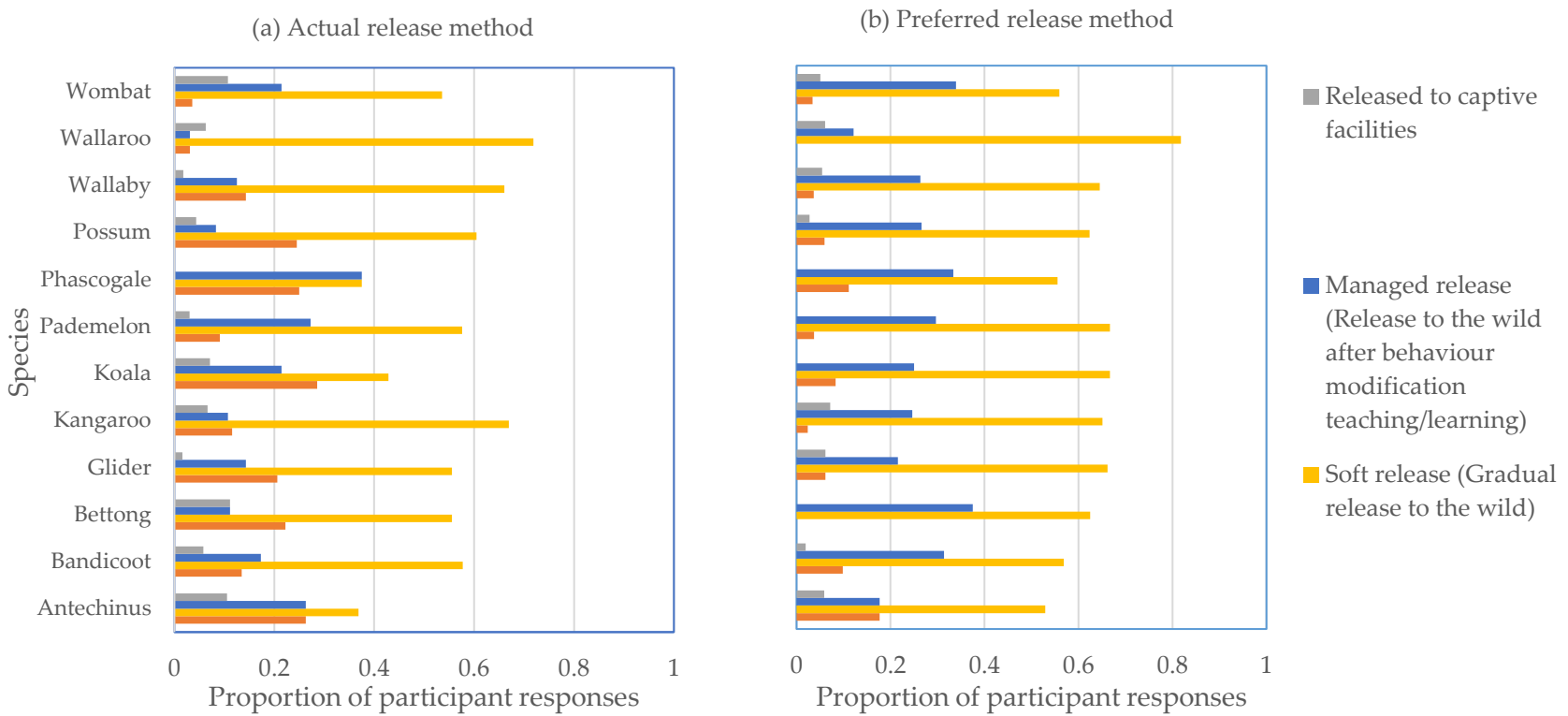
Figure 3. Animals by species and the reason for their rescue.

A total of 4608 joeys were raised by the respondents (n = 305) in the 12-month reflective reporting period 2016–2017, with an average of 15 animals.carer^{-1}.year^{-1}. The species raised were predominantly possums (*Trichosurus* sp.; n = 1681, 36.48%), kangaroos (*Macropus* sp.; n = 1009, 21.89%), wallabies (*Macropus* sp.; n = 614, 13.31%) and wombats (*Vombatidae* sp.; n = 327, 7.10%). Those four species accounted for nearly 80% of all animals reared. Gliders (*Petauridae* sp.; n = 290, 6.29%), bandicoots (*Peramelemorphia* sp.; n = 205, 4.44%), wallaroos (*Macropus robustus*; n = 127, 2.75%) and pademelons (*Thylogale* sp.; n = 124, 2.69%) were the four next represented, totaling 16%. Least represented were *Antechinus* (*Antechinus* sp.; n = 88, 1.9%), koalas (*Phascolarctos cinereus*; n = 72, 1.56%), phascogales (*Phascogale tapoatafa*; n = 23, 0.4%) and bettongs (*Bettongia* sp.; n = 22, 0.47%), followed by Tasmanian devils (*Sarcophilus harrisii*; n = 7, 0.15%), quolls (*Dasyurus* sp.; n = 7, 0.15%), potoroos (*Potorous* sp.; n = 6, 0.13%), dibblers (*Parantechinus apicalis*; n = 3, 0.06%), dunnarts (*Sminthopsis* sp.; n = 2, 0.04%) and a quokka (*Setonix brachyurus*); n = 1, 0.02%). Seven Australian native species were not represented: bilby (*Macrotis lagotis*), kowari (*Dasyuroides byrnei*), marsupial mole (*Notoryctes typhlops*), mulgara (*Dasycercus* sp.), ningaui (*NingauI ridei*), numbat (*Myrmecobius fasciatus*) and planigale (*Planigale maculata*).

A total of 722 joeys died whilst in care of the respondents (n = 276) in the 12-month reflective reporting period, with an average of 2.6 animals.carer^{-1}.year^{-1}. When compared with the average number of animals reared and released, i.e., 15 animals.carer^{-1}.year^{-1}, this represents a success rate of over 85%. Wildlife carers were asked about how they implemented pre-release training or conditioning of animals. Over 70% of wildlife carers reported that they taught the rescued animals two lessons: learning how to find their own food (forage; n = 238, 82.6%) and how to interact with their conspecifics (n = 206, 71.5%). Four other lessons were taught by approximately half the carers: finding water (n = 168, 58.3%); avoiding pet animals (n = 165, 57.3%); avoiding humans (154, 53.5%); and, seeking shelter or digging a den (n = 143, 49.7%). Few carers claimed that they trained the animals to avoid or evade predator species (n = 84, 29.2%) or to avoid motor vehicles (n = 40, 13.9%), and 65% (n = 188) had received no training in animal behavior modification techniques.

For all animals reported by the current respondents, the primary method used to release them was a soft release while the second most popular method used was a hard release (Figure 4a). Significantly, for releasing wombats, hard releases were the least used, with managed releases being the second most

popular method. Given a choice to select a method of release for all animals, a managed release was selected as the second most popular method, rather than a hard release (Figure 4b).



415

416 **Figure 4.** Required and preferred methods used to return rehabilitated wildlife to the wild. (**a**) The actual method required by legislation and (**b**) the method preferred
417 by wildlife carers.

Rehabilitated animals may require extra time in captivity for various reasons including behavioral and health assessment, integration with conspecifics, inappropriate weather or season for release and availability of suitable habitat. However, depending on the species in question, wildlife carers preferred to retain the animals on their property and so use facilities that were relevant to the physical size of the animal and its behavioral needs of space. Least favored was to use a zoo or wildlife park to house the animals (see Figure 5).

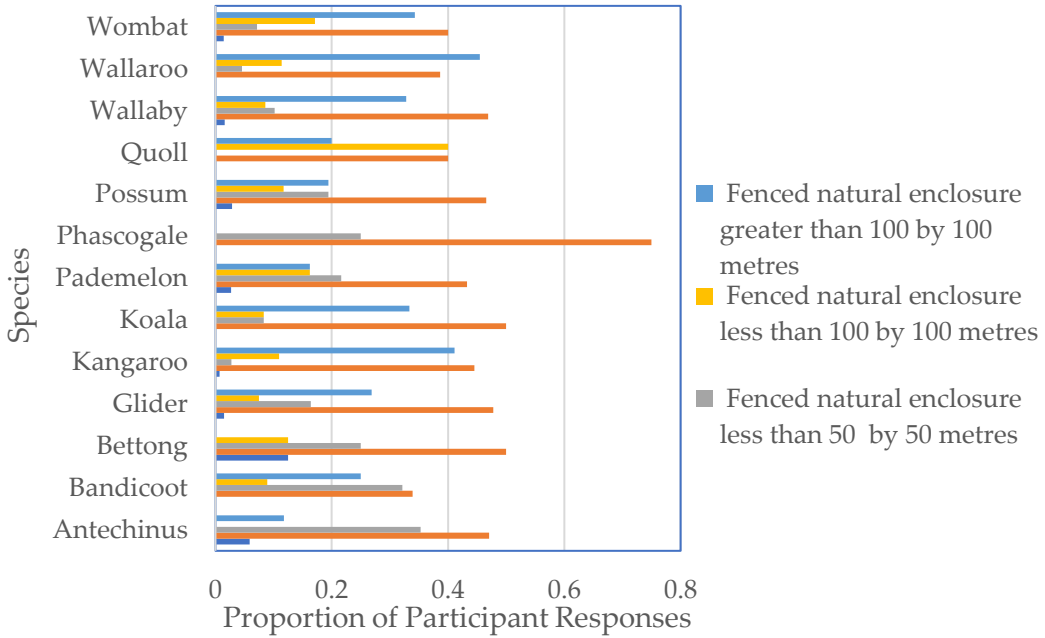


Figure 5. Preferred place of temporarily housing rehabilitated animals prior to release.

In deciding the preferred option for animals deemed unsuitable for release, two-thirds of the wildlife carers (n = 202, 67%) preferred the options of the animal being kept on their private property or at a secure enclosure, rather than euthanasia. However, some respondents also indicated that they would prefer an animal to be euthanized (n = 51, 17%) rather than be retained in a zoo or wildlife park (n = 45, 15%).

Information was sought on methods used to identify animals in rehabilitation and whether any feedback was received after release. An estimated 3900 animals were reported to have been released by the respondents (n = 289). Upon release, only 4.7% of the animals released were identifiable in an objective manner; by microchip (2.2%), ear tag (1.5%) or tracking collar (1%), i.e., one that did not rely on the knowledge or expertise of an individual carer. Respondents reported that

they were able to identify individual animals by visual appearance (32%), by individual behavior traits (20.2%) and other unspecified means (3.7%). Thus, 39.4% of released animals were not identifiable.

Feedback on the fate of animals that had been released was species-dependent, with the smaller species (*Antechinus*, bandicoot, glider, phascogale and quoll) generating the least amount of feedback. Over 50% of the respondents received feedback about the eight larger species, which aligns with the reported 57% of animals being identifiable (as shown in Table 5). However, even if visual (32%) and behavioral observations (20.2%) are accepted as reliable, there still remain 39.4% of animals for which there is no record of outcome after release.

437 **Table 5.** Feedback that was received by wildlife carers relevant to the species they released to the wild.

	Total Animals Released	Feedback Received		SE
		Yes	No	
Antechinus	15	2	13	0.18
Bandicoot	46	17	29	0.14
Glider	58	3	4	0.13
Kangaroo	117	31	27	0.09
Koala	10	7	3	0.29
Pademelon	27	17	10	0.19
Phascogale	8	0	8	0.00
Possum	189	102	87	0.07
Quoll	5	0	5	0.00
Wallaby	103	66	37	0.09
Wallaroo	32	23	9	0.16
Wombat	57	37	20	0.13

438 Approximately half of the carers (n = 155, 49%) agreed or strongly agreed that the current methods of release were the best that could be achieved. However,
439 given that slightly over half of the respondents were either being unsure (n = 92, 29%), disagreed or strongly disagreed (n = 70, 22%), this is an area that merits
440 attention.
441 The mental health of wildlife carers was examined by the type of loss they had experienced. The loss of an animal in care through death was the most frequently
442 reported event that caused distress in the previous 12 months (n = 212, 72.4%),
443 An assessment of the grief experienced by wildlife carers in the year 2017–2018 was conducted with 30 of the 316 total respondents declining to answer
444 questions on loss and grief, and 12 did not complete all the questions in this section. The collated data on participants' responses to 16 grief-related questions
445 revealed that six respondents reported having experienced no grief (n = 274, 2.2%), 119 reported minimum grief (n = 274, 43.4%), 72 reported mild grief (n = 274,
446 26.3%), 30 reported moderate grief (n = 274, 10.9%) and 47 reported severe grief (n = 274, 17.2%).
447 For the 274 respondents who completed all 16 questions, Cronbach's alpha was calculated using the R-library ltm [33] at 0.94 with 95% bootstrap confidence
448 limits ranging from 0.93 to 0.95. This demonstrates a very high degree of reliability of the GDI scores. The R-library mgcv was used to fit the GAM and obtain
449 predictions, and for residual histograms and qq-plots [31]. The outputs from the fitted GAM, presented in Figure 6 Figure 7 Figure 8 Figure 9, show that the 60 years
450 and over age category had the lowest average GDI score, after accounting for the other significant predictor variables, while the 16–30 years age category had the
451 highest average, with the intermediate age categories showing intermediate levels of average GDI score. The difference between the 16–30 years and 60 years plus
452 age category GAM parameter estimates was highly significant (p = 0.025) as obtained from the corresponding t-statistic, while for the other two age categories
453 parameter estimates were not significantly different from the estimate for the 16–30 years age category (Figure 6). As years of caring increased up to a peak at
454 around 20 years, the average GDI score also increased, after accounting for the other significant predictor variables, and plateaued afterwards (Figure 7a).

Average GDI score increased linearly with the number of joeys that had died in care in the previous year, after accounting for the other significant predictor variables (Figure 7b).

Average GDI score increased as financial cost in the previous year increased, but plateaued when cost reached AUD$8000 for the 2016–2017 year, after accounting for the other significant predictor variables (Figure 7c). Financial cost was the most significant of the predictor variables, with its spline term highly significant ($p < 0.005$), and the precision of the estimated curve was much higher than the predicted trends for the other two continuous predictor variables.

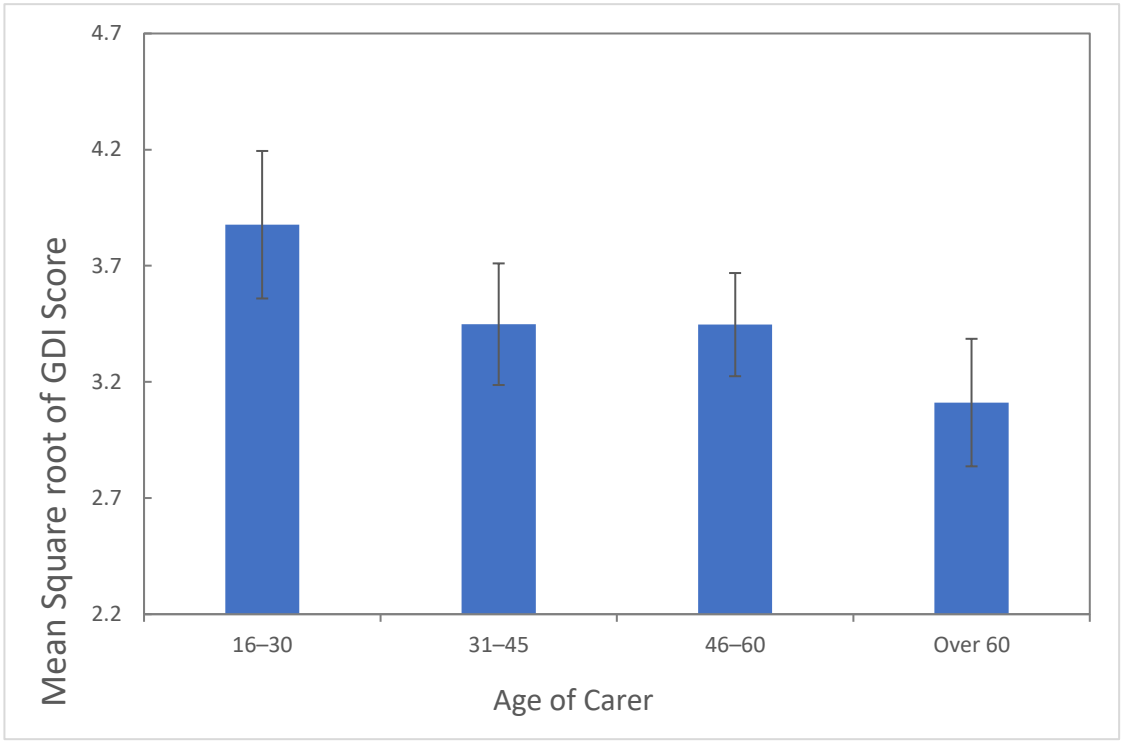


Figure 6. Predictions of grief diagnostic instrument (GDI) score on the square root scale versus Age Category of Carer for Years as Carer fixed at 5 and Number of Joeys Died set to the average of 2.6 and log of costs set to 9 (i.e., AUD8103). SE bars are shown.

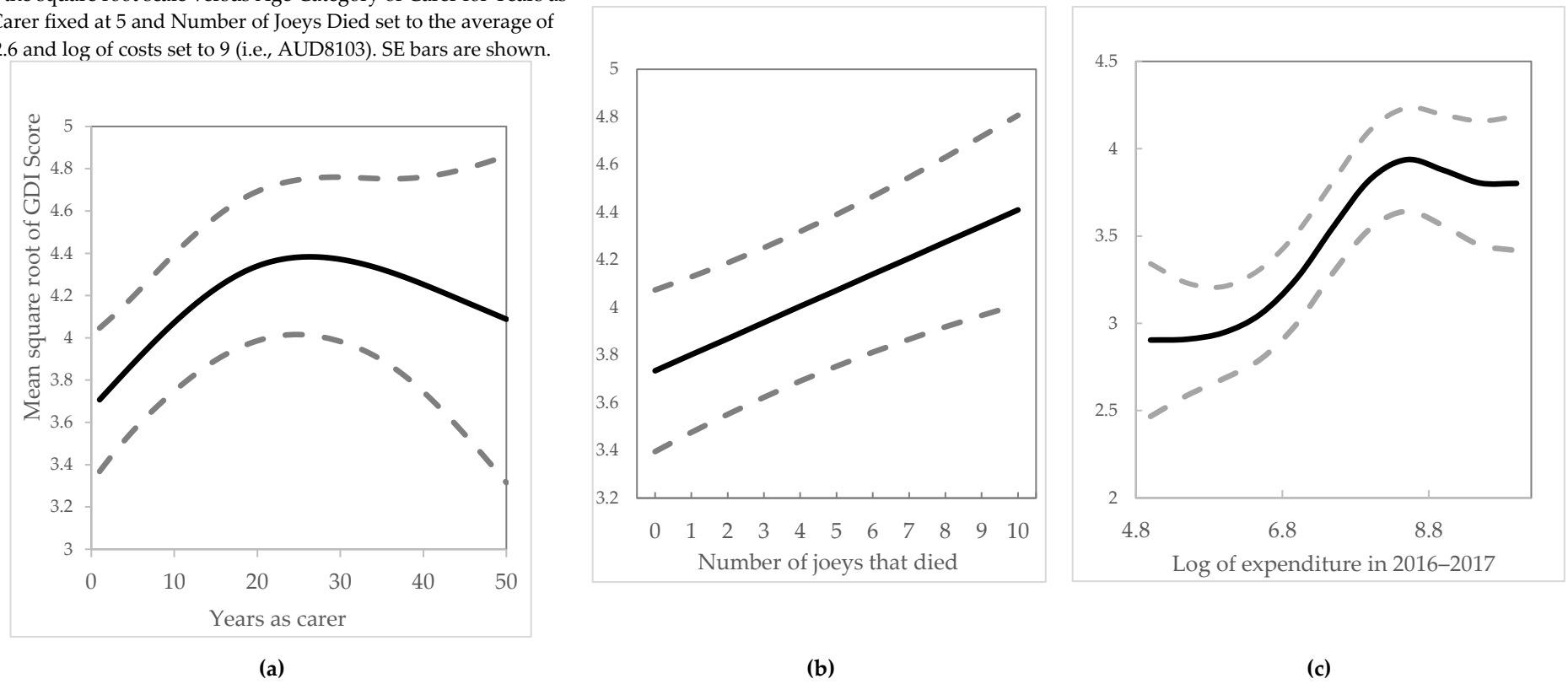


Figure 7. Predictions of GDI score on the square root scale for age category of carer fixed at 16–30 versus (**a**) years as carer for the number of joeys died set to the average of 2.4 and log of costs set to 9 (i.e., AUD8103), (**b**) number of joeys died for years as carer set to 5 and log of costs set to 9 and (**c**) log of the financial cost in the previous year for years as carer set to 5, and number of joeys died set to the average of 2.4. Standard error bounds are shown as dashed lines.

Of the respondents who responded to questions about the support they received and their own wellbeing, 95 (32.1%) stated that they felt supported enough as a wildlife carer at the organizational/government level to achieve optimal outcomes for the animals they reared, 107 (36.0%) stated that they felt supported enough for their own wellbeing and 221 (74.9%) stated that immediate support was available should they become ill or injured.

There were 152 responses (n = 316, 48.1%) to the open question inviting details on the role carers play in rescuing, rehabilitating and releasing native marsupial wildlife. Fourteen different types of concerns were expressed by more than one individual respondent. The greatest concerns expressed were lack of government support (n = 47), financial pressures (n = 32), mental health problems (n = 25) and lack of public care (n = 17). Other concerns were lack of wildlife carers and interpersonal issues (n = 12), the release of animals to the wild with no ongoing monitoring (n = 11), destruction of habitat and lack of knowledge about wildlife rescue among the public (n = 10), abuse by the public (n = 8), physical health problems (n = 7) and burnout (n = 6). Lack of knowledge (n = 5) and lack of veterinary input (n = 3) were the least considerations.

A more detailed analysis of the three main concerns (lack of government support, financial pressures and mental health problems) suggests that wildlife carers felt they have a general lack of central government support and that their work is not valued or even understood by government agencies. The issues detailed were: general lack of central government support (n = 22); lack of understanding and work not valued (n = 9); lack of government financial support (n = 5); poor regulation of wildlife carers (n = 5); lack of agency support; e.g., Parks and Wildlife (n = 4) and culling of wildlife by government agencies (n = 2). They also considered that some of the public have no regard for animal welfare and that some do not understand or appreciate their work. The issues detailed were: a no care attitude of the general public (n = 8); the need to educate the public about the work wildlife carers undertake (n = 6); lack of recognition (n = 5); lack of general understanding (n = 4); lack of public support (n = 2) and intolerance from the public (n = 2). A specific example was a lack of understanding that wildlife carers are volunteers who contribute their time, are self-funded and do not receive any financial remuneration for the service they provide. Some members of the public expect them to attend at a moment's notice and clear up the mess of a roadkill victim. Nearly two-thirds of the respondents who cited mental health issues (n = 16) as a problem had feelings of depression, disillusionment, grief, isolation, emotional issues or mental stress. The issues detailed were: emotional issues and mental stress (n = 9); feelings of depression, disillusionment, isolation and grief (n = 7); feelings of guilt and failure at having to refuse to take in additional animals (n = 4); managing non-core stressors, e.g., administration tasks (n = 3) and not knowing what happens to animals after they are released (n = 2).

4. Discussion.

Wildlife carers from all states and territories of Australia were represented in this survey, with adult respondents of all ages and both genders reporting on their care for injured and orphaned native wildlife. South Australia was significantly under-represented perhaps because a large South Australian wildlife carer organization stated it was unable to participate in this research project due to their members' high workload and so declining to distribute the survey to its carers. Therefore, caution is warranted as this may have hindered the power of statistical analysis when assessing whether stress, caused by a high workload, was a factor in possible compassion fatigue and carer burnout.

As the number of animals needing rescue and rehabilitation increases with external threats, the workload for the wildlife carers is likely to increase, exacerbating workload-related stress. Of particular concern for 25% of the respondents is the lack of emergency support. There are severe consequences for those animals requiring feeding at regular intervals, should their carers become incapacitated for any reason. The perceived lack of government or agency support at 68% and 64%, respectively, is also of concern. A similar dissatisfaction rate has been reported from other studies both nationally and internationally [2,15,34–36].`

Among the reported motivations for becoming a carer, animal welfare, conservation of the environment and contributing to the community were the top three reasons for participating in wildlife rehabilitation given in a survey of wildlife rehabilitators in New South Wales [2] and the top two in a survey of wildlife carers in Queensland, New South Wales, Victoria and South Australia [7]. These motivations reflect a one welfare approach [37] contributing to the common good of animals, humans and the environment. From an international perspective, similar results were presented in studies in other countries [26,34,36,38]. However, in the current survey, there were significant gender differences, in particular for the two dominant motivations across all respondents of "having an affinity with animals" and "conserving the environment". The ratio of female:male wildlife carers was 85.6:14.4 in the current study; similar to those obtained in other surveys of wildlife carers, with 79:21 in NSW [39], 77:23 in New Zealand [24] and 87:13 for Australia [15]. However, when describing wildlife carers, Tribe and Brown [15] reported that in 1997 the mean age in Victoria was 44 years and in New York 42 years. These figures contrast with the current study where, in Australia in 2018, 70% of wildlife carers were over the age of 46 years and 25% were over the age of 60 years, with a mean age of 49.6 years, indicating an increasingly aging demographic. In a recent survey of wildlife carers in NSW [2], the average duration of participation in the wildlife sector was nine years, with the longest period serving as a carer being 65 years. Furthermore, that study reported flux in the membership of organizations that ranged from 25% to 60%. These average durations, coupled with 70% of wildlife carers being over the age of 46 in the current study, suggest that over the next ten years there will be a need for large-scale recruitment even to retain the current number of wildlife carers.

In many ways, nursing can be considered a profession that is similar to wildlife caring. Roche et al. [40] reported that, in an Australian hospital setting, the annual turnover of nurses with more than 5 years' experience was 20%. International comparison of turnover rates reveals similar figures. A survey of the turnover of nurses in a major medical centre in America [41] highlighted that staff turnover was greatest in the first year of recruitment at 39%, falling to 17% by the end of the fifth year and then remaining at 17% for the following years. If applied to the 32.8% of wildlife carers in the current cohort with less than five years' experience, such annual turnover rates emphasize the need for an increase in recruitment. One significant area where recruitment might be boosted is in the older male population where, as revealed in our results, 57% of male wildlife carers are over 60 years old. It may be that, given the opportunities that financial stability, spare time and reasonable health can provide at retirement age, males could be encouraged to take up an animal caring role as a new interest. It is worth noting that conserving the environment was cited as the main motivation for becoming a wildlife carer among male respondents in the current study.

Internal threats to the people who volunteer to take on the responsibility of caring for injured and orphaned wildlife are financial, physical and mental. From the financial perspective, the average annual expenditure of AUD5307 incurred by wildlife carers surveyed in this study contrasts with figures quoted in a report on wildlife volunteers in New South Wales in 2017 [2], in which primary animal carers estimated an average annual expenditure of AUD4000. This difference in expenditure probably reflects the NSW study's inclusion of all wildlife carers, whereas the current study focused only on those caring for marsupials. Birds, reptiles and bats account for more than half of the animals rescued, generally require less time in rehabilitation and accordingly attract lower feed and infrastructure costs [2,3,25]. Some wildlife carers contribute up to AUD800,000 over a lifetime of caring for wildlife, considerably more than the 2015 average individual Australian superannuation fund balance at retirement (AUD292,500 for men and AUD138,150 for women [42]). These monetary values demonstrate the high financial burden placed on anyone volunteering to be a wildlife carer. In all jurisdictions, regulations state that wildlife carers themselves must meet all the costs involved in rescue and rehabilitation [4].

The physical effort required of wildlife carers is considerable. The Australian Bureau of Statistics gives an average of 2.5 $h.week^{-1}$ as the contribution of general volunteering in 2014 [43]. Average actual hours worked per week by Australians in all jobs generally decreased over the 32 years from 1978 until 2010, from approximately 35.5 h in early 1978 to approximately 33 h in 2010 [44]. A

significant number of wildlife carers (n = 141, 45%) devote more hours than these average 33 working hours and over 90% of wildlife carers volunteer more than 2.5 h.week^{-1}. Some respondents invest up to 100 h.week^{-1} caring for animals, much more time than if they were in full-time paid employment. The current survey did not reveal how many carers were in full or part-time paid employment. However, the physical burden of working such long hours is considerable and likely involves sleep deprivation.

The cognitive challenges of rescuing and rehabilitating animals need clarification. Wildlife carers are obliged to constantly update their knowledge base and navigate complex legislation and also manage detailed record-keeping [4]. They are mandated in the way that they have to keep and release animals, although these may not correspond with their knowledge, experience and preferences. Slightly more than half of the current respondents did not believe that the methods currently used to release animals are optimal and, given a choice, they would alter the method of release, with behavioral modification of the animals being granted a much higher priority than is currently the case, while acknowledging that they would need professional training to accomplish this.

From the perspective of an animal, motor vehicles could be regarded as a reasonably novel threat to wildlife. Like historic threats, such as predators, they kill or injure, strike randomly, move at high speed and with little warning. This motor vehicle threat must appear alongside the other predatory threats to which, as hand-reared orphans, these animals are naïve. In a survey of rehabilitation practices in 2010, Guy and Banks [7] noted that antipredator training was conducted by only 20% of respondents. In contrast to their report, the current study reveals an increase to 29.2% in anti-predator training by wildlife carers over the ensuing seven years. However, even at this percentage, there remains a significant gap in this aspect of behavioral training. This could indicate that carers realize there is a shortfall in professional training in animal behavior modification techniques but, given training, would apply these techniques to animals prior to releasing them. These methods could be applied to considerable advantage with those animals that may have inadvertently been desensitized to humans, vehicles and companion animals through habituation during the rehabilitation process.

That it is possible to successfully introduce aversion behavior modification techniques has been demonstrated in the reintroduction of quolls [45], prairie dogs [46] and possibly greater bilbies [47]. Without antipredator training prior to release, the attrition rate among prey species can be high, as happened with eastern quolls recently reintroduced to mainland Australia; 70% were dead within three months of being released [48]. It may be that the highly successful release of 54 hand-reared wombats monitored over a period of eight years and released using seven release pens, as a soft/managed release protocol, set a precedent for the release of wombats and perhaps other species [49]. Although reintroduction biology emerged as a new science in Australia and New Zealand only some 20 years ago [50], the understanding of the process of reintroduction has advanced [51]. This is an area where research and practical application could address a persistent void in behavior modification training for wildlife carers.

From the time they rescue or accept an animal for rehabilitation, wildlife carers face various emotional challenges. They commonly confront the necessity to have rescued animals euthanized or witnessing them die during the rehabilitation process. Approximately 86% of wildlife carers in the current study were female and they reported that the primary motivation for becoming a wildlife carer was having an affinity with animals, whereas for males this was the secondary motivation, the first being conservation of the environment. Previous research indicates that having an affinity with animals makes the euthanasia or death of animals a significantly distressing event [52–54]. Thus, female wildlife carers who cited having an affinity with animals as their main motivation for becoming a wildlife carer seem especially vulnerable to distress that may lead to feelings of anger, sadness, guilt, fear, depression and helplessness [54–58] when confronted with the possibility of having an animal euthanized.

In the current study, euthanasia was the third preferred option for an animal deemed unsuitable for release into the wild, which contrasts with the findings of a survey of rehabilitation practices in the eastern states of Australia, conducted in 2012 [7], where euthanasia was the most preferred option. This presents a dilemma for carers, most jurisdictions mandate euthanasia for animals

deemed unsuitable for release [4]. In Queensland and the Australian Capital Territory, there are mechanisms whereby the animal may be placed in a wildlife park or zoo. However, this option was the least favored option among the current respondents, who preferred the animals to be euthanized (n = 51, 17%) rather than be retained in a zoo or wildlife park (n = 45, 15%). This appears to be a surprising finding, given that these are institutions where expertise is likely available to ensure the animal will have a life worth living, if remaining in captivity.

An animal may require temporary housing before release, for reasons that may include integrating a gregarious animal with conspecifics, undertaking a behavior modification program, quarantining for health reasons, recovery from injury or so that an assessment can be made as to its suitability to be returned to the wild. Again, it is interesting to note that, of all the options for temporary housing, the places where professional input for all the above factors would be available, i.e., a zoo or wildlife park, are the least favored by carers (Figure 5).

CF is a condition resulting from a decline in compassion among those caring for others, whether human or animal. It was defined by Figley as the "cost of caring" [17]. This may well be the outcome for wildlife carers, with 77 of the current respondents (n = 274, 25%) reporting moderate to severe grief at constantly being faced with seeing dead animals, euthanizing animals or having them die during rehabilitation. Further confirmation of this outcome came with evidence that the grief they experienced when animals died in their care increased in direct correlation with the mortality rate. Given that a general grief instrument was used for the current study, caution is required when interpreting the grief results. The grief measured may not be entirely due to distressing events during the process of caring for wildlife, and may include other life events. However, the GDI does enable the variety of loss variables to be taken into account. Wildlife carers in the current study also mentioned the culling of wildlife by government authorities as a concern, which highlights another dilemma they face. In one context, wallabies and kangaroos are seen as pest species to be culled [59–61], whereas in another context wildlife carers work to save members of these species, only to acknowledge that these animals would possibly be killed after release; a conflict that may be philosophically disquieting for the carers, and compound stress.

Respondents to the current survey also reported on the predicament that, after spending up to two years rearing an animal, they have to release it into the wild with no reliable method of identification, so that they are unlikely to ever know what happened to the animal after release. Given that conservation of the environment was stated as the highest motivation for male respondents and the second highest for female respondents, it is difficult to believe that carers' motivations are being met satisfactorily, especially given that the current lack of reliable identification makes it unfeasible to monitor what happens to animals post-release.

Respondent wildlife carers reported bullying, intimidation and a lack of concern and compassion from peers, wildlife organizations and the general public. Similarly, Haering et al. [2] reported 'infighting and bullying' in a survey of wildlife carers in NSW and Carleton [25] stated that bullying was experienced by 39% of participants in an international survey of wildlife carers.

The effect of all the above factors, in combination, on the mental health of the wildlife carers merits consideration. The current analyses reveal that, regardless of the number of hours worked, the lower the age of wildlife carers, the longer the length of service in wildlife caring, the greater the financial input and the greater the number of joeys that died during rehabilitation, the more susceptible the respondents were to a GDI score ranging from moderate to severe. Clark et al. state that a GDI score of 18 or above for a patient in a general medical practice setting indicates a need for further assessment [28]. In a review of roadkill rescue that includes a section on mental health, Englefield et al. [45] posit that wildlife carers could be susceptible to 17 of the 20 established types of grief. The current study reveals 77 respondents scoring a GDI of 18 or above (n = 274, 28%), indicating moderate to severe grief. This suggests the need for wildlife carers to have access to counseling from a professional mental health practitioner as well as to programs that build resilience. Unfortunately, two-thirds of respondents reported that their personal mental health was not supported.

Caregiver burnout is a manifestation of physical, emotional and mental exhaustion. Caregivers who are burned out may experience fatigue, stress, anxiety and depression [62,63]. Wildlife carers

who are physically challenged by experiencing sleep deprivation through 4-hourly feeding of joeys and working over 40 h a week (26%), who are emotionally challenged by animals dying or being euthanized (92%), experiencing bullying, isolation and mental challenges (16.5%), who are under financial pressures (38%) and experiencing moderate to severe grief (28%) are likely to suffer caregiver burnout, given that they are subject to a combination of physical, emotional and mental stress. This could result in wildlife carers having to stop rehabilitating animals. Unless replacements can be recruited and are trained to be resilient and receive appropriate support, this will lead to an exponential spiraling effect where the workload will increase for the remaining wildlife carers, leading to more of them burning out and more animals going without care. Within this emerging domain of understanding compassion fatigue, burnout, resilience building, psychosocial and organizational support and PTSD, there is much to be studied providing an avenue for future research.

5. Conclusions

Australian wildlife rescue, rehabilitation and release all rely heavily on the voluntary effort of wildlife carers. Carers provide a valuable service to the community, through organizations and as individuals, but many feel unappreciated, undervalued and lacking in support. The work they undertake is demanding physically, financially, emotionally and mentally. External and internal threats to the Australian wildlife rescue sector are increasing. It is likely that, in the immediate future, unless changes are made to reduce roadkill and to increase the recruitment of carers with financial, emotional and mental support given to them, viable animals will need to be euthanized in larger numbers, more roadkill victims will remain unattended and animal and wildlife carer wellbeing will be compromised.

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Chapter 5. Trial of a 'virtual fence' system to mitigate roadkill

5.1. Preamble. Modifying animal behaviour to mitigate roadkill, using recently introduced electronic devices

Road agencies have a legal, social and environmental obligation to implement measures that minimise the ecological and environmental effects when roads are constructed or upgraded [46, 47]. An important element of this obligation is the mitigation of wildlife vehicle collisions and the resultant roadkill. This has benefits for the environment in allowing movement of species, fewer animals and humans killed or injured, and less roadkill to be cleaned up.

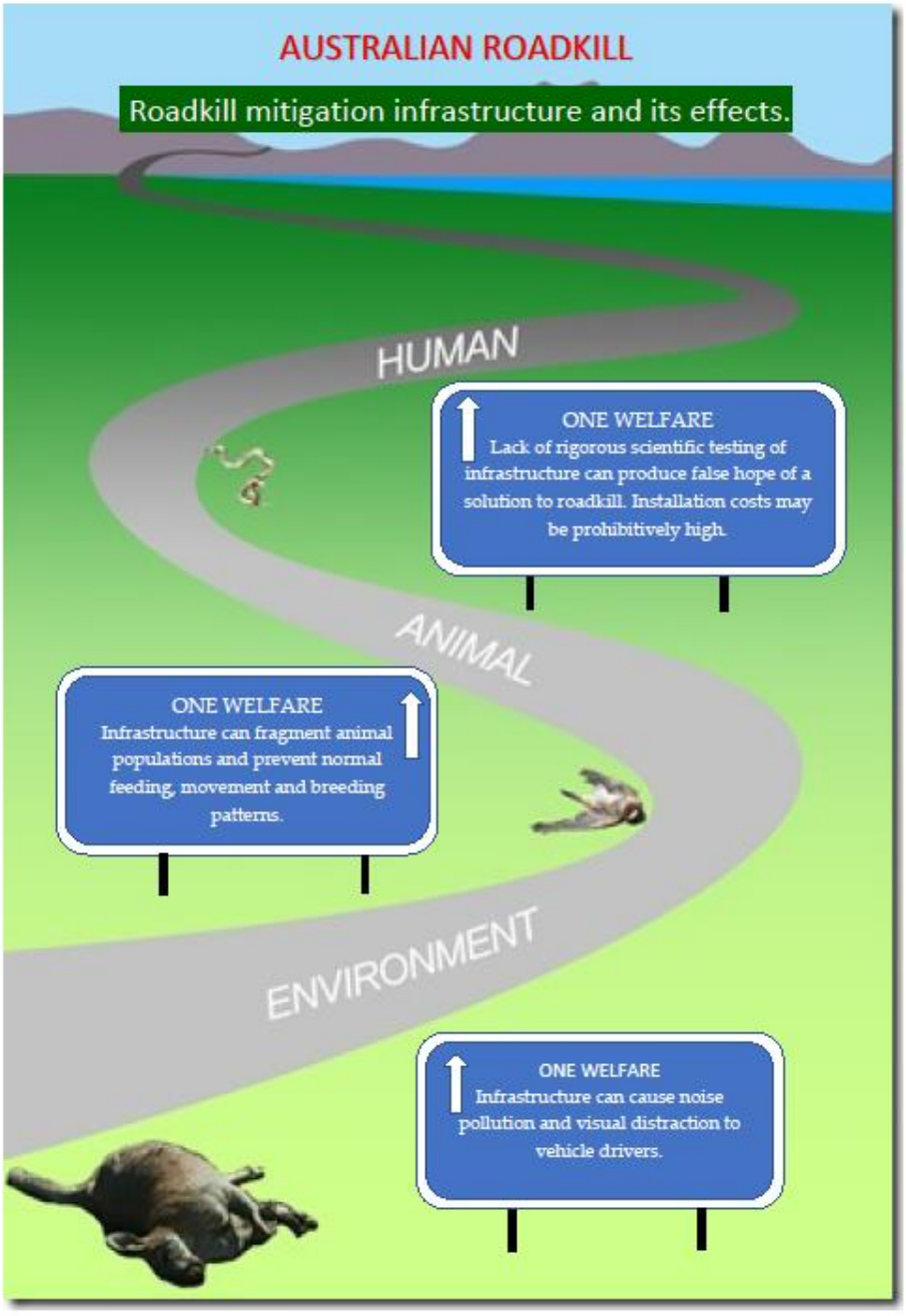
There are two main approaches used to mitigate roadkill: changing human behaviour and modifying animal behaviour. Unfortunately, it has proved difficult to confirm compliance with strategies that change human behaviour, for example driving more slowly, not travelling at night, and observing warning signs and speed bumps [13, 48, 49]. Changing animal behaviour by the use of infrastructure, warning signals or behaviour modification techniques has had some success but can be expensive. The land-bridge overpasses developed by Professor Daryl Jones in Queensland Australia, for wildlife to traverse the highway, are proving highly successful when used in conjunction with fencing along the highway [50]. They have been used as a model by several countries. [50-52]. However, they cost over AUD2,000,000 each to build, including the 500 metres of fencing each side of the bridges, needed to funnel the animals to the bridge.

Measures that hold promise for mitigating roadkill and are considerably cheaper than the current gold standard (of land-bridges and fencing) are extremely attractive to stakeholders such as local council authorities. Their lower cost may lead to the hope that they will be adopted at a higher rate, thus providing meaningful mitigation measures for places that currently have none. However, research on the *Shu Roo* and *Rooguard*, Australian devices alleged to prevent WVC, would suggest that such devices merit rigorous scientific scrutiny before they are implemented on a large scale [53-55]. Without such scrutiny large expenditure can result in financial wastage and false confidence among vehicle drivers, who believe animals will remove themselves from the highway.

A device was introduced to Australia in 2013 with a claim from the manufacturer that it would reduce roadkill by 90% on Australian roads [56]. The device produces acoustic and visual stimuli, triggered by the headlights of vehicles. It operates between dusk and dawn when crepuscular and nocturnal mammals are most active. These stimuli are said to alert the animals and make them feel so uncomfortable that they leave the roadway. When mounted on posts alongside a highway, the devices produce what is termed a 'virtual fence' (VF). The VF was subjected to a trial by scientists working for the Department of Primary Industries, Parks, Water and the Environment on a highway on the west coast of Tasmania. It was concluded by Fox et al. that the VF reduced roadkill by over 50% [57]. However, the scientific rigour of the trial and its findings were questioned [58]. Coulson and Bender, the authors, questioned the conceptual basis of the 'virtual fence' and also identified 'a total of eight methodological flaws ranging from imprecise measurements, confounding effects of treatments, low statistical power, violation of test assumptions and failure to consider habituation'. Also the experimental design of the trial employed pseudo-replication [59]. This casts doubt on whether the VF alone was responsible for the reduction in roadkill or whether there were other variables that had an effect. The VF system was to be installed on a section of the Huon Highway, near Hobart, Tasmania by the Department of Infrastructure Energy and Resources. This provided an opportunity to subject the manufacturer's claims to more rigorous scientific scrutiny.

Study 4 details a field trial of the VF system on a Tasmanian highway, conducted daily over 126 days. The resultant report investigates whether previous claims for the efficacy of this commercial device are upheld and how its functionality might be improved.

Figure A5. One Welfare. Roadkill mitigation and infrastructure



5.2. Published paper: A Trial of a Solar-Powered, Cooperative Sensor/Actuator, Opto-Acoustical, Virtual Road-Fence to Mitigate Roadkill in Tasmania, Australia

The following article has been published in *Animals*, as part of a special issue on Behaviour and Management of Urban Wildlife.

A Trial of a Solar-Powered, Cooperative Sensor/Actuator, Opto-Acoustical, Virtual Road-Fence to Mitigate Roadkill in Tasmania, Australia

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Simple Summary: The Australian state of Tasmania has a high rate of roadkill, so any method that reduces roadkill in this state deserves attention. A commercial roadkill mitigation device, which combines an auditory warning signal with flashing blue and amber lights in linked units to form a so-called virtual fence, is said to reduce roadkill by up to 90%. For the current trial, a virtual fence was installed on a 4.5-km segment of Tasmanian highway south of Hobart and roadkill was monitored on a daily basis for a period of 126 days. Sections of the virtual fence were switched on or off, according to a predetermined experimental design. Bennett's wallabies, Tasmanian pademelons, and common brush-tail possums accounted for most of the total roadkill of 174 animals over the study period. For these three species, four complementary methods of analysis failed to reveal any significant effect of the virtual fence in reducing roadkill. This study does not confirm previously reported estimates of reduction in roadkill rates of 50%–90%.

Abstract: When wildlife and motor vehicles collide, the result for the animals is often death (roadkill). A commercial roadkill mitigation device that forms a so-called virtual fence (VF), is said to reduce roadkill by up to 90%. A field trial to test its effectiveness was undertaken along a 4.5-km segment of a Tasmanian highway subdivided into 6 equal sections. A total of 126 days of monitoring of roadkill by species was conducted, with alternate sections being switched on or off, according to a variation of Crossover and Multiple Before-After-Control-Impact experimental designs that divided monitoring into five periods. From the six sections over the five periods, the 30 aggregated values of daily counts of roadkill for each species were modelled. Bennett's wallabies (BW) *(Notamacropus rufogriseus)*, Tasmanian pademelons (TP) *(Thylogale billardierii)* and common brush-tail possums (BP) *(Trichosurus vulpecula)* accounted for most of the total roadkill of 174 animals. Although initially there appeared to be an effect, linear model fits to standardised roadkill rates were not statistically significant for each of BW, TP, and BP using each of the Crossover, Multiple Before-After-Control-Impact, and simple On versus Off comparisons.

Adjustment for spatial and temporal trends using a Generalised Additive Model with Poisson error also failed to detect a significant VF effect. A simulation study used to estimate the power to detect a statistically significant reduction in roadkill rate gave, for median estimates of reduction of 21%, 48%, and 57%, estimates of power of 0.24, 0.78, and 0.91, respectively. Therefore, this study failed to confirm previously reported estimates of reduction in roadkill rates claimed for this VF of 50%–90%, despite having adequate power to do so. However, point estimates obtained for these three species of reductions ranging from 13% to 32% leave open the question of there being a real but modest effect that was below statistical detection limits.

Keywords: wildlife vehicle collisions; roadkill; One Welfare; virtual fence; avoidance learning; animal welfare

1. Introduction

Road infrastructure is expanding rapidly on a global scale as industrialisation and urbanisation increase [1,2]. One consequence is a global rise in animals being killed or injured (roadkill) in wildlife vehicle collisions (WVC) in Europe [3], the Americas [4] and Australia [5], and a rising number of injured and orphaned animals being rescued by welfare organisations [6-9]. WVC can have serious consequences for animals and humans, i.e. death or injury, and can affect the environment through species decline [10-14]. The 'One Welfare' concept [15,16], where animal health, human well-being, socio-economic development and environmental sustainability are inexorably linked, might suggest that any reduction in WVC could have strategic value, such as fewer car insurance claims and less roadkill for tourists to encounter. In Australia, a 10% reduction in WVC would mean up to 40,000 fewer mammalian roadkill victims each year [5].

Three main approaches to mitigating the problem of WVC, which can be undertaken individually or in combination, are infrastructure management, changing human behaviour and changing animal behaviour. Infrastructure management includes building roadside fences, culverts and land-bridges, but these are costly, ranging from AUD 50,000 for a culvert to AUD 2 million for a land-bridge [17,18]. Changing human behaviour focuses chiefly on attempts to change the behaviour of drivers but this has proven to be elusive [12,19,20], although studies are starting to merge the biological and social sciences in a bid to address this [21,22]. Changing animal behaviour to reduce WVC may involve the use of devices that trigger a flight response and alert wildlife to approaching traffic. However, when subjected to scientific evaluation, few of these methods have been shown to be effective and have little value [23-27].

Since the studies that have attempted to change animal behaviour in the proximity of roads have met with limited success, there is a need for further exploration in this area. The recent development of lithium-polymer rechargeable batteries and thin film solar-cells, combined with very low power consumption, has enabled the production of active roadside systems. These produce auditory and optical signals to alert animals to approaching vehicles, which are intended to trigger a response from animals of avoiding or departure from the road. One such device is a solar-powered, transport system sensor/actuator manufactured in Austria (iPTE Traffic Solutions Ltd. 8054 Graz/Austria Mantscha-Wald-Weg 48, Austria). The units are designed to produce a virtual fence (VF) along the roadway and to work from dusk to dawn, alerting crepuscular and nocturnal animals. An internal light sensor in the unit detects an approaching vehicle's headlight at a standard threshold of 150 lux. This causes an optical/acoustical alert system to be triggered. According to the manufacturer, *'The acoustic sound of the warning sequence raises the attention of the animals and the flashing lights makes the animals feel uncomfortable and leave the road area'.* A five-year trial of these devices in Austria concluded that there was a 90% sustainable reduction of roadkill resulting from WVC [28,29]. Subsequently, a three-year

trial in Tasmania, Australia, concluded that roadkill was reduced by over 50% and that *'these devices have enormous potential to substantially reduce roadkill rates'* [30]. However, at other test sites in other countries, these results could not be reproduced. For example, in the UK, USA and Hungary, the devices did not have any significant effect in reducing roadkill from WVC [28]. The aim of this study was to assess the operation of the VF and to explain the divergent findings on its effectiveness. The experimental design employed true spatial and temporal replication.

2. Materials and Methods

2.1. Study Site and Data Collection

In April 2018, the Tasmanian Department of State Growth, Infrastructure, Energy and Resources erected a VF system (Virtual fencing unit DD430-B Gen_3) produced by IPTE Traffic Solutions Ltd, (8054 Graz, Austria, Mantscha-Wald-Weg 48) along a 4.5-km section of the Huon Highway, a single carriageway road with three lanes, from Lesley Vale (42^0 58' 0.49S 147^0 14' 51.57E) to Sandfly (42^0 58'49.18S 147^0 12'26.53E). (See Supplementary Material for operating characteristics of the VF system including Figure S1 showing sound levels for general traffic, the VF unit, and background noise).

The 4.5-km section of highway [31] is mostly straight with some sweeping bends. Proceeding north-east to south-west, the highway has a gentle decline but becomes steeper over approximately the last 1.5 km. Rough pasture abuts the highway, with intermittent copses of eucalypts and light undergrowth either side of the clear gravel and grassy verges, ranging in width from approximately 5 to 15 m. The vegetation becomes more dense woodland alongside the steeper south-western sections. A traffic counter (Vehicle Classifier System, MetroCount, 15 O'Connor Close, North Coogee, W.A. 6163 Australia) was located 910 m west of the Huon HWY-Leslie Road intersection. The counter accurately approximates the number of vehicles using the full extent of the trial site, their speed, and time of their passage on a continuous basis. The operational status of the VF units was checked on a weekly basis.

The VF was divided into six, equal in length, segments, using Google Maps to measure and allocate exact GPS co-ordinates of the start and end of segments [31]. A buffer section of 750 m at each end of the virtual fence resulted in eight segments being monitored. The eight segments were searched for roadkill on a daily basis over 18 weeks. Each roadkill was photographed with a time, date, and GPS stamp and then left in place (see Supplementary Material). Removing carcasses would have reduced the likelihood of scavengers such as Tasmanian devils (*Sarcophilus harrisii*) and quolls (*Dasyurus maculatus* and *Dasyurus viverrinus*) becoming roadkill [32] (also see Figure S2, Supplementary Material).

2.2. Experimental Design Including Treatment Allocation (VF Off vs On) and Data Aggregation/Standardisation

The experimental design, drawing data from only the six sections (i.e., Sections 2 to 7) that had the VF installed, was a combination of a replicated BACI [i.e., Multiple Before-After-Control-Impact (MBACI)] design [33], and a Crossover design. Crossover designs are commonly used in medical research [34,35] and, in our application, the VF switched Off and On are analogous to a placebo and a medical intervention, such as administering a drug, respectively. The Crossover design involved dividing the six sections (contiguous Sections 2 to 7, proceeding westward) into two blocks, with Block 1 composed of Sections 2, 4, and 6, and Block 2 composed of Sections 3, 5, and 7 (Table 1).

2.2.1. Monitoring Periods

A period of 42 days of monitoring of roadkill by species, starting 26 March 2018, was conducted prior to the VF being switched on, being split into a pre-trial monitoring period of 14 days, starting from the 26 March, then a 28-day period from when the VF was installed (on 1 May) but not switched on. For the next 28 days, the three sections in Block 1 had their VF switched on. The road was then monitored for a 14-day period with the VF switched off on all sections. In the following 28-day period,

only the three sections in Block 2 had their VF switched on. Monitoring proceeded for a subsequent and final 14-day period with the VF switched off for all sections. Total days of monitoring were 128, half of which were with the VF switched on (Table 1).

Table 1. Periods with On-Off Periods 3 and 5 disaggregated to On and Off Blocks excluding buffer sections 1 and 8 (Block 1: Sections 2, 4, 6; Block 2: Sections 3, 5, 7). The periods when the virtual fence (VF) was switched on are italicised.

Period label	Pre-trial	Pre_All_Off	Block1_On	Post1_All_Off	Block2_On	Post2_All_Off
Period No	1	2	*3*	4	*5*	6
Start Date 2018	26/03	1/05	*28/05*	25/06	*9/07*	6/08
End Date 2018	8/04	28/05	*25/06*	9/07	*6/08*	20/08
Period (days)	14	28	*28*	14	*28*	14

2.2.2. Aggregation and Standardisation of Roadkill Rates

Counts were aggregated across the Pre-trial (14 days) period and the subsequent Pre_All_Off (28 days) period within each section into a single count for the statistical analyses. Period 2 in the statistical analyses refers to this combined period of 42 days. The Period factor, therefore, takes the five labels of Period 2 to Period 6. From the six sections over the five periods, 30 aggregated values of daily counts of roadkill were obtained separately for each species and used for formal statistical analyses. The roadkill counts were analysed as raw counts or standardised rates depending on the statistical model that was fitted (see Supplementary Material).

2.3. Statistical Analyses

We present four different analyses because they each exploit different aspects of the experimental design and, in doing so, they either use different subsets of the data or, in the case of the omnibus LM test and the GAM, the same full dataset. The latter attempts to improve the precision of estimates of reduction in roadkill rates due to the VF by adjusting for any spatial or temporal smooth trends in roadkill rates. The Crossover analysis compares On vs Off across different spatial units for a fixed temporal unit within Block, while the MBACI analysis contrasts Before and After (i.e., Off vs On, respectively) within, effectively fixed spatial units (i.e. this contrast removes spatial effects).

2.3.1. Crossover, MBACI, and Omnibus Off vs On Analyses

For these analyses the response variable used was standardised roadkill rate using either period length (days) or traffic count standardisations. The Crossover analysis compares roadkill rates in Block 1 between On (Period 3) and Off (Period 5) and in Block 2 for Period 5 and Period 3 with this order of periods reflecting VF On vs VF Off. Means were graphed by block and visually overlaid. If there was a substantial reduction in mean standardised roadkill rate due to the virtual fence operating, then the lines connecting means within blocks should intersect (i.e., when graphed with order on the abscissa scale of Period 3 then Period 5 the slope should be positive for Block 1 and negative for Block 2). A formal test of the significance of the effect of the VF used a linear mixed model (LMM) [36] with period factor (two levels) as a main effect, a single degree of freedom contrast between Off and On, and a random section effect for the data restricted to just these two periods. Carry-over effects [34] were assumed to be negligible due to the 14-day period with the VF switched off on all sections. The *lmer* function from the R-software [37] library *lme4* was used to fit the LMM. Note that this crossover test can also be considered a contemporaneous comparison since it equivalently tests the difference between On versus Off within each period.

The MBACI design consisted of pairing adjacent On/Off sections in that order: For Period 3, these were Sections 2/3, 4/5, 6/7, while for Period 5, these were reversed as 3/2, 5/4, and 7/6. So, in each case the On sections within a block are considered the Impact (Treatment) site, and the Off sections the Control site. Further, for Block 1, Period 2 represents the Before versus Period 3 as the After in BACI notation, so that subtracting the mean rate for the former from that of the latter gives the contrast estimating the effect of the VF. Similarly, for Block 2, Period 4 represents the Before versus Period 5 as the After comparison. Therefore, there were six spatial replicates of the BACI design; three in each block. Similar to the crossover, the means, and their standard error (SE) bars are graphed separately for each block. However, for the MBACI components, there were four means and two connecting lines for each block with means within each block graphed, with abscissa being the factor levels in the order of Before and On (denoted After for a MBACI design where On or After for the Control section/periods simply means that they were observed contemporaneously with the adjacent VF On section within the pairs). If there was a substantial reduction in mean standardised roadkill due to the virtual fence operating then there should be a substantial decline in mean from Before to On for the Impact section/periods, consistently, across the two blocks, and a similar decline should not be seen for the Control section/periods. For the Control sections, a minor decline, no change, or an increase might occur in mean standardised roadkill. To take into account any change within the Control section/periods contemporaneously with the Impact section/periods, this change was subtracted from the mean change for the Impact section/periods and the corresponding estimate of the standard error of this adjusted mean contrast was obtained.

A final general or omnibus test was conducted for all periods, using a simple On versus Off factor (denoted as VF_On_vs_Off) fitted in a linear model (i.e., with no Block main effect or interaction with VF_On_vs_Off included). The percentage change (i.e., 100 times the Off minus On mean divided by the Off mean) was calculated and the standard error of this percentage approximated by a second order Taylor series approximation [38] and these values are given in the Results.

2.3.2. Generalized Additive Models

To account for differing length of time periods, length of sections, or traffic count for the period, the raw counts were used as the response variable in a Generalized Additive Model (GAM) [39] with Poisson error, Naperian log link function, and an offset of the sum of the log of the number of days in the period (or alternatively the log of the traffic count for the period) and the log of the length of the section. The GAM allows trends to be modelled as empirical smooths (such as a cubic smoothing spline). GAMs were used to adjust estimates of the effect of the VF by effectively removing the effect of smooth trends in each of the Section sequence number (i.e., a very close approximation to using the distance to the spatial locations of section midpoints from the start of Section 2) and the period midpoints (i.e., where for Period 2 this was taken as the days from March 25 to May 1), using cubic smoothing splines with basis dimension of three for both splines. These splines were added to the linear predictor of the GAM and combined with the general On versus Off contrast (i.e., effect). The percentage reduction in standardised roadkill was obtained directly from the GAM as 100*{1-exp(par_est)} where par_est is the parameter estimate for the contrast in question with corresponding standard error approximated using a first-order Taylor series approximation by 100*exp(par_est)*SE(par_est) where SE(par_est) is the estimated standard error. The GAM was fitted using the *gam* function from the R *mgcv* library [39]. Comparison of the fit of the GAM that used the component of the offset of log of Period length compared to that using the log of traffic count was made using the Akaike Information Criterion (AIC) [40]. The Dunn-Smyth type quantile residuals [86] were plotted against the standard normal quantiles to assess whether the assumption of Poisson distributed counts was reasonable using functions from the R *statmod* library.

2.3.3. Simulation Study to Estimate Power to Detect a Statistically Significant Reduction in Roadkill

To deliver true replication of On and Off treatments (i.e., section by period combinations), the study design focused on providing a valid comparison, both spatially and temporally. Accordingly, there was less replication of the On treatment (six replicates) than the Off treatment (24 replicates). To investigate the power to detect a statistically significant reduction in roadkill using this study design, using an assumed Poisson distribution for counts and the spatial and temporal trends estimated by the GAM, the fitted GAM for the BW data was used as a baseline operating model.

To investigate a range of percentage roadkill reductions, including the estimate from the fit to the actual data, the On and Off predicted means for section by period combinations obtained from the GAM were jointly deflated and inflated, respectively, each by 0%, 10%, 20%, and 30%. For each of these four sets of predicted values, a random Gaussian estimation error from the fit of the GAM was added to the linear predictor scale. A random Poisson value for each section by period combination was drawn based on these scaled and perturbated linear predictor values to generate 1000 sets of these 30 random values for each deflation/inflation percentage. The GAM was refitted to the 1000 datasets and the percentage reduction estimated for each deflation/inflation percentage. Using these 1000 estimates of percentage reduction, the median estimate was determined and used as the assumed alternative hypothesis value to test (i.e., where the null hypothesis is a zero value for percentage reduction). The critical value for the null hypothesis test was taken to be the upper 95% quantile of the 1000 sample estimates minus the median estimate (i.e., a one-sided test). The power of the test (i.e., probability of rejecting the null hypothesis when the true value of percentage reduction is as estimated by the median value corresponding to one minus the probability of a Type II error) was calculated as the proportion of the 1000 simulations where the estimated percentage reduction was greater than the critical value. Thus, the power was estimated for each median estimate of percentage reduction corresponding to each deflation/inflation percentage, since these last percentages were more straightforward to use as inputs than trying to manipulate percentage reduction directly.

The above analysis of power considers only one study site (i.e., the site of this study). So, the power analysis was extended to consider a random sample of S sites with the same study design. To do this, transformations of the percentage reduction that gave a constant standard error across the range of estimated percentage reductions in the above simulation study were examined empirically. Once this transformed estimate was obtained, the corresponding average standard error was divided by the square root of S to give an estimate of the standard error of the mean transformed reduction estimated across S replicates of the design. Classic power analysis (Steele and Torrie, 1960) was carried out using a t-distribution for the transformed estimate, a target value for percentage reduction, and the value of S in order to obtain an estimate of the power to detect the target reduction in roadkill rate due to the VF. Note that the power will be under-estimated to a degree dependent on the magnitude of between-site variance in the site-specific estimates of percentage reduction since, in the absence of any reliable estimate of its magnitude, this variance was assumed to be negligible in the extended power analysis.

3. Results

3.1. Vehicle and Roadkill Data.

Daily total traffic count (i.e., across axle types and speed classes) for the dusk to dawn period separately for westbound and eastbound traffic on the 6 km stretch of road studied are shown in Figure S3 (Supplementary Material). Average (± one standard deviation) and maximum daily traffic speed between dusk and dawn for the study period are shown in Figure S4 (Supplementary Material). Over the period 25 March to 20 August, 388,595 vehicles were recorded between dusk and dawn (combining eastward and westward traffic) with 25.4% recording a speed above the limit of 100 $km.h^{-1}$ and 3.5% recording a speed in excess of 120 $km.h^{-1}$.

Bennett's wallabies (BW) *(Notamacropus rufogriseus)*, Tasmanian pademelons (TP) *(Thylogale billardierii)* and common brush-tail possums (BP) *(Trichosurus vulpecula)* accounted for most of the

total roadkill of 174 animals. The total roadkill disaggregated by species and period is shown in Table S1 (Supplementary Material). Spatial locations for BW, TP, and BP found dead are given by Figure S5 (Supplementary Material) and show an even spread along the trial 6 km road section for each of these three species. Table 2 gives estimates of standardised roadkill rates by the status of the fence as Off versus On for the three most prevalent (see Table S1) roadkill species. Rates were calculated using counts for trial sections 2 to 7. Raw rates were obtained by standardisation after aggregating raw counts across these sections and periods within Off versus On treatment status.

Table 2. Roadkill rates for three most prevalent roadkill species. Number killed for VF "On" versus "Off" and standardised rates using a general VF "On" vs "Off" factor/contrast estimated using all periods and Sections 2 to 7. Raw standardised rates and resultant percentage reduction based on total counts appear in the second, third, and fourth columns followed by model-based standardised rates and resultant percentage reductions with corresponding standard errors. For raw rates, the standardisation used total days of VF "On" apportioned by the fraction of sections switched on in Periods 3 and 5 and similarly for VF "Off". Vehicle counts were similarly apportioned.

Species	**Total counts**			**Rate (number.month^{-1}km^{-1})**			**Rate (number.100kVeh^{-1}km^{-1})**		
	VF Off	**VF On**	**Total**	**VF Off**	**VF On**	**%Reduction**	**VF Off**	**VF On**	**%Reduction**
Wallaby (BW)	58	10	68	3.946	2.381	39.66	4.986	2.612	47.60
Pademelon (TP)	48	10	58	3.265	2.381	27.08	4.126	2.612	36.69
Possum (BP)	23	5	28	1.565	1.190	23.91	1.977	1.306	33.93
				Rates from LM estimates (SE) for general VF_On_vs_Off factor					
Wallaby (BW)				3.194 (0.490)	2.381 (0.980)	25.5 (39.3)	3.951 (0.649)	25.5 (39.3)	33.1 (41.7)
Pademelon (TP)				3.492 (0.411)	2.381 (0.823)	31.8 (29.9)	4.614 (0.595)	31.8 (29.9)	42.2 (32.4)
Possum (BP)				1.389 (0.367)	1.190 (0.734)	14.3 (68.6)	1.818 (0.483)	14.3 (68.6)	27.3 (67.9)
				Rates from GAM estimates (SE) for general VF_On_vs_Off factor					
Wallaby (BW)						23.0 (27.7)			31.3 (24.7)
Pademelon (TP)						32.2 (23.9)			29.4 (26.0)
Possum (BP)						12.5 (45.5)			21.5 (40.8)

3.2. LM, LMM, and GAM Outputs

The mean rates of roadkill for both standardisations obtained from the LM estimates for the general VF_On_vs_Off factor are given in Table 2, along with corresponding estimates of the percentage reduction in roadkill rate due to the VF being switched on. Note that the raw mean rates are different from the LM-estimated rates since the raw rates are effectively weighted versions of the modelled rates with weights corresponding to the standardisation variable for each period and section. Similarly, the GAM estimates of the percentage reduction in roadkill rate are given after adjusting for the smooth trends (described in 2.3.2) estimated across periods and sections. These smooth trends are shown for the "month^{-1} km^{-1}" standardisation (i.e., corresponding to the appropriate offset term in the GAM) for BW, for smooths in section number and period midpoint in Figures S6 and S7 (Supplementary Material), respectively.

The percentage reduction estimate, its estimate of standard error, and the formal test of significance obtained from the GAM for VF_On_vs_Off are shown in Table 2. The VF_On_vs_Off parameter estimate (par_est) for the default "set to zero" parameterisation (i.e., giving an On minus Off estimate) was not statistically different from zero ($p > 0.1$) for each of the three main species for both the LM and GAM fits. Dunn and Smyth quantile residuals from the GAM fit indicated that the Poisson assumption was reasonable for each of the three species modelled. The results for the GAM, using the log of traffic count as offset, were very close to those presented above and gave a difference in AIC statistic between these two models that was less than 1% across all three species. Therefore,

either model can be used for inference since they have very close to the same goodness of fit to the data. The percentage reduction in rate, using the general On versus Off contrast (Table 2), was generally higher (by between 7% and 12%) when using the traffic count-based standardisation but the standard errors were also in general higher. Using either standardisation, the GAM parameter estimate (par_est) was not significantly different from zero ($p > 0.1$). We concentrated on the results using the log of period length as offset (i.e., effectively the month-based standardisation) as it was easier to interpret (i.e., in the absence of traffic counts).

Table 3 gives the contrast means (using the month-based standardisation) corresponding to On minus Off mean for the crossover, MBACI and general VF_On_vs_Off analyses by Block. For the MBACI contrast of Impact: Before versus After, the adjustment for the MBACI Control contrast is also given in Table 3. For the crossover, the combined (across blocks) means and standard errors were obtained from the LMM. As seen from Table 3, none of the contrasts that quantify the effect of the VF were statistically significantly different from zero ($p > 0.1$). Figures 1 and 2, respectively, are presented for BW to demonstrate the crossover and MBACI analyses graphically for species observed. Similar graphs could be constructed for TP and BP from Table 3 (which have not been shown for brevity).

Table 3. Roadkill rates for three most prevalent roadkill species. Linear model contrast parameter estimates for standardised roadkill rates (number.month^{-1}km^{-1}).

	Crossover contrast[a] (Periods 3 and 5)					
	Block 1		**Block 2**		**LMM/LM estimate**	
Species	Estimate	SE	Estimate	SE	Estimate	SE
Wallaby (BW)	-2.381	1.615	-0.952	1.615	-1.667	1.080
Pademelon (TP)	-1.429	1.782	0.476	1.782	-0.476	1.230
Possum (BP)	0.476	1.166	-1.905	1.166	-0.714	0.673
	MBACI contrast 1[b] (Impact: Before vs After) (all periods except 6)					
Wallaby (BW)	-1.905	1.495	-0.952	1.495	-1.429	1.057
					(-1.032)	(1.495)
Pademelon (TP)	-0.794	1.611	0.476	1.611	-0.159	1.139
					(0.794)	(1.611)
Possum (BP)	0.635	1.365	0.476	1.365	0.556	0.966
					(0.238)	(1.365)
	MBACI contrast 2[c] (Control: Before vs After) (all periods except 6)					
Wallaby (BW)	-3.175**	1.495	2.381	1.495	-0.397	1.057
Pademelon (TP)	-0.952	1.611	-0.952	1.611	-0.952	1.139
Possum (BP)	0.159	1.365	0.317	1.365	0.317	0.966
	On vs Off contrast[a] (all periods)					
Wallaby (BW)	-0.833	1.575	-0.794	1.575	-0.813	1.095
Pademelon (TP)	-1.508	1.271	-0.714	1.271	-1.111	0.920
Possum (BP)	0.516	1.047	-0.913	1.047	-0.198	0.821

[a] A negative estimate indicates a reduction in standardised roadkill when the fence was switched on. [b] A negative estimate indicates a reduction in standardised roadkill when the fence was switched on. Values within brackets

are the Impact values after subtracting the corresponding Control estimates. [c] A negative estimate indicates a reduction in standardised roadkill when the fence for the control comparison for the sections of the block switched off both before and during the periods corresponding to the On block. [ns] Not significantly different from zero (i.e., no effect of VF detected) ($p > 0.1$). If no probability level is given then $p > 0.1$. [*] Probability level 0.05–0.10. [**] Probability level 0.025–0.05.

The fitted GAM with effective standardisations of either period length in days or as dusk-to-dawn traffic counts, showed no significant reduction in roadkill rate (Table 3) ($p > 0.1$) for the VF. Point estimates obtained from the fitted GAM indicate a moderate reduction of between 23% and 32% for BW and TP, respectively, albeit with large uncertainty bounds as reflected in the large relative standard errors (Table 2). For Bennett's wallaby, there was an indication of a reduction in roadkill rate (month^{-1}km^{-1}) due to the VF (Figure 1). Both Blocks 1 and 2 had an apparent drop in mean roadkill rate when the VF was switched on, so that overlaying the trends for the two blocks revealed a cross-over effect that confirmed point-estimates of a reduction in rate. However, these reductions were not significant ($p > 0.1$) either when considering blocks separately or when averaging across blocks (Table 3). The MBACI analyses showed (Figure 2) that, for Block 1, the decrease in rate from Before to After the VF was switched on for the Impact sections (2, 4 and 6) was exceeded by a greater decrease for the Control sections (3, 5 and 7), which infers there was no reduction due to the VF switched on relative to being switched off. For Block 2, a relatively minor decrease in rate for the Impact sections can be inferred to be an underestimate due to the increase in rate for the Control sections. However, averaging across blocks, the overall reduction was only minor. None of the above effects were statistically significant, with the exception of the reduction for the MBACI analysis and the Control sections for Block 1 ($p < 0.05$, Table 3), where this last effect clearly is not the result of the VF being switched on.

Figure 1. Standardised roadkill (month^{-1}km^{-1}) for Bennett's wallaby for Periods 3 and 5 for Crossover means by Block. Single Standard Error (SE) bars are shown.

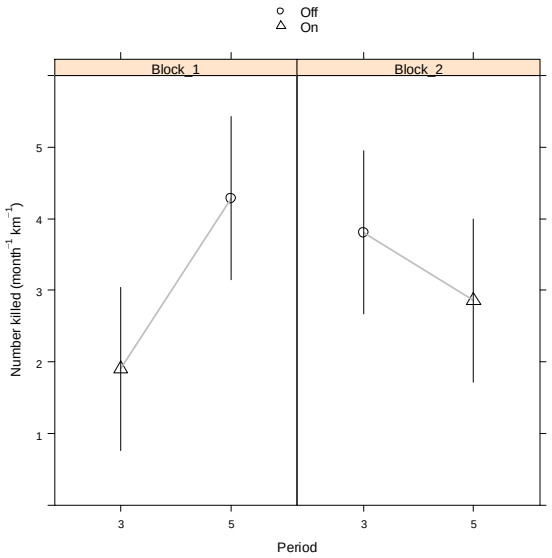
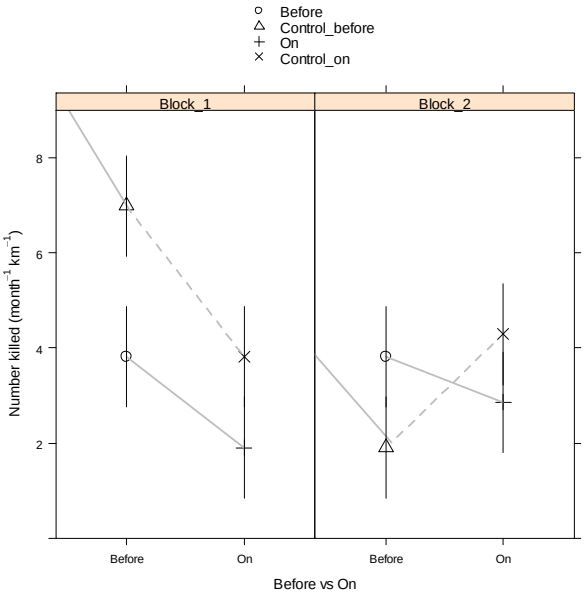


Figure 2. Standardised roadkill rate (month^{-1}km^{-1}) for Bennett's wallaby for Periods 2 to 5 for MBACI means by Block. Single SE bars are shown.



3.3. Simulation Study of Power

Table 4 gives the results of the simulation study investigating the power of the study design and the GAM fitted to the BW data to detect a statistically significant reduction in roadkill rate of Bennet's wallaby.

Table 4. Power calculation using 1000 simulations for each artificial inflation/deflation level of the VF Off/On Bennett's wallaby counts. Simulations used Poisson random draws for each of the 30 combinations of Period by Section using Poisson means calculated as predicted means from the GAM fitted to the original data with an additional Gaussian random prediction error added on the linear predictor scale.

"On" deflation "Off" inflation (%) of prediction	Median[a] Percentage Reduction (%)	Standard[b] deviation (%)	Median Standard[c] Error (%)	Power[d] to detect Median Reduction (P-level)
0	20.96	34.32	27.87	0.242
10	36.69	28.75	23.25	0.543
20	48.08	22.60	19.69	0.783
30	57.32	20.38	16.94	0.907

[a] Median of 1000 simulations and GAM estimate of VF_On_vs_Off factor On effect (Par_est), with percentage reduction due to the VF switched on, given by 100*{1-exp(Par_est)}. The GAM also included cubic smoothing splines as fitted to the original data. [b] Standard deviation of the 1000 simulation estimates of percentage reduction. [c] Median of 1000 simulation estimates of the standard error of the percentage reduction given by 100*exp(Par_est)*SE_Par_est, where SE_Par_est is the standard error of the estimate (Par_est). [d] 1-Probability of a Type II error, where a Type II error is accepting the null hypothesis that the percentage reduction is zero when it is false as given by the alternative hypothesis that the true percentage reduction is given by the median percentage reduction. Note that the critical value of the null hypothesis test was the difference between the 95% quantile of the 1000 estimates of percentage reduction minus the median of these 1000 estimates.

For the extended power analysis, the transformation log(1-%reduction/100) gave standard error of estimates, corresponding to median percentage reductions (Table 4), ranging from 0.43 to 0.48 with a mean of 0.45. The untransformed median percentage reductions (Table 4) and alternative transformations of log(%reduction/100) gave much larger ranges in standard error estimates. Using the above mean of 0.45 for the standard error and the estimated residual degrees of freedom from the fit of the GAM of 25.7 giving the required t-distribution with degrees of freedom 25.7 × *S*, a target of a 25% reduction, a value of *S* of 16 (i.e., our study replicated at each of 15 other sites), the power to detect such a reduction was 0.82. For a target of 10% reduction and *S* of 100, the power was 0.76.

4. Discussion

4.1. Efficacy of Virtual Fencing

Mitigating the effects of wildlife vehicle collisions, and subsequent roadkill, is a world-wide goal for road managers. Many different measures have been studied and trialled to determine their effectiveness [2,42]. However, only two studies, Schalk et al. [28] and Fox et al. [30], have reported on the efficacy of a virtual fence of the design tested in the current study. The rigour of the science of the latter has been questioned, with Coulson and Bender [43] stating that 'there are a total of eight methodological flaws ranging from imprecise measurements, confounding effects of treatments, low statistical power, violation of test assumptions and failure to consider habituation'. The design of the current study employed true spatial replication as well as temporal replication (i.e., in contrast to the lack of spatial replication in Fox et al. [30]) of On and Off treatments (i.e., section by period combinations) and therefore provided a valid comparison both spatially and temporally. To confirm the power of the study to detect a statistically significant reduction in roadkill using this study design, the simulation study showed that for true reductions of 21%, 48%, and 57% the power (i.e., 1-probability of a Type II error, often denoted as a false negative) was estimated at 0.24, 0.78, and 0.91, respectively. Thus, a marginal reduction of around 20% (i.e., of similar magnitude to the point estimates obtained for BW and TP in the current study) had low power to reveal such a reduction as statistically significant but, for the reported reduction of 50% for TP reported by Fox et al. [30] (and with reference to European estimates in the order of 80%–90%), it had high levels of power.

A limitation of our study is the relatively short monitoring period of 18 weeks compared to that of Fox et al. [30]. The much higher roadkill rate in our study compared to that of Fox et al. [30] compensates to a considerable degree for the shorter monitoring period since the relative standard deviation for a Poisson count variable decreases as the mean rate increases, thus improving the power to detect differences in mean rates. However, the current study, on its own, was reliably estimated to have sufficient statistical power to confirm estimates of a substantial (i.e., >50%) reduction in roadkill rates due to operation of the VF, but failed to do so.

The main research question investigated by the statistical modelling and hypothesis testing in the current study was whether the VF reduced the roadkill rate for any of the species that had sufficient numbers of roadkill within the study period to give reliable estimates of average roadkill rates. A linear model for standardised rates of roadkill using either days combined with section length, or dawn-to-dusk traffic count combined with section length, did not indicate the VF would significantly reduce the rate of roadkill for any of the three major roadkill species ($p > 0.1$). These tests were obtained from three separate analyses using crossover, MBACI, and general VF On versus Off comparisons. The fitted GAM with effective standardisations of either period length in days or as dusk-to-dawn traffic counts, showed no significant reduction in roadkill rate for the VF.

These four analyses are complementary and none of the tests they imposed could detect any significant effect of the VF on roadkill rates. Combined with the simulation study that estimated the power to detect effects of a range of magnitudes, the current study indicates that if the true effect size corresponded to any of the point estimates (i.e., an approximate 20% to 30% reduction) obtained from the different analyses, it was below that detectable using the study design and dataset obtained. There was a reasonably uniform distribution of roadkill along the entire section of highway studied (Figure

S5). Any "spill-over" effects are likely to have been minor and would not have affected our estimation of the effect of the VF to any practical degree. Had there been any significant “spill-over”, combined with a substantial reduction in roadkill due to the operation of the VF, then roadkill in the start or end of the sections with VF turned off would have been lower than expected due to operation of the VF in the adjacent "On" section. However, this is not apparent from Figure S5.

A notable feature of the results from this study is the high rate of roadkill for BW and TP, by a factor of nine and five times, respectively, compared with the rate for the unfenced road section reported by Fox et al. [30] of 0.347 month^{-1}km^{-1} for BW and at 0.627 month^{-1}km^{-1} for TP. However, while common brush-tailed possums (BP) were a major component in the current study, with a rate of 1.39 month^{-1}km^{-1}, they were a minor component in the Fox et al. study, with a rate of 0.026 month^{-1}km^{-1}. The much higher roadkill rates for Bennett’s wallaby (BW) and Tasmanian pademelon (TP) in the current study could possibly be attributed to much higher dusk-to-dawn daily traffic counts, although no traffic count data were reported in the Fox et al. study. BW, TP, and BP are crepuscular/nocturnal feeders, so most roadkill occurs between dusk and dawn [19]. There is also the possibility that the two habitats contained differing animal population densities. The Arthur Highway on the west coast of Tasmania runs through coastal scrubland whereas the Huon Highway of South East Tasmania runs through farmed grazing land combined with native bushland. Larger population sizes due to the greater availability of nearby pastureland and native bushland for grazing would be expected at the Huon Highway site [44].

4.2. Limitations of the Virtual Fence and Future Research

The secondary purpose of the trial was to explore why the VF may be ineffective at this location. In correspondence with the current research team, the manufacturers state: *‘effectiveness* [of the devices] *is speed dependent’*. That speed can be of significance with roadkill as demonstrated by Hobday and Mistrell [19] although they measured driver reaction times rather than animal reaction times. Night-time driving speeds needed to be below 80 km/h^{-1} if roadkill rates were to be reduced. Average driving speeds recorded during the VF trial were well above the 80 km/h^{-1} suggested by Hobday and Mistrell, with the highest recorded speed being 189 km/h^{-1}. At 189 km/h^{-1} and with the VF being triggered by headlights from 150 m, the latency for the vehicle to reach an animal at the VF site would be 2.85 s, whereas at 140 km/h^{-1} it would be 3.85 s. Both delays are likely to be insufficient for an animal to react and leave the road, so we estimate that the VF would be ineffective in these circumstances. This is confirmed in the VF manufacturer’s literature. In the section ‘Constraints on the vehicle speed’, they give the time to react and then leave the road as 6 s with the sensor-range at 150 m on low beam and maximum vehicle speed of 90 km/h^{-1} [28,29]. At the speed recommended by Hobday and Mistrell (80 km/h^{-1}) it would take 6.8 s for a vehicle to reach an animal, sufficient time for the animal to react and leave the road. However, for the VF to be effective, it must produce a response from the wildlife and it is unclear whether the sound and light stimuli of the VF produce such a reaction. No research has been undertaken in Tasmania to study the reaction of native animals to stimuli such as the sound and light emissions of the VF. This would be a useful addition to understanding possible limitations of the VF. Further possible limitations are described in the supplementary material.

Further, well-designed field trials with valid spatial and temporal replication and sufficient numbers of roadkill are required to obtain an acceptable relative precision of the estimate of reduction in rate, particularly if the anticipated reduction is substantially less than 50%. The extended power analysis indicates that our study would need to be replicated at an additional 15 sites to be able to detect a 25% reduction with adequate power. To detect a 10% reduction, an additional 99 sites would be required. Note that this number of sites is conservative since we were unable to take into account between-site variance in estimates of percent reduction in the extended power analyses. Before conducting any extensive program of field trials to more precisely estimate the degree of mitigation corresponding to reductions of the order of 20% or less, this needs to be balanced against the cost of such a program as well as the cost/benefit of such moderate reductions compared to alternative

roadkill mitigation strategies. In addition, there are concerns about the theory of how the VF operates (see Appendix). We recommend that investigation of potential improvements in the effectiveness of the VF be conducted before more research trials or operational implementation of the VF are undertaken.

5. Conclusions

In contrast to comparable studies of this virtual fence (Fox et al. [30), the current study site and observation period, combined with the experimental design, strict measurement protocols, and data analysis methods used did not reveal a statistically significant reduction of the order of 50% or greater even though there was adequate power to do so. For the three dominant roadkill species of BW, TP, and BP, all four complementary methods of analysis failed to find any statistically significant positive effect of the virtual fence in reducing roadkill.

The data and R-code used in this study can be provided on request from the senior author.

Supplementary Materials: The following are available online at www.mdpi.com/xxx/s1, Figure S1: Measurement of sound volume for general traffic, general background, and virtual fence, Figure S2: Possum feeding at the base of a VF post, Figure S3: Daily traffic counts between dusk and dawn, Figure S4: Average and maximum daily traffic speed, Figure S5: Spatial locations of roadkill for three most prevalent species, Figure S6: Predicted roadkill rate of Bennett's wallaby from GAM versus Section, Figure S7: Predicted roadkill rate of Bennett's wallaby from GAM versus Period midpoint, Table S1: Roadkill numbers by all species and period.

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Supplementary Material

1. VF operation
2. Monitoring regime
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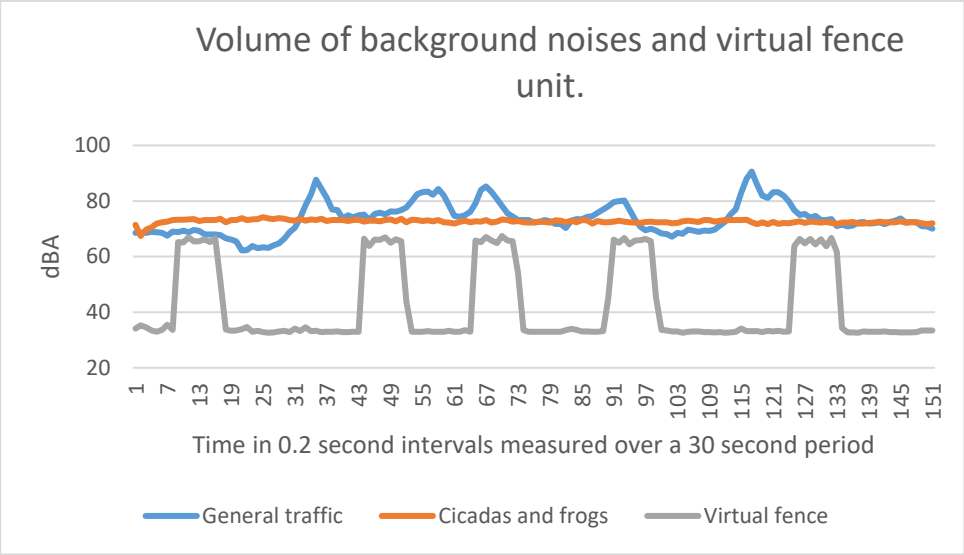
1. VF operation

There are concerns about the theory of how the VF operates. The manufacturers state: 'The acoustic sound of the warning sequence raises the attention of the animals and the flashing lights makes (sic) the animals feel uncomfortable and leave the road area'. Pavlovian (classical) conditioning requires that the unit's alerting signal of the 4 kHz sound, a conditioned stimulus, would have to precede the flashing lights that produce the conditioned response of feeling uncomfortable, i.e., an aversive stimulus. In reality, sound travels at a slower rate, approximately 343

$m.s^{-1}$, than light, 300,000,000 $m.s^{-1}$. For animals positioned 30 m from the units, there is a delay of approximately 0.1 s. Thus, the sound arrives at the animal after the flashing lights, the opposite of what is required for the manufacturer's explanation to have credence. Habituation, a form of non-associative learning in which an innate (non-reinforced) response to a stimulus decreases after repeated or prolonged presentations of that stimulus [91] could also be of concern, but could only be measured if the VF actually produces an innate response. Since in our study the VF produced no apparent effect on roadkill, i.e., the VF did not produce an innate response to make animals leave the roadway, habituation is not discussed as a reason for this effect.

Another concern is the loudness of the VF signal. Coulson and Bender mention that the sound frequency of the VF units is suitable but the intensity is unknown [88] . We used a Decibel X Pro-Sound Meter dBA (SkyPaw Co. Ltd, Hanoi Vietnam, Mobile app) noise detector, with unlocked frequency weighting A, to record the loudness (intensity) of the auditory output signal from the VF, the loudness of road traffic and the loudness of cicadas *(Cicadoidea)* and frogs *(Anura)* signalling at a distance of 12.5 m from the VF units and on the road verge between two VF units. The measurements of the road traffic and the cicadas and frogs were taken during the period from dusk into night-time at the trial site and measurement of the virtual fence unit in a secluded indoor area with low background noise, of approximately 32 dBA, and measured at a distance of 12.5 m from the unit.

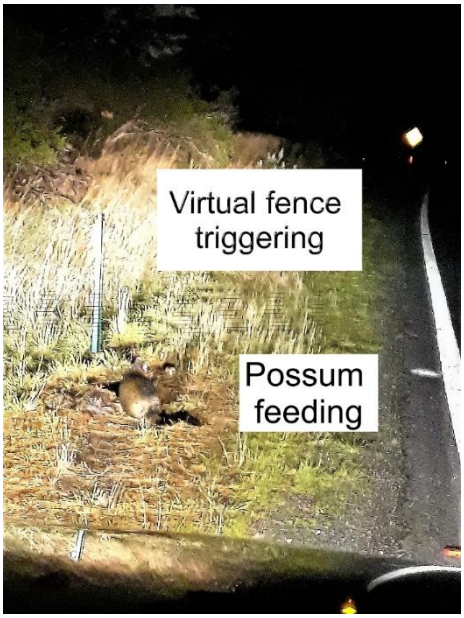
Figure S1. Measurement of sound volume for general traffic, general background, and virtual fence.



As illustrated, the peak output volume of the VF unit (68dBA), is approximately 5dBA below the noise of cicadas, frogs, and general background noise (73dBA). This is well below the sound volume of vehicles (91dBA) shown as peaks in the graph in Figure S1. It may well be that competing background noises do not allow the animals to hear or discriminate the 4 or 8 kHz audio signal of the VF units. Human hearing is most sensitive in the 2–5 kHz range and the Pro Sound measuring instrument was weighted for this. Although there is a paucity of auditory research on Tasmanian native animals, the hearing sensitivity of kangaroos and wallabies is in a similar range, 2 kHz–3.5 kHz, to that of humans [53]. Thus, discrimination would be extremely difficult when the background noise level is higher than that of the VF units. In correspondence with the current research team, the manufacturers stated: *'VF systems work most effective at low to medium dense traffic. If the traffic becomes nearly uninterrupted, the system cannot guide animals safely'*.

A further reason why the VF may be ineffective is that animals may ignore the auditory and visual stimuli of the VF units if there is an attractive resource that is more salient to them than the aversive LED flashing lights proposed by the manufacturers. This was illustrated to the researcher when carrying out night-time checks on the serviceability of the VF. A possum was seen feeding and to be ignoring the sound and flashing LED's of the VF unit (Figure S2).

Figure S2. Possum feeding at the base of a VF post, whilst the unit is operating.



Similarly, on four separate occasions whilst checking the operation of the VF at night two Tasmanian pademelons, a Bennett's wallaby and a common brush-tailed possum were seen to be stationary on the roadside verge ignoring the sound and light emitted by the VF whilst it was triggering. It is recommended that further research investigates the efficacy of an increase in the volume of the auditory output of the VF units. Additionally, there seems to be merit in replacing the current 4 kHz and 8 kHz tones with a biologically significant auditory signal and adding a delay so that it follows the LED flashing lights by one to two s. A similar method was trialled in Poland to reduce the risk of train collisions with animals [92]. It was observed that 85%–93% of wild mammals escaped an approaching train. Biedenweg et al. [93] also reported that utilising both biologically significant sounds, such as a kangaroo alarm foot thump and Australian Raven (*Corvus coronoides*) call, and artificial sounds such as a bull whip crack, solicited an aversive reaction in western grey kangaroos (*Macropus fulliginosus*).

Operating the VF units in the trial

Switching is managed by the use of a paperclip inserted into a small entrance hole on the underside of the unit. There are three modes that can be engaged: Off, a 4 Khz so-called rural tone and an 8 Khz urban tone. Identification of the mode is by a series of different auditory beeps emitted by the unit. After switching off all the units, a subsequent check of the units the next day revealed that three were in the incorrect mode and so were only then switched to the off position. Discriminating the three modes is difficult when there is heavy traffic because the sound of the VF is compromised by the loudness of the vehicles.

The method of checking recommended by the manufacturers was to attend in daylight and place a black bag over the unit, to wait two min and then expose the unit to natural light to trigger the auditory and visual signals. This proved unreliable and time-consuming. The method eventually adopted was to undertake the checking at night with the use of vehicle headlights on the researcher's car. The car was driven onto the verge and stopped. After a ten s pause, with the headlights switched off, and with the car windows down, the units were triggered by a flash of the headlights enabling the sound and light emitted by the units to be checked for efficacy.

2. Monitoring regime

A car was furnished with a flashing amber safety lamp on its roof and a Caution, Frequently Stopping Vehicle sign on its rear. The researchers wore Australian Standard 1742.3 devices, i.e., Class D – Outdoor daytime use only, fluorescent high-visibility material vests.

There is no verge at the buffer zone sites, so the researchers walked the 4 km of these sections always facing the oncoming traffic and keeping 3 m from the roadway. For the rest of the study zone, the car was driven off the road along the wide verge at low speed which enabled roadkill to be spotted. When roadkill was observed, the car was stopped and used to protect the researcher as they left the vehicle to record a photograph, GPS, time, and date. The sightlines on the roadway provided a minimum of 10 s to see approaching vehicles with the speed of vehicles at the maximum speed limit of 100 km/$^{h-1}$. The work was classed as of short duration so was undertaken without delineation (e.g., warning signs, advisory speed signs, or line marking). The vehicle was positioned so that drivers travelling in either direction could see its flashing amber light as an extra safety precaution when the researcher was out of the vehicle.

In the unlikely, but possible, event of an animal roadkill victim being seen on the roadway that appeared to have live pouch young (joey), the victim was dragged to the verge at least 3 m from the roadway for examination. This procedure to remove the roadkill animal took no more than five to eight s, so a 20 s gap in the traffic was used for this purpose. Under the regulations, Standards Australia AS 1742.3-2009 Manual of uniform traffic control devices Traffic control for works on roads, '*a lookout person may be dispensed with if the work will not take more than ten seconds and approaching traffic can be seen for a distance away equal to 20 seconds travel time*' [94] (Pt 3).

3. Data manipulation details

Standardisation of rates was carried out as either a "month" standardisation given by 30*raw count/{(number of days in period)*(section length, km)} giving roadkill rate number.month^{-1}.km^{-1}, or as a "traffic count" standardisation given by 10^5*raw count/{(traffic count in period)*(section length, km)} giving roadkill per 100,000 trips.km^{-1} where the traffic count was all vehicles travelling between the hours of dusk and dawn. Times of sunset and sunrise for each day between 25 March and 20 August, used to calculate daily traffic counts between dusk and dawn (i.e., taken as the time of sunset and sunrise, respectively), were obtained using the Geoscience Australia National Mapping Division's sunrisenset program (version 2.2) for latitude -42°58'00" and longitude 147°14'00" and adjusted from Australian Eastern Standard Time to Australian Eastern Daylight-saving Time where necessary [95].

4. Traffic data

Figure S3. Daily traffic counts between dusk and dawn showing eastbound and westbound 26 March until 20 August 2018.

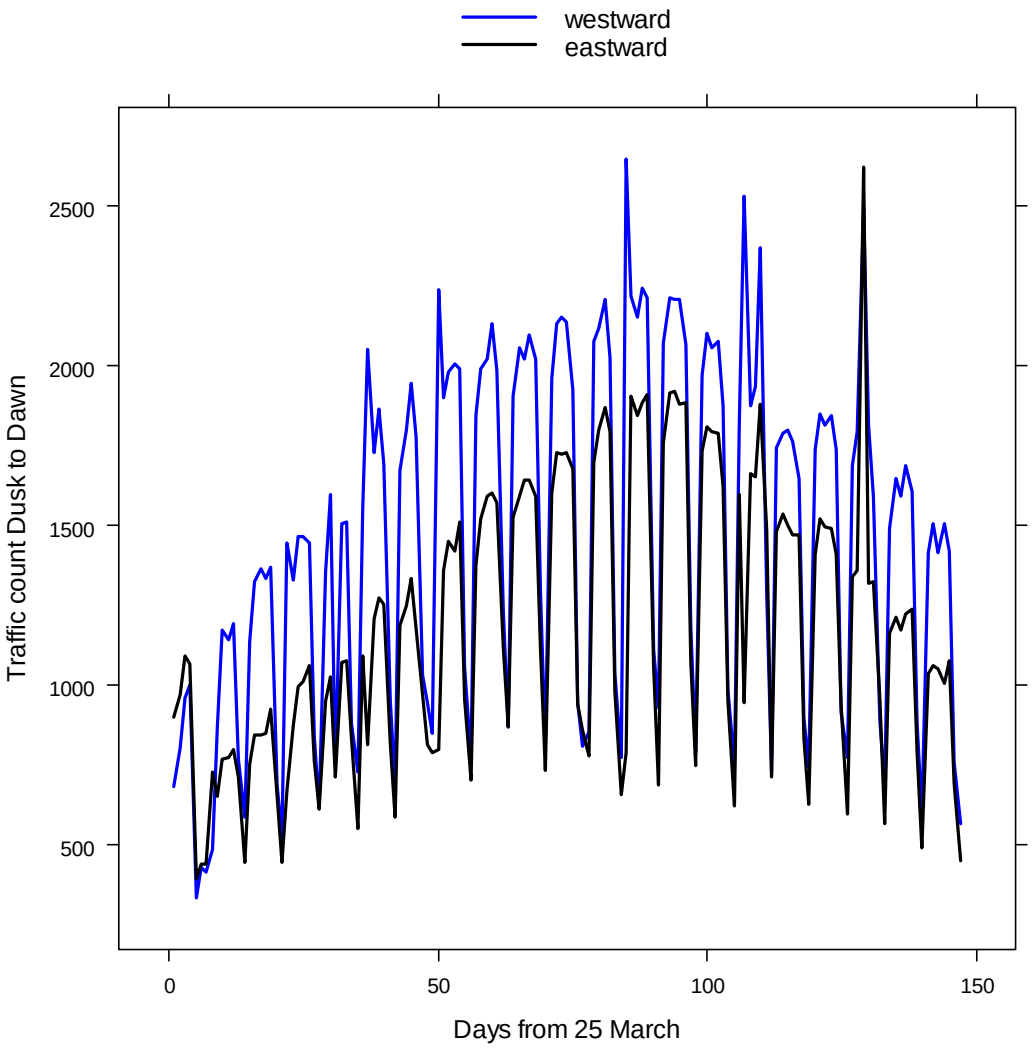
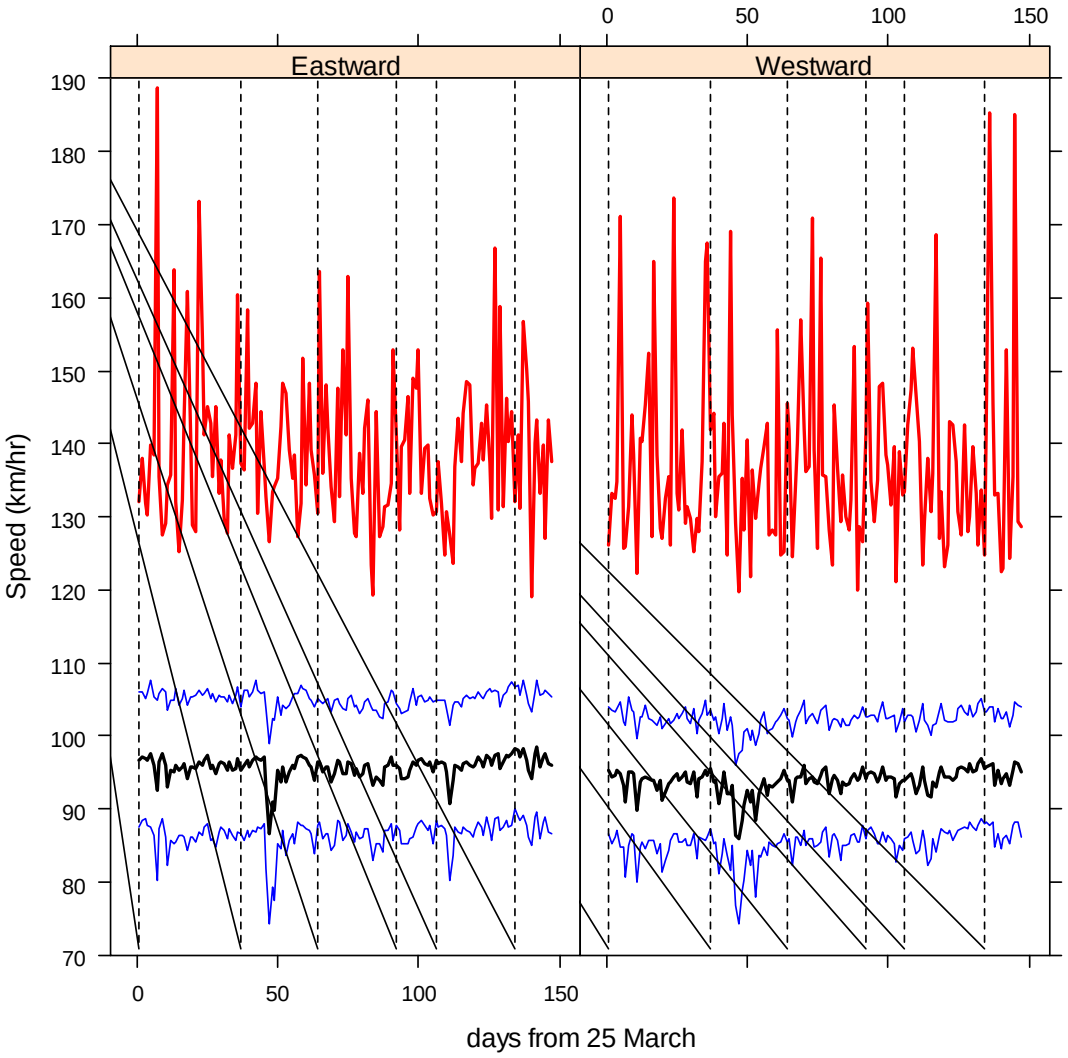
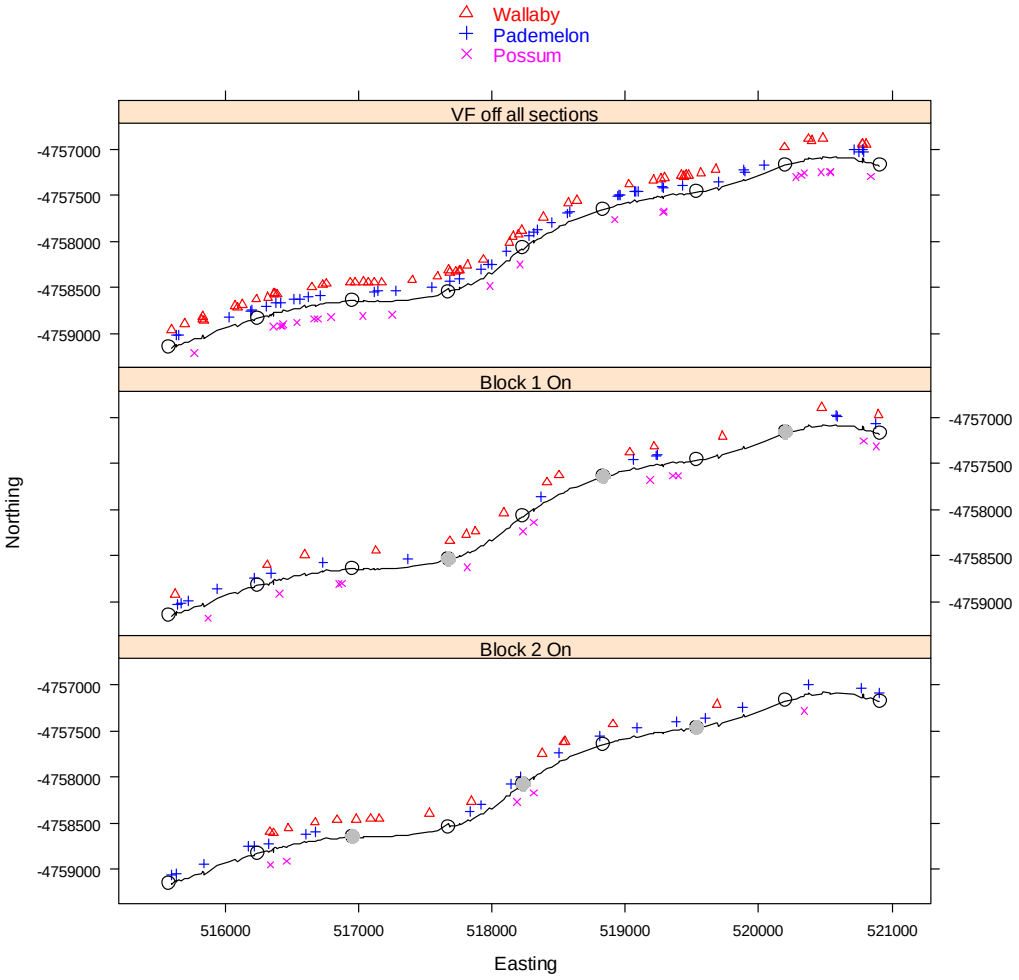

westward
eastward
Traffic count Dusk to Dawn
2500
2000
1500
1000
500
0
50
100
150
Days from 25 March

Figure S4. Average and maximum daily traffic speed between dusk and dawn by direction from the start of the trial. The average daily speed (thick black line) plus or minus one standard deviation (calculated daily) (thin blue lines) and daily maximum speed (thick red line). The thin vertical dashed lines represent the start date for each of the six periods shown in Table 1. The strong dip in speeds starting at Day 47 and lasting for three to four days corresponds to the extreme rainfall event (twice the previous record for a single day) that occurred for Hobart and surrounds starting on the evening of 10 May and resulted in erosion and large amounts of gravel washed on to the highway.



5. Raw spatial locations of three most prevalent roadkill species

FigureS 5. Spatial locations of roadkill for three most prevalent species of Bennett's wallabies (BW) *(Notamacropus rufogriseus)*, Tasmanian pademelons (TP) *(Thylogale billardierii)* and common brush-tail possums (BP) *(Trichosurus vulpecula)* .



Panels show the locations (Northing, i.e., latitude and Easting, i.e., longitude) for Periods 1, 2, 4, and 6 (total of 70 days) when the VF was switched off on Sections 2 to 7 and for each of Period 3 (Block 1 sections On) and Period 5 (Block 2 sections On) with each a period of 28 days. Sections 1 to 8 are delineated by the circles with the start of sections (proceeding westward) when switched on shown as filled circles. The Northing of roadkill locations have been offset slightly for clarity. The right-hand side of the graph starts at the eastern buffer Section 1.

6. Roadkill numbers by all species and period

Table S1. Roadkill numbers by all species and period with On-Off Periods 3 and 5 disaggregated to On and Off Blocks excluding buffer sections 1 and 8 (Block 1: Sections 2, 4, 6; Block 2: Sections 3, 5, 7). The periods when the VF was switched on are italicised.

Period label	**Pre-trial**	**Pre_All_Off**	**Block1_On**		**Post1_All_Off**	**Block2_On**		**Post2_All_Off**	
Period No	1	2	*3*		4	*5*		6	
Start Date 2018	26/03	1/05	*28/05*		25/06	*9/07*		6/08	
End Date 2018	8/04	28/05	*25/06*		9/07	*6/08*		20/08	
Period (days)	14	28	*28*		14	*28*		14	
VF on vs off			**Block2_Off**	**Block1_*On***		**Block1_Off**	**Block2_*On***		
Species									**Species Total**
Bennett's wallaby (*M.rufogriseus*)	16	18	8	4	6	9	6	1	68
rufous-bellied pademelon (*T.billardierii*)	6	13	6	3	7	6	7	10	58
common brush-tailed possum (*T.vulpecula*)	7	4	7	2	1	1	3	3	28
bettong (*Bettongia aimardi*)	0	2	0	0	0	0	2	1	5

bandicoot (*Perameles gunnii*)	0	1	0	1	0	2	0	0	4
kookaburra (Dacelo)	2	1	0	0	1	0	0	0	4
Tasmanian native_hen (*Tribonyx mortierii*)	0	1	0	0	1	0	0	1	3
ringtail possum (*Pseudocheirus peregrinus*)	0	1	0	1	0	0	0	0	2
Tasmanian devil (*S. harrisii*)	0	1	0	0	0	0	1	0	2
Period Total	31	42	21	11	16	18	19	16	174

7. GAM fitted cubic smoothing splines

Figure S6. Predicted roadkill rate of Bennett's wallaby from GAM versus Section for VF_on_vs_off set to "Off" and Period_mid set to 100. Single SE bounds shown as dashed lines.

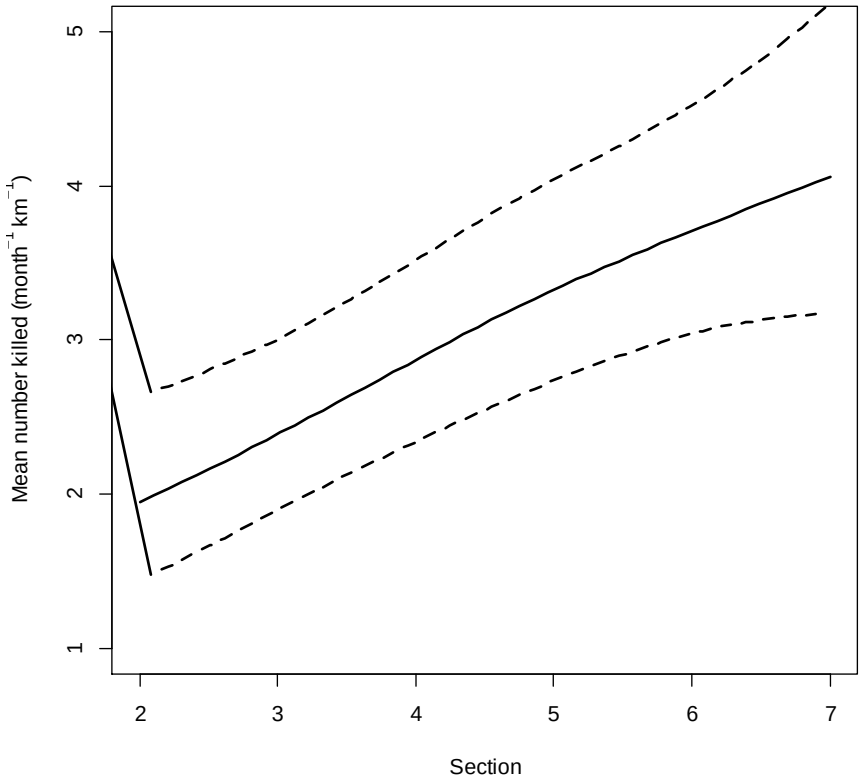
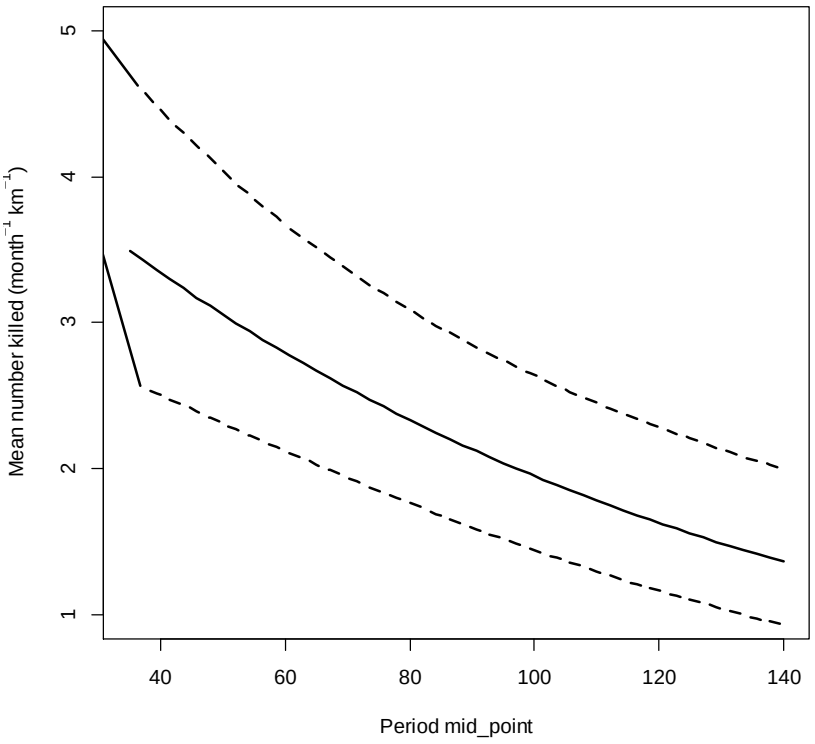


Figure S7. Predicted roadkill rate of Bennett's wallaby from GAM versus Period midpoint for VF_on_vs_off set to "Off" and Section set to 2. Single SE bounds shown as dashed lines.



Chapter 6. Australian roadkill project

6.1. Preamble. Mapping Australian roadkill by the use of citizen science

There are currently no national data on Australian roadkill. An Australia-wide database of roadkill would confer many benefits. The data would potentially:

a. identify roadkill hotspots. This then facilitates the prioritisation of expenditure into the areas where it will have the greatest mitigation effect. The hotspots can be targeted for measures such as fencing, verge clearing and culverts. A before-and-after cost/benefit analysis would be expected to show more animals saved from becoming roadkill in a sustainable way.
b. determine where new technological devices might operate successfully to reduce roadkill.
c. give an indication of any increase in roadkill numbers and enable a projection of the increase in future workload for wildlife carers. This enables recruitment to be targeted proactively, rather than reactively. Having sufficient wildlife carers trained and equipped in the right place geographically could lessen the likelihood of physical, financial and mental stressors mentioned in previous chapters. An understanding of the distribution and number of rescuers and carers in relation to injured and orphaned wildlife would allow training and outreach to be targeted around specific seasonal peaks, species and causes of injury [132]
d. monitor animal decline or change in animal movement patterns.
e. indicate the success or otherwise of roadkill mitigation infrastructure.

There are approximately 823,000 km of roads throughout Australia, and the cost of providing the required researchers and resources to monitor all of them for roadkill would be prohibitive. Projects in other countries around the world have approached this problem by recruiting volunteers to record data. These non-professionals do not necessarily have a formal science background but contribute their time, effort and resources toward scientific research, usually in collaboration with professionals. They collect data and sometimes analyse them and have become known as citizen scientists. Citizen science projects focused on roadkill include in the UK, 'Project Splatter' [96], in Belgium 'Dieren onder de wielen' [97], in Austria 'Roadkill' [98], in South Africa 'Endangered Wildlife Trust' [99] and in the USA, 'California Roadkill Observation System' [100].

A Roadkill Reporting application (RRApp) was developed to enact an Australian project similar to those mentioned above. This was made available for free download from Google Play and iTunes by Australian citizen scientists. Emerging technologies were embraced to produce a mobile application that eliminated operator error, as far as possible, to ensure the data collected would withstand scientific

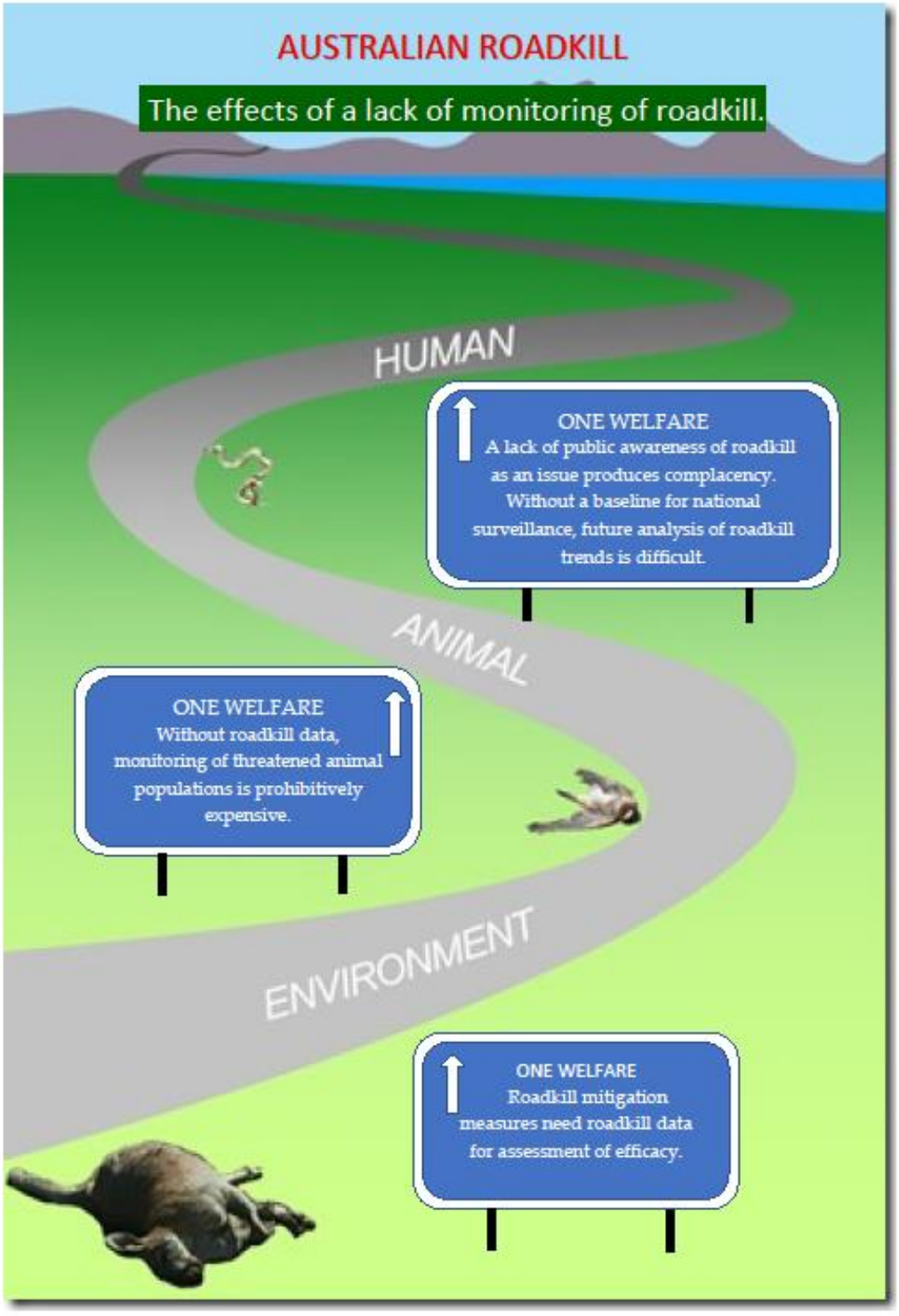
scrutiny. This was achieved by having the citizen scientists take a photograph of roadkill while, at the same time, a GPS tag location, the date and time are automatically superimposed onto the photograph. Thus, spurious items that do not show roadkill can be discarded. Also, any reported road-kills that seem to have similar GPS locations can be examined by photograph so that duplicate recordings of individual roadkill can be eliminated.

The power of the media was used to recruit volunteers. The federal Minister for the Environment, the Hon Sussan Ley MP, accompanied by the Hon senators Eric Abetz and Claire Chandler, launched the RRApp on national television. Also, the author attended interviews on national breakfast television and on 34 regional radio stations. Articles in newspapers and on social media added to the publicity. Information was also sent to the state wildlife carer networks and individual wildlife carers.

The advantage of this citizen science approach is that a large dataset can be gathered cheaply, allowing timely data collection to continue relatively inexpensively for many years. Disadvantages include the need to manage a large cohort of volunteers and maintain the integrity of the data collection. Chapter 6 reveals how these two disadvantages can be mitigated within a project to create a map of Australian wildlife roadkill. The project is intended to be open-ended and in the next year it will be managed by the author. It is hoped that this research project will be managed in the future by a university or the Australian Museum and will attract government funding to enable this.

Roadkill is a problem produced by humans. Humans can help mitigate the problem to the benefit of animals and the environment by being involved as citizen scientists and raising awareness of the conservation issues resulting from roadkill, collecting data for a national database that can be used for identifying hotspots of roadkill and which species are observed as roadkill. This is a concept embodied in the One Welfare philosophy.

Figure A6. One Welfare. Monitoring of roadkill by citizen scientists



6.2. Article

The Australian Roadkill Reporting Project—Applying Integrated Professional Research and Citizen Science to Monitor and Mitigate Roadkill in Australia

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Simple Summary: Australia has no database of national roadkill. The current research project fills that knowledge gap by developing a roadkill reporting application that enables professional and citizen scientists to record photographs of roadkill with location, time and date. This embodies the concept of 'One Welfare' as it affects humans, animals and the environment. Uploaded to a website, these data can identify roadkill hotspots, tabulate species of animals killed and potentially be used for ecological studies of roadkill numbers, species distribution, population trends, animal behaviour and disease. Initial results indicate that mammal roadkill mostly occurs at night and that of birds and reptiles during daytime. Mammals make up three-quarters of the roadkill recorded and this includes endangered species. Two examples of roadkill hotspots are shown in Queensland and Tasmania. These will enable further research to suggest how roadkill mitigation measures may be optimally employed.

Abstract: The expansion of roads throughout the world has led to increases in vehicular collisions with wildlife and subsequent deaths (roadkill) and injury. Monitoring of roadkill is essential to devise strategies to minimise the human, environmental and animal welfare costs of roadkill. Unlike many other developed countries, Australia has no national roadkill monitoring scheme. Emergent technologies on mobile telephones provide potential for national monitoring. To address this gap in knowledge, a roadkill reporting application (app) was developed to allow members of the public to join professional researchers in gathering Australian data. The app is used to photograph roadkill and simultaneously records the GPS location, time and date. These data are uploaded immediately to a website for data management. We review the opportunities and value-added information, such as roadkill hotspots, population dynamics, animal behavior and disease, and species distribution mapping that can be produced from such data. To illustrate the capacity to facilitate cost-effective mitigation measures our article focuses on two roadkill hotspots—in Queensland and Tasmania. In total, 1609 reports were gathered in the first three months of the project. They include data on mammals (n = 1203, 75%), birds (n = 125, 7.8%), reptiles (n = 79, 4.9%), amphibians (n = 4, 0.025%), unidentified (n = 189, 11.8%) and unserviceable ones (n = 9). A significant finding is variance in the distribution of mammals and birds at different times of day. These findings reflect diurnal variation in the activity levels of different species and underline the need for data on a targeted species to be collected at appropriate times of day. By continuing to facilitate roadkill monitoring, we anticipate that the data generated by the app will directly increase knowledge of roadkill numbers and hotspots.

Keywords: roving reports; mobile application; ecological studies

1. Introduction

Roadkill is a term that encompasses all mammals, birds, reptiles and amphibians that are killed by vehicles on roads. Although Australia lacks national data on roadkill, there have been state-based studies, notably in New South Wales [1–5], Tasmania [6], Victoria [7] and South Australia [8]. Dating from 1972 through to 2008, these were conducted by professional researchers. However, emergent technologies, such as mobile phones, mean that the collection of reliable roadkill data has become possible for volunteer citizens. Such volunteers have become known as citizen scientists, a term coined in 1995 by Rick Bonney in the USA [9] and Alan Irwin in the UK [10]. These technologies bring with them many advantages, particularly when the area of environment to be studied is as geographically broad as Australia. Citizen scientists can gather data on a scale beyond the reach of most research budgets.

It is acknowledged that there are problems and pitfalls with the recruitment, motivation and retention of volunteers, ensuring safe working practices and the quality of the data produced [11,12], but these challenges can be overcome [13]. An increase in citizen science projects from 100 in 2011 to 700 in 2018 is reported by SciStarter [14] and this is reflected in the increasing number of peer-reviewed papers published annually that use or include data from citizen scientists [15,16].

Examples of projects that use such data include the *doglogbook*, developed by the dogmanship team at the Sydney School of Veterinary Science, University of Sydney, Australia [17]. This aims to support evidence-based assessments of dog quality of life and enhance dog welfare in clinical practice. Outside the realm of domesticated animals, other examples are Birdata, run by Birdlife Australia [18], and eBird, run by Cornell University [19]—both of which collect information on birdlife supplied by citizen scientists. FeralScan (from which WomSat is a derivative) is another example.

Records made whenever and wherever an observer observes a target, such as roadkill, are called roving records [13]. National roadkill monitoring projects using roving reports are in place across the globe, but some, being fairly undersubscribed, are limited in their scope. Examples include 'Project Splatter' [20] and 'Mammals on Road' [21]—both of which are in the UK—'Roadkills' in India [22], 'Roadkill Observation Network' in Japan [23], 'Animals under Wheels' in Belgium [24], Srazenazver.cz in the Czech Republic [25] and 'Projekt Roadkill' in Austria [26]. Similar national reporting capacity is required in Australia.

The challenge of obtaining reliable data gathered by volunteers with only basic training can be overcome by emergent smartphones and tablets and the development of applications (apps) that can be downloaded to them. Many smartphones have in-built GPS receivers and can access environmental information through open-source databases such as Google Maps that offer satellite imagery, aerial photography, street maps, 360° interactive panoramic views of streets, real-time traffic conditions, and route planning for traveling by foot, car, bicycle, air or public transportation. The technological advantages of obtaining accurate data through apps is that no knowledge of computer technology is required to use them and they are mobile. It has been shown that these apps can be used to improve data collection by both professional researchers [27] and citizen scientists [28,29].

Potential shortfalls in the collection of accurate ecological data include the inaccurate recording of time and location, recording duplication of data, incorrect identification of subjects and even loss of data records. For roadkill studies specifically, numerous constraints on the collection of data reflect temporal, spatial and taxonomic challenges that can cause gaps in knowledge. If citizen scientists are to be used in a roadkill monitoring research programme, the method of data collection must be as fail proof as possible and produce data that are incontrovertible, or at least have known and expounded caveats.

In essence, this means that a citizen scientist engaged in recording roadkill needs no skills other than to be able to spot roadkill and then use the app as directed. A photograph of the roadkill that is digitally stamped with GPS location, time and date and then uploaded to a website would meet these criteria. In contrast to relying on paper hard copy, the use of a tablet or smartphone has the advantage over paper hard copy that the observations are recorded swiftly and accurately, and are automatically uploaded to a website. If no mobile coverage is available, the app can record the GPS via satellite and

then upload recorded roadkill data to a website once mobile coverage is obtained. The user does not need to separately record a GPS from a tracker, the time and date from a secondary source and then write the results into a field notebook—no easy task in wind, rain or snow. These notes have then to be copied to a website, a process that may allow errors to creep in. Each transcription step increases the human error.

The current project followed the exemplar of mobile application for recording roadkill reported in other countries including the UK [20], USA [30], Belgium and Austria [31] Columbia [32] and Tanzania [33]. The functional specifications were for the current Australian Roadkill Reporting Project (ARRP), using the Roadkill Reporter App (RRApp), to track Australian roadkill by means of a smartphone or tablet by a researcher or citizen scientist.

The initial aims of our project were to provide a set of requirements for recording Australian roadkill with locational metadata and to review the suitability of available mobile phone-based apps against those criteria. If none were fit-for-purpose then we aimed to develop, introduce and test a custom-made app for Australians to report roadkill and to assess the value of the app for determining hotspots to aid the allocation of remedial or mitigating actions.

The long-term aim of the project was to provide an estimate of the scale of the national roadkill problem and produce a baseline of annual roadkill data with which future roadkill data could be compared.

2. Materials and Methods

2.1. A Review of Roadkill Reporting Applications

A review of the functionality of existing roadkill reporting apps, undertaken in early 2017, demonstrated that all were inadequate for the purpose of this study or suffered from prohibitive leasing costs. A new app, named the Roadkill Reporter App (RRApp), was, therefore, developed locally.

2.2. Features and Functionality of the RRApp

It was important that the RRApp fulfilled the following criteria:

Must be easy to use, requiring no technical knowledge to operate;
Freely available;
Portable and accessible;
Functional on Apple iOS and Android™ mobile operating system;
Data on the app must be able to be uploaded to a secure website via a web facility to collate, store and allow access through a website interface to data submitted through use of the mobile application;
Data collected must include a photograph and GPS, time and date;
Users must have the ability to review trends by accessing the website;
Users must be able to make notes to accompany a roadkill photograph;
Users must remain anonymous.

The website for monitoring the app was required to be hosted in Australia if possible.

2.3. Usage and Testing of the RRApp

The app allows the use of personal mobile devices to take a photograph of roadkill and submit it to a cloud-based database of sightings along with the time of day and GPS location.

Once developed, the app was subjected to testing by a mixed group of volunteers (n = 14) of different gender, age and familiarity with the use of information technology. Feedback from this testing resulted in refinements being introduced to ensure the RRApp was user-friendly for citizen scientists of different abilities. The participants were required to take a photograph of the roadkill, categorise it as mammal, bird, other or splat (unidentifiable roadkill) and then submit the report. The app ensures that the GPS coordinates and time of day are automatically submitted.

The RRApp was further tested in a pilot study during a research trial of a virtual fence system [34]. Duplicate data were recorded on 10 occasions over three consecutive days, using both the RRApp and independent GPS tracking and a digitally stamped time and dated photograph. A concordance correlation coefficient and standard deviation were calculated for the GPS data and the time and date separately compared to confirm the accuracy of the RRApp was within required spatial limits.

2.4. Ethics Exemption and Launch of the RRApp

The project was exempted from ethics review and a Roadkill Reporter Exemption certificate was issued by the Research Integrity and Ethics Administration, Human Research Ethics Committee of the University of Sydney (Appendix A).

The RRApp was launched in September 2019 through national and local television, radio, newspapers, social media and wildlife carer networks. The data are used to produce a map of recorded roadkill that can be accessed at www.roadkillreporter.com.au/reports. The map is updated in real time and is open access, so that any individual can view their uploaded sightings as well as those reported by others.

2.5. Analysis of Roadkill Data

The normal choice for count data is a Poisson model but, in this case, the data were overdispersed (i.e., the variance was greater than the mean), so a quasi-Poisson model was used. Modelling was performed in the R stats package. The results were wrapped in an ANOVA wrapper using the car package [35]. To determine what time of day roadkill was being recorded using the RRApp, the quadrant of day (QOD) coefficients were graphed in R with a 95% confidence interval using jtools [36]. The emmeans package [37] was used to calculate estimated marginal means and asymptotic confidence intervals, and attain pairwise p-tests. The level of significance was set at $p \leq 0.05$

Locations were categorised into states and territories manually using the state borders from the Australian Bureau of Statistics [38]

The reporting time was recorded as Universal Time Coordinated (UTC). This is the time standard commonly used globally and recorded by the RRApp. This was converted to local time based on the state boundaries and daylight-saving protocols specific to each state.

The time of day was divided into 4 quadrants rather than either 24 h or as a continuous variable due to low numbers of data. The quadrants were ordered in six-hour blocks starting from midnight: 00:00–05:59 Quadrant 1 (Q1), 06:00–11:59 Quadrant 2 (Q2), 12:00–17:59 Quadrant 3 (Q3), and 18:00–23:59 Quadrant 4 (Q4).

Experts in herpetology (n = 4), ornithology (n = 4) and zoology (n = 5) examined reported photographs and used the location and respondents' notes, together with their own expertise, to subjectively produce a percentage certainty between 1% and 100% confidence of the accuracy of their species identification of each image. Only where two or more experts had a confidence of greater than 90% was identification ratified; otherwise, the species was recorded as unknown. Sensitivity and specificity were analysed by comparing the user identification of a photograph as a mammal to the expert identifying a photograph as a mammal. The purpose of these identifications was to assess the accuracy of user identifications. Future users of data collected by this app may find such statistics useful for estimating sample sizes and efficiently allocating research resources such as availability of taxonomic experts.

Using the webmap of Australian roadkill produced from the citizen scientist observations of roadkill from the RRApp, it was possible to identify transects where there was a high density of roadkill. These transects were extracted and individually analysed.

Two clusters of roadkill on transects in Queensland and Tasmania, where roadkill was apparently high, were selected for detailed examination using the webmap produced by the RRApp website.

A Fishers Exact was used to examine time of day and roadkill taxonomic class reporting.

3. Results

3.1. Assessment of the Accuracy of the Roadkill GPS Using the RRApp

Ten GPS locations of roadkill were recorded on both the SMGPS app and the RRApp and compared for concordance (Appendix B). Both the standard deviation (1.739) and product moment

correlation coefficients (Easting 0.999999231 and Northing 0.999997278) demonstrate that the RRApp performed well within the accuracy required in the specifications.

3.2. Overview of the RRApp from September 1st to November 30th 2019, Installation of the RRApp and Subsequent Reports

In total, 2318 people installed the RRApp (n = 1445 on iOS and n = 873 on Android). There were 1459 reports made, with 322 users making at least one report. Approximately 14% of people who installed the app made at least one report. There was an average of 4.5 reports per user and 39% of users made just one report. Ten per cent of the users were responsible for 50% of reports.

The purpose of this analysis was to uncover diurnal patterns in the reports. This was useful to explore the time of day users are using the app and also when roadkill deaths are reported. The importance of this was to understand whether roadkill, particularly avian and reptilian, that occurred during the daytime was being recorded before being removed by anthropogenic means or by animal scavengers.

Reports from 24/9/2019 to 5/12/2019 are more common in the later morning (Q2, 6:00 a.m. to midday) and in the afternoon (Q3, midday to 6:00 p.m.), with a moderate number of reports in the evening (Q4, 6:00 p.m. to midnight) and the fewest in the early morning (Q1, midnight to 6:00 a.m.) (Figure 1).

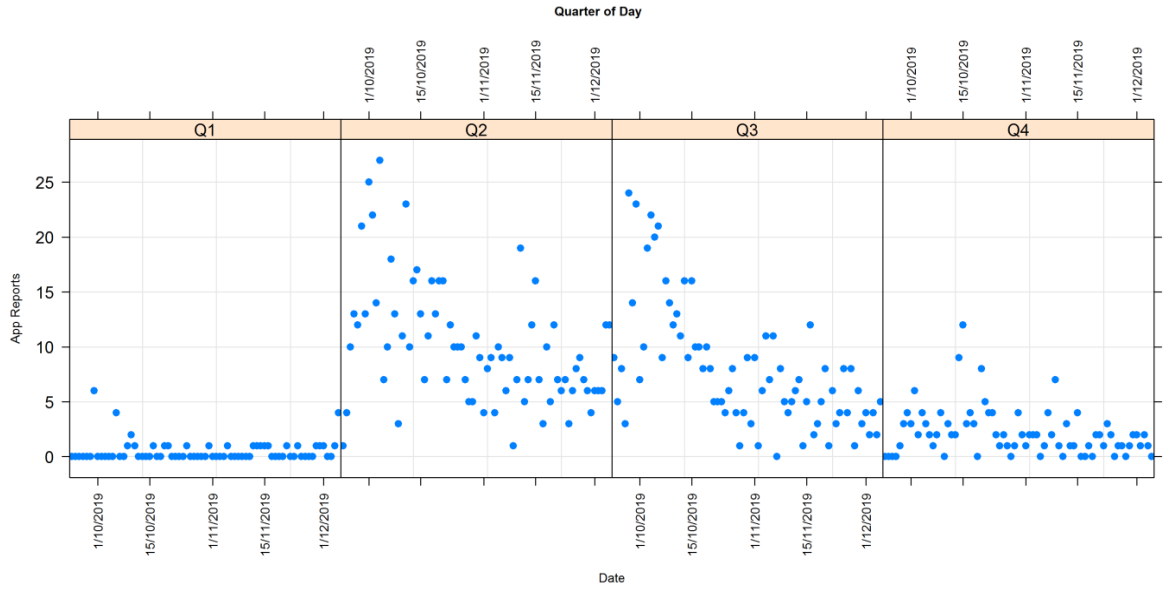


Figure 1. Distribution of roadkill reports by time of day, divided into four quadrants of day.

Both the quadrant of day (QOD) (LR χ^2 = 482.38; $p < 0.001$) and date (LR χ^2 = 66.8; $p < 0.001$) were significantly associated with the number of reports made. The number of reports declined towards the end of the study period.

Q1 (early morning before 6:00 a.m.) was associated with the lowest number of reports, significantly lower than Q2 (6:00 a.m. to midday) (r = −3.046, z = 12.013; $p < 0.001$), Q3 (midday to 6:00 p.m.) (r = −2.804, z = 10.992; $p < 0.001$), and Q4 (6:00 p.m. to midnight (r = −1.520, z = 5.557; $p < 0.001$). Q4 was also significantly lower than Q2 (r = −1.526, z = 11.937; $p < 0.001$) and Q3 (r = −1.284, z = 9.810; $p < 0.001$). Of the two daylight quadrants, Q2 had significantly more reports (r = 0.242. z = 2.967, p = 0.016) than Q3.

The location of reports by the local QOD suggests that the RRApp is widely used but that the Northern Territory is under represented (Figure 2). This is demonstrated on a state by state basis (Table 1)

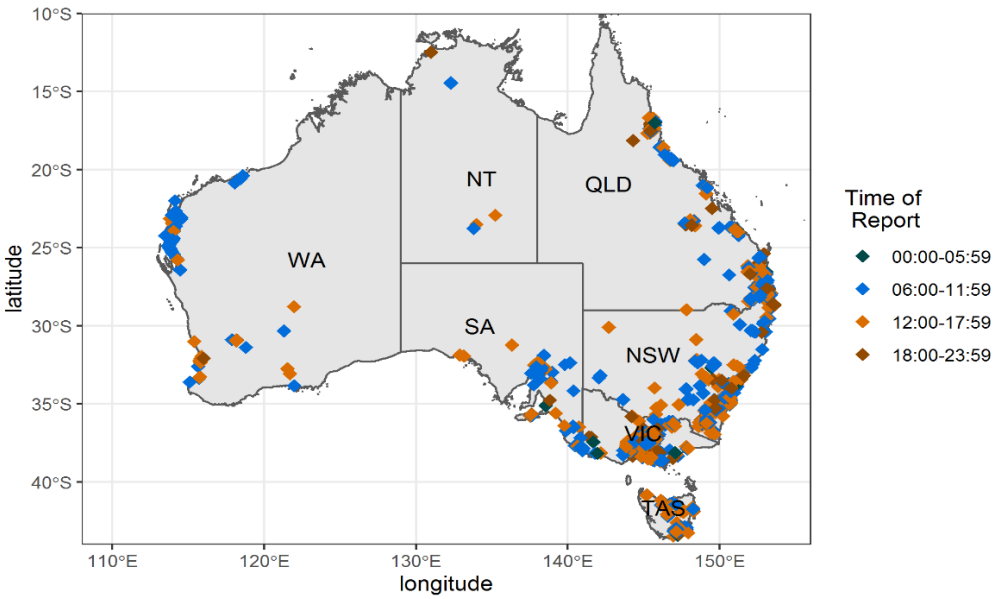


Figure 2. Location of roadkill reports by the quadrant of day (QOD), superimposed on a map of Australia. WA—Western Australia, NT—Northern Territory, SA—South Australia, QLD-Queensland, NSW—New South Wales, VIC—Victoria, and TAS—Tasmania.

Table 1. Total number of reports by state between 24/9/2019 and 5/12/2019 UTC.

STATE	Mammal	Bird	Other	Splat	TOTAL
NSW	295	28	39	6	368
NT	3	1	2	0	6
QLD	389	41	35	7	472
SA	63	10	7	1	81
TAS	147	8	18	7	180
VIC	245	35	4	3	287
WA	85	12	15	3	115
TOTAL	1227	135	120	27	1509

3.3. Reported Class of Roadkill during Study Period

Data from 1509 reports of roadkill were classified by RRApp users into mammal (class Mammalia), bird (class Aves), other (any other class) or splat (unidentifiable remains). Data for 1509 reports taken between 24/9/2019 and 5/12/2019 UTC are shown in Table 2.

Table 2. Taxonomic class of 1509 roadkill reports from RRApp users according to the local quadrant of day of the report.

Time of Day	Mammal, *n* (%)	Bird, *n* (%)	Other, *n* (%)	Splat (Unidentifiable), *n* (%)
00:00–05:59 (Q1)	30 (85.71%)	3 (8.57%)	1 (2.86%)	1 (2.86%)
06:00–11:59 (Q2)	639 (86.82%)	50 (6.79%)	35 (4.76%)	12 (1.63%)
12:00–17:59 (Q3)	445 (76.99%)	58 (10.03%)	63 (10.90%)	12 (2.08%)
18:00–23:59 (Q4)	113 (70.63%)	24 (15.00%)	21 (13.13%)	2 (1.25%)
TOTAL	1227	135	120	27

The most commonly reported class of roadkill were mammals, comprising 81.3% (1227/1509) of reports. The next most common were birds, accounting for 9.0% (135/1509) of reports. Other identified remains accounted for 8.0% (120/1509) of reports and unidentified reports as the remaining 1.8% (27/1509). The taxonomic class of roadkill were strongly associated with the time of day that the report was made (Fishers exact $P<0.001$) with the significant differences between the second and third Q2 and

Q3 (P=0.001) and between the first and fourth quadrants (P=0.013). Other differences were not significant. Post-hoc tests showed that the proportion of roadkill by class differed across all pairwise comparisons, including between the two "daylight" quadrants (Q2 and Q3; z = 2.967, p = 0.016) and between the two "night' quadrants (Q1 and Q4; z = 5.557, $p < 0.001$)

3.4. Accuracy of Observers Recording Mammal, Bird, Other, and Splat

Experts were able to characterise all reports recorded as splat (n = 25) as either mammalian, bird or other (Table 3).

Table 3. Experts' identification of roadkill from photographs and how RRApp users identified those same roadkill.

Expert Identification	User Identification				
	Class	Mammal	Bird	Other	Splat
mammal—macropod	907	894	2	5	6
mammal—non-macropod	284	275	1	5	3
nocturnal bird	13	0	13	0	0
reptile	79	4	0	74	1
not otherwise classifiable	175	126	18	18	13
bird	110	0	108	1	1
exotic	25	14	2	8	1
amphibian	3	0	0	3	0

The roadkill (n = 175) identified as not otherwise classifiable represent a combination of results from experts who were able to identify the class but not the species (therefore, unable to classify whether it was native or exotic, macropod or not macropod, nocturnal or diurnal, etc.)

3.5. Accuracy For Mammals, Birds and Other

The sensitivity with which a user 'mammal' report predicted a mammalian roadkill photo was 0.980 and the specificity with which a user report of 'mammal' predicted a mammalian roadkill photo was 0.915. The sensitivity with which a user 'bird' report predicted an avian roadkill photo was 0.986 and the specificity with which a user report of 'bird' predicted an avian roadkill photo was 0.995. Finally, the sensitivity with which a user's 'other' report predicted an amphibian or reptilian roadkill photo was 0.944 and the specificity with which a user's report of 'other' predicted an amphibian or reptilian roadkill photo was 0.981.

3.6. Animal Species Represented in Roadkill

Images from 1609 reports between 28 September 2019 and 31 December 2019 were examined to identify the species of roadkill reported. In total, 13 reports were immediately discarded as duplicates (n = 6) or miscellaneous (n = 7). The main species represented in the roadkill photographs were kangaroos (n = 415, 26.74%), wallabies (n = 360, 23.19%) and wombats (n = 181, 11.66%), which collectively represented 61.59% of the total roadkill (Table 4).

Table 4. Principal roadkill taxa between 28 September 2019 and 31 December 2019 by group, identified by experts using photographic RRApp data.

Group/Species		Number Recorded	Percentage of Total Roadkill
Mammals	Kangaroo (*Macropus*)	415	26.74
	Wallaby (*Macropus*)	360	23.19
	Wombat (*Vombatus ursinus*)	181	11.66
	Wallaroo (*Macropus robustus*)	77	4.96
	Brushtail possum (*Trichosurus vulpecula)*	48	3.10
	Koala (*Phascolarctos cinereus*)	17	1.09
	Ringtail possum *(Pseudocheirus peregrinus)*	9	0.58

Birds	Forest raven (*Corvus tasmanicus*)	16	1.03
	Brush turkey (*Alectura lathami*)	15	0.97
	Magpie *(Gymnorhina tibicen)*	14	0.90
Reptiles	Eastern brown snake (*Pseudonaja textilis*)	15	0.97
	Blue-tongue lizard (*Tiliqua*)	14	0.90
	Lace monitor lizard (*Varanus varius*)	11	0.71
	Red-bellied black snake (*Pseudechis porphyriacus*)	9	0.58
Exotics	Hare (*Lepus*)	6	0.39
	Cane toad (*Rhinella marina*)	4	0.26

Only the main taxa in each category are shown in Table 4. A full analysis of the roadkill represented in the report photographs is available in the Appendix (Table C1).

3.7. Roadkill Hotspots

Two transects of highways, where high roadkill was apparent in the records, were selected for detailed examination: one in Queensland and the other in Tasmania. On the Queensland transect, the roadkill total (n = 38) on 600 km of road observed (20 km × approximately 30 days of monitoring by citizen scientists) was calculated to be a rate of 0.063 roadkill·km^{-1}·day^{-1} (Figure 3).

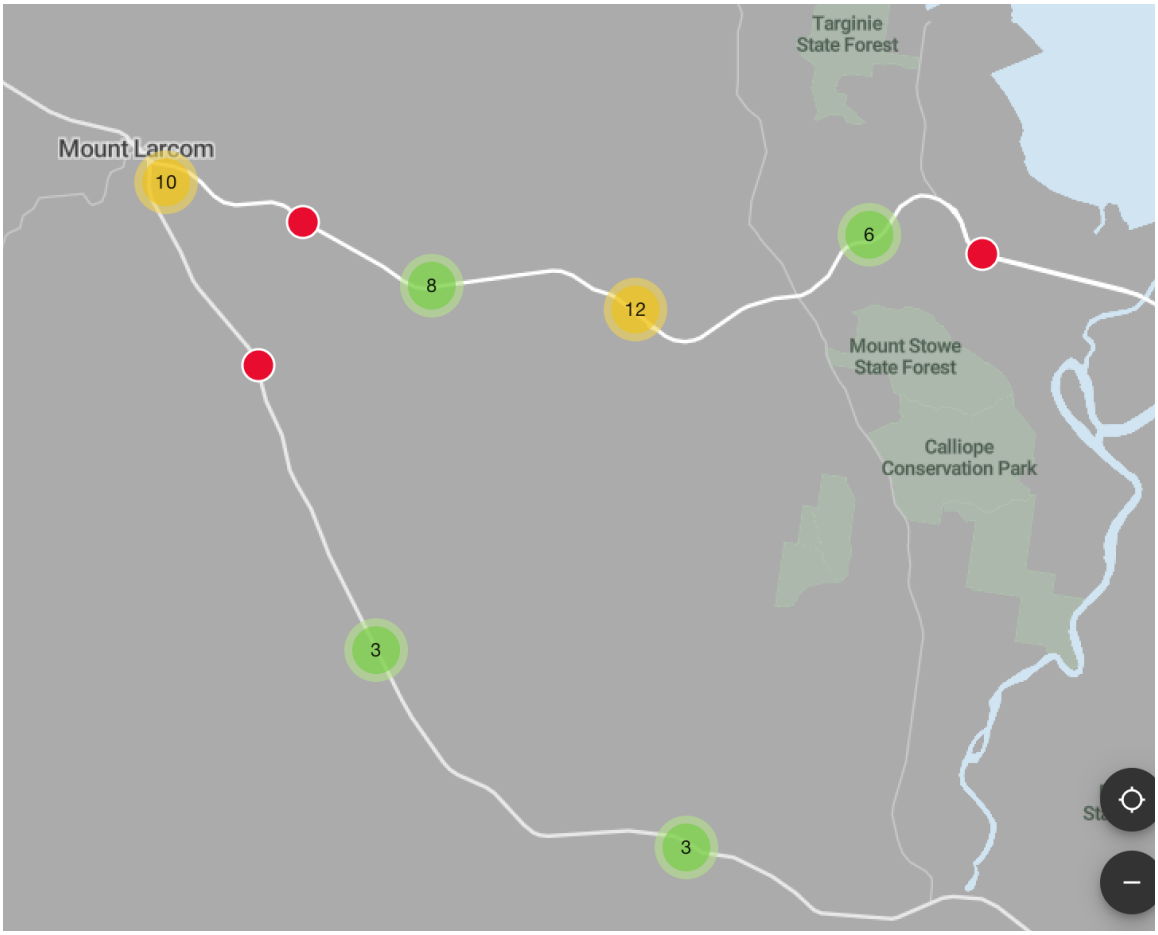


Figure 3. Roadkill hotspot on 20 km of highway between Mount Larcom and east towards Mount Stowe State Forest, Queensland.

The red circles represent one roadkill, the green dots represent between three and nine roadkill and the yellow dots represent upwards of ten roadkill.

On the Tasmanian transect, the roadkill total (n = 25) on approximately 360 km of road observed (6 km × approximately 60 days of monitoring by citizen scientists) was calculated to be at a rate of 0.069 roadkill·km^{-1}·day^{-1} (Figure 4).

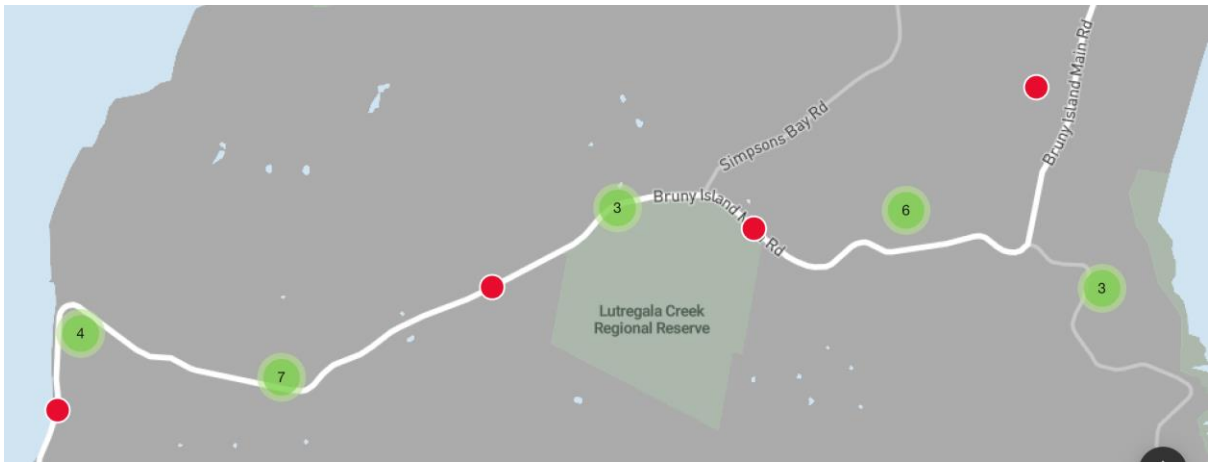


Figure 4. Roadkill hotspot on 6 km of highway between Alona on the west side and Adventure Bay on the east side of Bruny Island Tasmania.

The red dots represent one roadkill and the green dots between two and nine roadkill.

4. Discussion

There are several ecological and human reasons for gathering data on Australian roadkill. 'One Welfare' is an emerging term. The concept refers to animal welfare, human welfare and environmental sustainability. It serves to highlight the interconnections between animal welfare, human wellbeing and the environment, and promotes direct and indirect links [39,40]. Roadkill can be placed in the concept of 'One Welfare,' as it affects humans, animals and the environment. Effectively monitoring roadkill through the RRApp enables the 'One Welfare' concept and philosophy, and can be used to encourage its adoption. The data obtained can be used to identify roadkill hotspots, raise the profile of roadkill in the mind of the public, identify the species killed and determine their distribution. This information can be entered into the database for future evaluation.

Engaging citizen scientists has highlighted the problems that roadkill presents in the mind of the general public through the publicity it has attracted. Evidence of this can be seen in the sudden increase in installations during September and reports at the beginning of November 2019 correlating with media exposure. As the project progresses, this public awareness is likely to increase. Community groups will be able to use the data from the RRApp to enhance research and general funding for mitigation measures and their implementation. The project has also shown how the time of monitoring by citizen scientists is dependent on the quadrant of day (QOD), i.e., mainly between 06:00 and 18:00. This would indicate that it might be advantageous to target members of those demographic groups, such as wildlife carers, so-called grey nomads (Australian retirees who travel within their own country for an extended time, usually in a caravan or motor home), local councils and environment networks who are travelling or working during this time (Q2 and Q3). The data gathered can also be used in ecological studies.

The data shown in Figure 2 are based on state boundaries, and this process draws in some minor idiosyncrasies. Specifically, it does not account for the small area in the southeast of Western Australia that borders South Australia that is on Australian Central Western Standard Time or the fact that Broken Hill (New South Wales) is on Adelaide time (Australian Central Standard Time).

Five areas of how monitoring roadkill can be used are explored in depth by Shilling and Perkins [41]. These areas are: monitoring wildlife roadkill numbers; species distributions; population trends

and impacts; animal behaviour (specifically movement); and contaminants and disease. In the context of the current project, it has been shown how these five areas are pertinent to Australian ecology.

The current mapping outputs (Figures 3 and 4) confirm that by citizen scientists monitoring wildlife roadkill numbers, transects where roadkill seems high can be identified as hotspots. Although the project is in its early stages, it has been possible to use some of the data to indicate important species distributions. One example is the reported distribution of cane toads *(Rhinella marina*, $n = 4$)—three reports of which were from Queensland and one of which was apparently from northern New South Wales—this latter report is at the limit of the known range of this introduced pest species. Future roadkill sightings could alert authorities to further spread of this species which has been predicted to stretch to the Australian west coast [42]. Should increasing global temperatures make habitat available to cane toads even further south than is currently predicted, it is important to use every method available, such as the simple and inexpensive RRApp, to detect this possible movement.

The current data indicate that over time, collated roadkill reports can highlight relative population trends. One example is that of the Tasmanian devil (*Sarcophilus harrisii*). Only a single roadkill was reported by RRApp users (Appendix C). This low number could reflect the ongoing impact of Devil Facial Tumour Disease (DFTD). This is an aggressive non-viral clonally transmissible cancer which affects Tasmanian devils which is associated with high mortality [43]. As DFTD has reduced the Tasmanian devil population, there has been a drop in Tasmanian devil roadkill numbers [6,44,45]. A higher reported number of roadkill devils in longitudinal data might indicate a species recovery, as other authors have used roadkill as an indicator of population size or density. For example, Baker et al. used road traffic casualties (roadkill) to monitor population changes in red foxes *(Vulpes vulpes)* [46], Canova and Balestrieri to monitor mammal species [47] and Gehrt to monitor raccoon abundance [48]. Similar data could be used as an index for detecting changes in feral cat distribution and abundance in Australia. The discovery of foxes *(Vulpes vulpes)* in Tasmania was triggered by a roadkill sighting, as happened in the early stages of the successful fox eradication programme in the island state [49–51]. In the re-introduction programme of the eastern quoll *(Dasyurus viverrinus)* to the mainland of Australia, many quolls became roadkill [52]. With over 2000 roadkill reports in three months from the RRApp, there exists an initial baseline against which longitudinal trends can be analysed in the future.

The current data hold some promise for revealing how species adapt their behaviour to the rapid environmental changes caused by bushfires, drought and flood [53,54]. Monitoring of roadkill can give an insight into behavioural changes, such as animal movement. In a country as large as Australia, RRApp data from citizens already on the move could provide a means to reveal behaviour changes that would otherwise be expensive and difficult to undertake.

Photographs of roadkill can also be used to monitor the incidence of disease in affected animals. Sarcoptic mange in wombats [55] and DFTD in Tasmanian devils [56,57], for example, can both be successfully detected using roadkill photographs. Such photographs, combined with their date- and time-stamped GPS can be used as sentinels to monitor wildlife disease. Sarcoptic mange is the biggest killer of wombats, and is reported to be present in 90% of bare-nosed wombat (*Vombatus ursinus*) populations. However, from examination of the photographs of reported roadkill wombats ($n = 181$), it was not possible to identify a single case of mange. This suggests that infected animals may not present on roadsides to feed and expose themselves to vehicles and become roadkill or that prevalence is not detectable by this method.

A further benefit of data from RRApp is the ability for Australia to join the other 12 countries undertaking roadkill reporting schemes and to submit data to the Global Biodiversity Information Facility (GBIF) (https://www.gbif.org/), a database that currently (March 2020) holds approximately 1.5 billion records and 49,050 individual datasets [41]. Australia is a voting participant on the GBIF. Publishing data from the RRApp, into the *Atlas of Living Australia* [58] the national node, should result in those data being published globally through the GBIF. The benefits of doing so are explored by Edwards [59], and include allowing users to search for georeferenced specimen and observational data, check for common and scientific names for organisms (including synonyms), plot maps showing the known localities of specimens in the system, and retrieve lists of taxa by country.

There are several limitations to using citizen scientists to gather roving reports and there is also concern regarding the rigor and usability of data collected by citizens who have not been formally trained in empirical science [60]. These merit detailed consideration. Recruitment and sustainability of volunteers can be a restraint on the quantity of data gathered. This is shown in this project by the reduction in reports logged per day over time, which infers a reduction in participants from the launch in September to November 2019. Media coverage was successful in driving initial uptake in Oct (n = 812), but usage dropped off considerably in November (n = 443). The number of reports seemed to be stabilizing, with approximately n = 380 in December 2019 and n = 368 in January 2020.

There was a drop off between the 2318 people who installed the RRApp between September 2019 and December 2019 and the 322 who produced a report. These data showed that approximately 10% of the users were responsible for approximately 50% of the reports. This indicates that there is a subset of users who are highly engaged with the app. In addition, potential reporters have to possess and operate a smartphone, iPad, tablet or similar device and to be able to download the RRApp. The population of Australia over the age of 14 years in 2020 is estimated at 20,500,000 [61]—of whom, 18,880,000 [62] have smartphones. However, despite 91% of the Australian population over the age of 14 owning smartphones, rates of engagement with the app remain an issue.

Another constraint on gathering data is that the time of day that the recording takes place is at the discretion of the user. Citizen scientists do not actively report the absence of roadkill and they may travel along a transect and take no photograph. This may skew results, as it is not possible to know the frequency with which any given transect is monitored. Thus, only an approximate calculation can be made of the roadkill rate per kilometre per week. The inability to calculate effort put in by each observer remains problematic for most citizen science projects, although the literature does suggest a number of ways that this can be addressed [63–65]. The data in Figure 1 indicate that most reports were made between 06:00 and 18:00 (Q2 and Q3). Mammal reports appear to account for a higher percentage of midnight to midday reports whereas bird reports appear disproportionately high between midday and midnight. "Other" reports (reptilian) appear most common in the afternoon and evening. The significance of these findings is that they align with the reality that most of Australian mammals killed on the roads are crepuscular or nocturnal feeders [2,6]. This means that the most advantageous time to record mammalian roadkill is after dawn and before carcasses are dispersed by daytime scavengers or anthropogenic means. In contrast, avian and reptilian roadkill tend to be active day feeders and reptiles that use the warming tarmac of roads to thermo-regulate before hunting, so most are likely to be killed between dawn and dusk. Thus, the most advantageous time to monitor avian and reptilian roadkill is before dusk and before nocturnal scavengers are active.

Another consideration is the in situ persistence of roadkill. The lower the mass of individual roadkill, the shorter the time it persists in the environment [66]. Therefore, avian and reptilian roadkill need to be recorded as soon as possible after death, i.e., evening time. In controlled studies of roadkill among targeted species, the observation and recording time would be chosen to suit the species being monitored; a provision unlikely to be available for citizen science projects in which volunteers are likely to monitor at a time suited to themselves. In the current project, most reports were taken in Q2 and Q3. Although the highest number of avian and reptilian roadkill reports occurred in Q4, the evening (Table 1), the overall percentage of avian and reptilian roadkill relative to mammalian roadkill was 14.75% to 85.25%, respectively.

A further limitation of citizen science could be the possible need to train the citizen scientists. However, the RRApp requires virtually no training to use it, so it is extremely efficient in use of respondents' time. However, it does require experts to donate their time for identification of animals but many organisations such as iNaturalist, the Australian and State museums as well as the National Parks and Wildlife service offer complimentary species identification services. These services could be utilised to make the ARRP sustainable.

It is also worth acknowledging that the current data may include some observer bias. People are more likely to see larger roadkill than smaller, and may be more interested in mammals than reptiles/birds. The local speed limit of the road might also have an impact on ability to observe, pull over and

record. It also likely affects the numbers and species of roadkill, particularly on secondary and tertiary roads.

The use of the RRApp whilst driving a vehicle would render the user liable to prosecution. So, a solo driver is unable to use the RRApp unless they are able to stop the vehicle in a safe place. This adds to one's journey time, so may discourage the driver from recording data. Clearly, if the vehicle carries on moving, a passenger must operate the RRApp. This presents a further limitation because the resultant photographs would be taken from some distance and could be blurred, which in turn makes identification of roadkill at a species level unreliable. Similarly, depending upon the speed of a vehicle, smaller animals may go unseen and unrecorded.

There are important differences in potential reporters' opportunity to pull over and record data. Some roads have no safe shoulder on which to pull over. Drivers may be unable to stop on busy roads because of freeway regulations mandating against stopping except in emergencies or under heavy traffic. It may be that roads with high speed limits [67], that are winding or uneven, are more likely to have high roadkill but are also more difficult for citizen scientists to pull over in a safe and timely fashion after spotting roadkill, so compromising the representativeness of data collection. However, data that are collected can be used to produce a general map that may help identify a possible roadkill hotspot transect. The data recorded are the minimum roadkill that occurred so produce a conservative estimate of roadkill. Nevertheless, some hotspots will inevitably be missed through lack of data. For the two hotspots analysed here, from Queensland and Tasmania (Figures 3 and 4), even the conservative results show a significantly higher rate of roadkill·km^{-1}·day^{-1}, at 0.063 and 0.069, respectively, than the average roadkill recorded in other projects such as 0.031 [69], 0.035 [1] and 0.042 [6]. However, these other studies revealed marked temporal and spatially differences in topography and road conditions and were not focused on identifying hotspots. Thus, it is reasonable to conclude that the two transects discussed in Queensland and Tasmania can reliably be defined as roadkill hotspots.

Another limitation is the possibility of double counts of roadkill but, by using an algorithm to identify proximate GPS locations and viewing individual photographs contained in the RRApp recorded on the website, this problem was mitigated during analysis and duplicates (n = 6) were discarded. However, there is a positive aspect to duplicates in that they could be used to estimate persistence of roadkill and duplicates may present an image that improves the potential for animal identification.

As indicated above, there are several possible avenues for future research. Once analysis of the RRApp reports has identified a cluster on a transect as a roadkill hotspot, it needs to be followed up by detailed research. Ideally, this should be supervised by professional researchers so that observer effort is standardised. Other areas that could be investigated, but would also require professional supervision on the use of the RRApp, include estimated latency between the roadkill event and the report, the relationships among day of the week, vehicle density and speed and roadkill [6,35,67,69]; the effectiveness of existing mitigation strategies (such as roadside fencing, wildlife bridges/crossings and culverts) [70–72]; and the role of normal species behaviour [73,74] (such as daily foraging/hunting, mating/nesting season and migration).

Several possible refinements to the RRApp merit consideration. When recruiting users of the RRApp, the current project targeted the general population through television, newsprint and social media. The problem of volunteer recruitment and retention of citizen scientists has been addressed in several studies [75–79]. Bil et al. [80] suggest that volunteers need to be motivated by the organisers to participate on a long-term basis and be provided with regular feedback on how their data are being used to produce new scientific knowledge. They also suggest that the target group for recruitment are road users who have an interest in animal welfare, nature conservation or road safety and have access to the internet and a smartphone. This group could include environmentalists, hunters and shooters, local council road maintenance crews, police, drivers, commuters, grey nomads, wildlife carer groups, students and walkers. To reach the different groups that Bil et al. [80] suggested, it may prove

necessary to recruit a broader cohort of citizen scientists. This could be achieved through editorial and talking point items in magazines as well as television and radio items and newspaper feature magazines. In addition, a comprehensive dedicated website that disseminates information gained may retain recruits and engage them, perhaps with feedback on reports received. There is also a need to engage a principal organisational entity such as a university, museum, or non-government organisation to take over the running of the ARRP. The danger of leaving it to one, aging private individual to oversee and who could experience fatigue and burnout, similar to that experienced by wildlife carers, could result in the rapid demise of the project that underpins the RRApp. One strategy to be explored would be to secure funding to build a team of professionals similar to that of the ebird project (n = 25) [19].

5. Conclusions

The ARRP collects substantial roadkill data that contribute to knowledge across a diversity of research needs. The current RRApp data report on roadkill in mammals (n = 1203, 75%), birds (n = 125, 7.8%) reptiles (n = 79, 4.9%) amphibians (n = 4, 0.025%) and unidentified (n = 189, 11.8%). A significant finding is variance in the distribution of mammals and birds at different times of day ($p < 0.001$). This paper also reveals how clusters of roadkill are identified in Queensland and Tasmania and how the presence of invasive species such as the cane toad, red fox and feral cat can be demonstrated through roadkill sightings. To reveal longitudinal trends, recruitment of citizen scientists and their retention over several years are required. Together, these can provide valuable data that could be used to instigate cost-effective roadkill mitigation measures, track animal behaviour and disease, monitor animal re-introduction programmes and the dispersal of invasive animal species.

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Conflicts of Interest: The authors declare no conflict of interest.

Appendix A-Human Research Ethics Exemption Certificate

Roadkill Reporter Exemption certificate
Research Integrity & Ethics Administration
Human Research Ethics Committee
Wednesday, 9 October 2019
Mr Bruce Englefield
Veterinary Science
The University of Sydney
Email: beng3578@uni.sydney.edu.au

Dear Bruce,

Project Title: Roadkill Reporter App

The NHMRC National Statement on Ethical Conduct in Human Research ("The National Statement") outlines circumstances where research that carries only negligible risk may be exempted from ethical review.

The National Statement defines negligible risk as follows:

"The expression 'negligible risk research' describes research in which there is no foreseeable risk of harm or discomfort; and any foreseeable risk is no more than inconvenience." (National Statement 2.1.7)

Further, the National Statement states that institutions may choose to exempt research from ethical review which meets the following criteria:

(a) "is negligible risk research (as defined in paragraph 2.1.7); and
(b) involves the use of existing collections of data or records that contain only non-identifiable data about human beings." (*National Statement 5.1.22*).

Based on what you have described in communication with the Ethics Office, your project is negligible risk and will utilise a dataset from the Roadkill Reporter App that does not contain any identifiable information about human beings. Therefore, your project can be exempt from ethics review.

Should any future work not comply with any of the above, the project must be submitted for ethical review prior to commencing research. Please note that retrospective ethical approval of research cannot be given by the HREC.

Please contact the Ethics Office should you require further information or clarification.

Sincerely,

Karyn Ridgway
Human Ethics Officer
Research Integrity & Ethics Administration
Research Portfolio
Level 3, F23 Administration Building
The University of Sydney
NSW 2006 Australia

Appendix B

Table B1. Standard deviation, Easting and Northing difference correlation co-efficient for comparisons (n = 10) of accuracy of RRApp taken by SMGPS app and the RRApp.

Location	Share My GPS_SEasting	Share My GPS_SNorthing	RRApp _SEasting	RRApp _SNorthing	Distance between Locations in Metres
1	520808.0227	−4757145.856	520806.3719	−4757146.984	1.999
2	520467.1664	−4757095.557	520465.2106	−4757097.795	2.972
3	516754.7728	−4758660.936	516756.1234	−4758658.762	2.559
4	516696.15	−4758687.444	516694.418	−4758688.85	2.231
5	517173.7803	−4758644.561	517175.3993	−4758639.456	5.355
6	517075.1475	−4758647.801	517075.6634	−4758646.892	1.045
7	518128.9989	−4758205.725	518130.5364	−4758204.019	2.297
8	520345.0792	−4757104.233	520350.832	−4757103.118	5.860
9	520483.7514	−4757074.752	520484.4861	−4757074.499	0.777
10	520777.6065	−4757136.758	520777.2204	−4757137.689	1.008
Easting Correlation Coef.: 0.999999231; Northing Correlation Coef.: 0.999997278; STD DEV: 1.739					

Location	Easting—Easting Point 1	Northing —Northing Point 1	RRapp_Easting —Easting Point 1	RRappNorthing—Northing Point 1
1	0	0	−1.650797029	−1.127687
2	−340.856288	50.29840563	−342.8121016	48.06106037
3	−4053.249922	−1515.079764	−4051.899284	−1512.906514
4	−4111.872771	−1541.588346	−4113.604726	−1542.994444
5	−3634.242451	−1498.704672	−3632.623418	−1493.600491
6	−3732.875241	−1501.945109	−3732.359305	−1501.035796
7	−2679.023828	−1059.868798	−2677.486356	−1058.162716
8	−462.9435304	41.62243735	−457.1906872	42.73805936
9	−324.2713339	71.10381988	−323.5365913	71.35703658
10	−30.41623254	9.09830529	−30.80235656	8.16666604
Easting difference Correlation Coefficient: 0.9999992; Northing difference Correlation Coefficient: 0.9933340				

Appendix C—Roadkill Photograph Report

Table C1. Animal roadkill recorded by the RRApp categorised by group and species.

Species	Oct	Nov	Dec	Total
Amphibians	4	0	0	4
eastern sedge frog (*Litoria fallax*)	1	0	0	1
spotted marsh frog, *(Limnodynastes tasmaniensis)*	1	0	0	1
marbled marsh frog (*Limnodynastes convexiusculus*)	1	0	0	1
stony creek frog *(Litoria wilcoxi)*	1	0	0	1
Diurnal Birds	65	29	16	111
black-faced wood swallow *(Artamus cinereus normani)*	0	2	0	2
black-shouldered kite *(Elanus axillaris)*	0	1	0	1
blue-billed duck *(Oxyura australis)*	0	1	0	1
brush cookoo (*Cacomantis variolosus)*	1	0	0	1
brush turkey *(Alectura lathami)*	11	2	2	15
butcherbird *(Cracticus)*	0	0	1	1
carrawong *(Strepera)*	1	0	0	1
crimson rosella *(Platycercus elegans)*	1	1	0	2
dusky moorhen *(Gallinula tenebrosa)*	1	0	1	2
eastern spinebill (*Acanthorhynchus tenuirostris)*	1	0	0	1
emu *(Dromaius novaehollandiae)*	2	0	0	2
figbird *(Sphecotheres)*	1	0	0	1
forest raven *(Corvus tasmanicus)*	11	2	3	16
galah *(Eolophus roseicapilla)*	5	0	0	5
green rosella *(Platycercus caledonicus)*	0	1	0	1
kookaburra *(Dacelo)*	2	2	4	8
little bronze-cuckoo *(Chrysococcyx minutillus)*	1	0	0	1
little wattlebird *(Anthochaera chrysoptera)*	1	0	0	1
magpie *(Gymnorhina tibicen)*	7	7	0	14
masked lapwing *(Vanellus miles)*	0	2	2	4
noisy friarbird *(Philemon corniculatus)*	1	0	0	1
noisy miner *(Manorina melanocephala)*	4	1	0	5
pied carrawong *(Strepera graculina)*	6	0	0	6
rainbow lorikeet *(Trichoglossus haematodus)*	1	0	0	1
silvereye *(Zosterops lateralis)*	1	0	0	1
spotted pardalote *(Pardalotus punctatus)*	1	0	0	1
sulphur crested cockatoo *(Cacatua galerita)*	1	0	1	2
Tasmanian native hen *(Tribonyx mortierii)*	1	1	0	2
wedge-tailed eagle *(Aquila audax)*	1	1	0	2
whistling kite *(Haliastur sphenurus)*	0	1	0	1
wood duck *(Chenonetta jubata)*	1	2	1	4
white ibis *(Threskiornis molucca)*	1	0	0	1
yellow-tailed black cockatoo *(Calyptorhynchus funereus)*	0	2	1	3
Exotic Species	15	5	5	24
boar pig *(Sus scrofa domesticus)*	1	0	0	1
cane toad *(Rhinella marina)*	2	1	1	4

cat *(Felis catus)*	2	0	1	3
cow *(Bos Taurus)*	2	0	0	2
deer *(Cervidae)*	0	0	1	1
goat *(Capra hircus)*	0	1	0	1
hare *(Lepus)*	3	2	1	6
rabbit *(Leporidae)*	2	0	1	3
red fox *(Vulpes vulpes)*	1	0	0	1
spotted turtle dove *(Spilopelia chinensis)*	0	1	0	1
call duck *(Anas platyrhynchos)*	1	0	0	1
Mammal—Macropod	412	260	232	904
agile wallaby *(Macropus agilis)*	19	15	16	50
bandicoot *(Perameles)*	16	2	3	21
Bennet's wallaby *(Macropus rufogriseus)*	5	6	9	20
bettong *(Bettongia)*	3	0	0	3
eastern grey kangaroo *(Macropus giganteus)*	181	107	89	377
long nose potoroo *(Potorous tridactylus)*	0	0	1	1
Lumholtz's tree kangaroo *(Dendrolagus lumholtzi)*	1	0	0	1
pademelon *(Thylogale)*	20	17	13	50
parma wallaby *(Macropus parma)*	0	0	1	1
potoroo *(Potorous)*	3	1	0	4
pretty-faced wallaby *(Macropus parryi)*	1	0	0	1
red kangaroo *(Macropus rufus)*	13	9	4	26
red necked wallaby *(Macropus rufogriseus)*	59	30	15	104
red-legged pademelon *(Thylogale stigmatica)*	0	2	1	3
rufous bettong *(Aepyprymnus rufescens)*	1	3	2	6
spectacled hare-wallaby *(Lagorchestes conspicillatus)*	0	1	0	1
swamp wallaby *(Wallabia bicolor)*	51	33	46	130
wallaroo *(Macropus robustus)*	24	22	31	77
western grey kangaroo *(Macropus fuliginosus)*	8	3	1	12
whiptail wallaby *(Macropus parryi)*	7	9	0	16
Mammal—Other	166	69	46	281
brushtail possum *(Trichosurus vulpecula)*	36	9	4	49
eastern quoll *(Dasyurus viverrinus)*	0	0	2	2
echidna *(Tachyglossidae)*	6	5	2	13
flying fox bat *(Pteropus)*	5	1	2	8
koala *(Phascolarctos cinereus)*	8	4	5	17
ringtail possum *(Pseudocheirus peregrinus)*	4	2	3	9
Tasmanian devil *(Sarcophilus harrisii)*	0	0	1	1
white-tailed rat *(Uromys caudimaculatus)*	0	0	1	1
Wombat *(Vombatidae)*	108	48	25	181
Nocturnal Birds	6	3	3	12
bush stone-curlew *(Burhinus grallarius)*	0	1	1	2
cassowary *(Casuarius)*	1	0	0	1
eastern barn owl *(Tyto alba)*	2	1	0	3
Papuan frogmouth *(Podargus papuensis)*	0	0	1	1
southern boobook owl *(Ninox novaeseelandiae)*	1	0	0	1
tawny frogmouth *(Podargus strigoides)*	2	1	1	4
Reptiles	52	16	11	79
bandy bandy snake *(Vermicella annulate)*	2	1	1	4
blotched blue-tongue lizard *(Tiliqua nigrolutea)*	7	4	3	14
bob tail lizard *(Tiliqua rugosa)*	0	2	0	2
broad headed snake *(Hoplocephalus bungaroides)*	0	1	0	1
brown tree snake *(Boiga irregularis)*	1	0	0	1
carpet python (*Morelia spilota)*	4	0	0	4
common tree snake *(Dendrelaphis punctulate)*	1	1	0	2
eastern bearded dragon (*Pogona barbata)*	1	0	0	1
eastern brown snake *(Pseudonaja textilis)*	13	1	2	16
eastern water dragon *(Intellagama lesueurii)*	2	0	1	3
goanna *(Varanus)*	1	0	0	1
keelback snake *(Tropidonophis mairii)*	1	0	0	1
lace monitor lizard *(Varanus varius)*	6	4	1	11
nobbi dragon lizard *(Diporiphora nobbi)*	1	0	0	1

red-bellied black snake *(Pseudechis porphyriacus)*	6	2	1	9
shingleback lizard *(Tiliqua rugosa)*	2	0	0	2
stumped-tailed lizard *(Tiliqua rugosa)*	2	0	0	2
tiger snake *(Notechis scutatus)*	1	0	1	2
western brown snake *(Pseudonaja nuchalis)*	0	0	1	1
yellow spotted monitor *(Varanus panoptes)*	1	0	0	1

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Chapter 7: Discussion and Conclusions

7.1 Overview

When the six separate British self-governing colonies of Queensland, New South Wales, Victoria, Tasmania, South Australia and Western Australia agreed to unite and form the Commonwealth of Australia in 1901, and establish a system of federalism in Australia, they did not delegate any authority to the Commonwealth for animal welfare. Thus, Australia lacks a national authority for animal welfare, making a national approach to the problem of roadkill extremely difficult. A national approach is seen as advantageous by other countries such as the UK with 'Project Splatter' [64] and 'Mammals on Roads' [65], India with 'Roadkills' [66], Japan with 'Roadkill Observation Network' [67], Belgium with 'Animals Under Wheels' [68] and Austria with 'Projekt Roadkill' [69]. An overview of global roadkill monitoring projects can be found at http://globalroadkill.net/.

It is the effects that WVC and subsequent roadkill have on humans, rather than the anthropogenic effect on animals and the environment, that receives political and media national attention. The challenges of obtaining a composite frame of reference for the national issue of roadkill in Australia are exacerbated by the current lack of a national initiative on animal welfare. This is particularly illustrated in Chapter 3, by the 'fragmented, complex, contradictory, inconsistent system of regulatory management'. It is unfortunate that in late 2013, the Abbott Government defunded Australia's Animal Welfare Strategy and disbanded the Australian Animal Welfare Advisory Committee. Subsequently it has been suggested that what is required is an Independent Office for Animal Welfare [70]. This would provide an avenue to bring together stakeholders with an interest in roadkill and embody the concept of a One Welfare approach. Possible stakeholders would be the public, state and territory authorities, veterinarians, wildlife carers, animal welfare groups, infrastructure manufacturers, the media and insurance companies. The One Health concept gained

traction in early 2000 [71] with a focus on zoonotic diseases and has since be joined by the One Welfare concept, which is a framework on animal, human and environmental welfare standards [72]. In the context of roadkill this can be seen to include health and wellbeing for animals, humans and the environment through the prevention of risks and the mitigation of effects that arise at the interface between humans, animals and the road environment. The current thesis uses a One Welfare approach and has integrated an assessment of roadkill data, animal welfare legislation, the operation of the wildlife carer network, roadkill mitigation measures and citizen science to reveal the size and nature of this issue from a national perspective.

The aims of a review of Australian roadkill were to produce an approximation of roadkill numbers, and to assess the effects on the volunteer wildlife carers who rescue injured and orphaned animals. An analysis of 14 peer-reviewed reports on roadkill in Australian states, together with data provided by manufacturers on the quantity of milk substitute supplied for the major species of animals being raised by wildlife carers, and data supplied by the major wildlife carer networks produced a composite overview on roadkill. These sources provided approximate annual numbers of roadkill victims, animals rescued and wildlife carers. Analysis also showed how the motivations that attracted people to become wildlife carer volunteers, as well as how their training and their financial commitments are managed. However, the review highlighted that there appeared to be no formal monitoring of how the financial, physical and mental welfare of the carers. Consequently, it was postulated that wildlife carers could experience many types of grief. This, in combination with financial, physical and mental personal stressors could compromise their welfare and their ability to care for animals in rehabilitation. These results led to the aims of the follow-up review of animal welfare legislation and to it being viewed through the One Welfare lens, rather than with an animal welfare focus alone. The aims were to analyse how legislation, regulation and codes of practice could affect outcomes for animal and

wildlife carer welfare and how releasing hand-reared and rehabilitated injured native wildlife could affect the environment.

A review of Australian wildlife legislation revealed fragmented, contradictory and maladaptive procedures that were largely produced before training and qualifications in animal law were available to law-makers. The many anomalies and contradictions within the regulatory processes can have profound effects on wildlife carers, the general public, wildlife and the environment. They result in rescued wildlife having no means of reliable identification either during the rehabilitation process or at the time of release to the environment. Furthermore, wildlife carers are mandated to release rehabilitated animals at the location where they were originally rescued. This means that, having cared for an animal for up to two years, wildlife carers have to release the animal at the location where its mother was killed or where it was injured, with no means of knowing what happens to it afterwards. This is likely to be maladaptive to the animal's future survival and offers no way of knowing its fate. It is also suggested that most hand-raised orphans will have habituated to the visual, auditory and olfactory stimuli of human presence and environment. This habituation runs counter to the animals' preparedness for release to the wild and ability to have a life worth living.

Thus, the effects of legislation on the native animals that are rescued as orphans or are injured as a result of WVC and roadkill are potentially profound. Animals must be euthanased if deemed unsuitable to be returned to the wild. If they are returned to the wild, it may be done with little desensitisation of their habituation to the human environment, a limited cognitive map of the location in which they are placed, and limited knowledge of the skills required to survive [12, 73, 74]. As legislation is enacted that recognises sentience in animals and that their mental welfare is as important as their physical welfare, placing animals into the wild could be seen as an

act of cruelty if they are likely to suffer fear, loneliness, frustration, predation and possibly starvation [37, 38]. As an emerging concept, the full impact of legally acknowledging sentience is yet to be examined in this context. However, new legislation enacted in the ACT [75] recognises *that animals are sentient beings with intrinsic value, deserve to be treated with compassion and have a quality of life that reflects their intrinsic value and that people have a duty to care for animals.* The legislation provides tough penalties for offenders. One example is that a fine of A$4000 can be imposed on any dog owner who confines their dog for 24 hours and does not exercise the dog within the next two hours, as this could be considered an act of cruelty.

The effects of legislation on wildlife carers who rescue, rehabilitate and release injured and orphaned native wildlife are multiple. The financial commitment to be a wildlife carer is substantial and it is mandated that they cover all the financial costs involved, even having to meet the cost of training courses required to obtain registration as a wildlife carer. Legislated record keeping, inspection assessments and visits to veterinarians add to the physical requirements of this form of care. Furthermore, being mandated to release animals back to the wild with no means of reliable identification and so not being able to ever know what happens to them can lead to ambiguous, disenfranchised and unresolved grief as mentioned in our review of roadkill rescue (Chapter 2). The constraints placed on wildlife carers by legislation, if combined with financial and physical stressors and the likely grief under these constraints, would suggest that their wellbeing could be compromised. The bushfires of January 2020, which decimated habitat in parts of the southern states, demonstrate that legislation requiring rehabilitated wildlife to be returned to the location of rescue would be maladaptive. Similarly, it would be inappropriate to return animals to a new location without a means of identification that allowed monitoring of the outcomes [73]. One positive result of this bushfire crisis has been the recognition that mental counselling is required for all involved, including the wildlife carers. Historically, wildlife carers

were likely to be told to 'toughen up' and be resilient, but recently those managing wildlife carers have recognised the need for counselling and funding has been made available [76]

The current survey of Australian wildlife carers revealed that 27% were reporting signs of moderate to severe grief, supporting the hypothesis mentioned in Chapter 4. The severity of grief increased with the number of animals that died in the rearing process, the time spent as a wildlife carer and as the financial input required increased. Also, the younger the carer the more severe the grief experienced. The survey also revealed that over 60% of wildlife carers felt unappreciated by government agencies and the general public, that there is an ageing demographic, the workload is increasing and that most rescued animals result from roadkill. These findings have highlighted the current threats to the future success of the wildlife carer network and indicate that, unless financial and emotional support is provided, this work is unlikely to be sustainable.

One way of reducing the workload for wildlife carers would be a reduction in roadkill. Many innovative solutions using hardware, such as the *Shu Roo*, *Roo Guard* and roadside reflectors [77] have been promoted by their manufacturers as being effective in reducing roadkill. However, roadkill mitigation has a long history of promising techniques that, when subjected to rigorous scientific scrutiny, have their efficacy disputed [52, 53, 57]. Authorities and wildlife carers could see a recent VF innovation as a 'silver bullet' solution to roadkill following the report by Fox et al. [56] of an over 50% reduction in roadkill in a trial on the west coast of Tasmania. However, when subjected to a novel experimental design on a highway near Hobart, Tasmania, the VF was shown to have produced little reduction in roadkill. An independent follow-up survey by GDH on the same stretch of highway, commissioned by the Department of State Growth, Tasmania, produced a similar result (Unpublished). A subsequent

rebuttal of the Fox et al. paper by Coulson and Bender [57] suggested there were 'eight methodological flaws' in their approach and that 'well designed long-term trials would be required to properly assess the virtual fence'. This prompted a response from Fox and Potts [78] who refuted some of the claims made regarding data collection and statistical analysis but stood by their assertion that the virtual fence 'devices hold great promise and are worthy of further research'. This would indicate that emerging technologies can be of benefit in modifying animal behaviour to mitigate roadkill. However, as indicated in our findings and recommendations, any progress in this domain requires greater understanding of how VFs work, what modifications can be made to improve their efficacy, and their impact on human behaviour and the environment. One solution would be to integrate professional researchers, citizen scientists, wildlife carers, road authorities and the equipment manufacturers under the One Welfare banner. This would enable multiple testing of technologies under differing landscape and roadway conditions.

A mobile device application to report roadkill, named the Roadkill Reporter (RRApp), was developed to be used by scientists, volunteers acting as citizen scientists and wildlife carers, Australia-wide, to gather reliable data on roadkill. The more Australians co-operating under the One Welfare concept and being involved in monitoring roadkill, the more the issue is brought into the public's mind. This could increase appreciation of the detrimental effects of roadkill on humans, animals and the environment and lead to mitigation action. The estimates by Englefield et al. [73] and Hobday [79] of annual roadkill on Australian roads of 4,000,000 mammals and 6,000,000 other vertebrates were conservative. It is probable that the roadkill monitoring project using the RRApp will produce more accurate estimations. Should legislation be updated such that all animals released to the wild must be identifiable by microchip or PIT, as recommended in previous papers, wildlife carers will be able to use the RRApp to record a photograph and GPS of roadkill in the area where they

have released animals. They could then match a microchip scan of the animal with known microchip data and the photograph to determine whether a roadkill was a hand-reared rehabilitated animal that had been released. This would be a useful way of knowing whether behaviour modification of animals prior to release was being effective in reducing mortality, and could also improve the availability of knowledge of what happens to animals released by the carers.

The findings of the current studies indicate several changes that would allow them to be integrated with a One Welfare approach. The provision of behaviour modification of animals undergoing rehabilitation to render them suitable for release is in need of an overhaul. The current survey of wildlife carers revealed that 65% had not received training in animal behaviour modification but that a managed release i.e. one that involved animal behaviour modification, would be a second most popular method for releasing animals. The significance of this is that an animal that has habituated to the human environment during rehabilitation, and not received behaviour modification, could be assessed as being unsuitable for return to the wild. One potential criterion of assessment could be that animals must show avoidance of humans and domestic pets [73], which is unlikely if they have habituated to their presence, as explained in chapter 3. Similarly, with social species such as kangaroos, an appropriate criterion would be that they must be able to interact with their conspecifics. Without behaviour modification prior to release it is unlikely that these criteria could be met. Even social hand-raised animals such as wallabies and kangaroos that have been bottle-reared in the presence of other conspecifics are likely to have lost their innate fear of humans. Wildlife carers and wildlife managers can benefit from a deep understanding of the processes involved in re-sensitising animals to humans and reversing habituation. Carers need to be aware of the possible transferability of habituation to humans to other potential predators such as domestic dogs and cats. Habituation, sensitisation and desensitisation involve single stimulus learning. Learning to avoid humans

involves fear conditioning, a more complex learning process. This conditioning is a method of associative learning that pairs an aversive unconditioned stimulus (US) with a neutral conditioned stimulus that predicts the US [80, 81]. Applying this technique requires considerable knowledge and expertise if it is to be successful. This lack of training and understanding highlights one of the problems of releasing non-identifiable wildlife and the failure to monitor the fate of released animals. The fate of any released animal requiring behaviour modification prior to release should be recorded to permit eventual assessment of the efficacy of the behaviour modification procedure. It is unknown whether this technique has been applied to hand-reared wildlife before release. What is required is the creation of an effective and ethically acceptable, welfare-friendly protocol on how this fear conditioning is to be applied and guidelines to determine suitable people to administer it.

If an assessor decides an animal is not suitable for release to the wild, in most jurisdictions this would mean euthanasia for the animal. However, it might be difficult for a veterinarian to euthanase an otherwise healthy animal and for the wildlife carer to accept it as necessary. This could therefore impact their mental wellbeing. It would also mean one less animal in the environment and life ended for the animal. This all points to the need for the emerging science of reintroduction biology [82, 83] to be used in the training of wildlife carers in behaviour modification techniques. It is reported that these techniques were used successfully in the conditioning of Tasmanian devils (*Sarcophilus harrisii*) to avoid vehicles when released to the wild and in the conditioning of quolls *(Dasyurus* spp.) and skinks *(*family Scincidae) to avoid cane toads (*Rhinella marina*) [84, 85]**.** In a circumstance where no behaviour modification was applied to reintroduced species, the survival rate was severely impacted. Fourteen of the 20 captive-bred and reared eastern quolls that were released on mainland Australia did not survive [86]. At present, there is no thread that connects roadkill rescue, rehabilitation and release to a successful outcome for the

wildlife carer, the animal and conservation of the environment. This calls for a national welfare authority to be established, one that embraces a One Welfare approach.

My studies have shown how, under the present system, many wildlife carers are experiencing grief. This is primarily because of animals dying while in care, and financial and physical stressors, but also because of not knowing what happens to the animals they have cared for. In addition, there may be inappropriate comments from others about toughening up. A programme to help build mental resilience needs to be integrated into carer training and an ongoing counselling service should be made available. Wildlife carers also need to be kept informed about how new technologies are being applied to roadkill mitigation, and their success or otherwise. Studies by Spencer and Renshaw suggest the best ways for knowledge to be disseminated through organisations such as wildlife carer networks [87, 88]. These include sourcing and abstracting the knowledge, honing this through development and refinement and then converting and diffusing it using social media and internal communication paths. The VF system featured in Chapter 5 is a pertinent example of where knowledge could be useful in campaigns to introduce mitigation measures at an identified roadkill hotspot. It might be possible to condition animals to the visual and auditory stimuli emitted by the VF units during a behaviour modification programme before the animals are released. This might greatly improve the efficacy of the units in deterring animals from the road through fear conditioning. Social learning is also worth considering. Gregarious animals, such as some macropod species, may even learn to avoid roads through warning signals from trained released animals, although this technique is yet to be tested. The converse could also occur – the trained, released animal may learn to ignore the VF from the wild individuals it associates with, if these have habituated to the stimuli emitted by the VF.

Integrating all the above reveals changes that are needed in the following areas: rehabilitated wild animal behaviour modification; the assessment of behavioural suitability prior to release; the identification and monitoring of rescued wildlife pre and post release; training of wildlife carers in resilience building and social interaction; monitoring of the mental health of wildlife carers; scientific assessment of new technologies applied to roadkill mitigation devices; and yearly monitoring of Australian roadkill. The One Welfare approach could be leveraged to obtain an understanding of best practice in implementing these changes. All veterinary colleges in Australia and New Zealand have come together to develop the One Welfare teaching platform [89]. If this ideal could be embraced by all stakeholders involved in roadkill and its mitigation then change could be successfully implemented nationally.

7.2. Limitations of research

7.2.1. Roadkill numbers

Fourteen roadkill studies employing systematic surveys were reviewed in Chapter 2 in an attempt to obtain the annual number of animals killed on Australian roads. Unfortunately, most of the studies are limited in scope. They were spatially and temporally disparate, conducted on different types of roadways with differing vehicular traffic, and in landscapes with differing topography, animal species and animal densities.

7.2.2. Self-reported measures of health outcomes relevant to caring for wildlife

Obtaining a representative sample when trying to investigate mental health is difficult, because the people suffering mental health problems are unlikely to be available to take part in a survey or they may decline to do so [90-92]. It was hoped that using a general survey of wildlife carer demographics, motivations and experiences, and incorporating a validated grief instrument to measure mental state

would mitigate this effect of bias. However, of the 316 participants who undertook the general part of the survey, 36 declined to take part in the grief survey section. This may have occurred because it involved too much effort, the context may have been difficult to understand, the purpose of the grief section may not have seemed legitimate or the information requested was too sensitive to answer. Respondent fatigue is a problem that occurs when survey respondents become bored, tired or uninterested in the survey and begin to perform at a substandard level. It is a problem with which researchers struggle [93]. However, there is information on how this problem can be avoided [94, 95] and this was considered in the preparation of the current survey of wildlife carers. What could not be avoided was that the wildlife carers in the largest wildlife carer network in South Australia being unable to participate in the survey. The person running the organisation made the decision to refuse to send on the survey. Persuasive discussion was unable to alter this decision. The person running the organisation stated that 'her carers were too busy and overworked'. Unfortunately, these are the very people who may be experiencing grief through loss of personal freedom, multiple animal deaths, lack of mental counselling and other personal stressors such as physical and financial ones.

Self-reported measures such as lack of sleep, tiredness, crying and anxiety are attractive to researchers because of their ability to capture an individual's perception of their own health, which is particularly important when examining psychological well-being [96, 97]. However, bias to conform within a group, such as wildlife carers, is likely to be of particular importance in mental health research [98, 99] due to the stigma and lack of understanding that surrounds mental illness [100, 101]. Thus, it was important to clearly state that all responses were anonymous and to use a validated grief diagnostic instrument. This instrument demonstrated a very high degree of reliability of the GDI scores. It would have added further reliability if stress, which has physiological markers, could have been incorporated in the study, but this was not possible as participants were anonymous. These markers are being implemented

to good effect in concussed athletes across recovery milestones [102], in improving body condition with moderate intensity exercise [103] and in those using mindfulness to mediate physiological signs of stress [104].

7.2.3. Statistical analysis of virtual fence trial

Detecting environmental impacts due to changed conditions, such as the installation of infrastructure to mitigate negative impacts, can be difficult, given that random spatial and temporal effects can mask or confound estimation of underlying impacts of interest. Sampling roadkill and testing for mitigation treatments is an example of such difficulties, since the sampling of continuous sections of road and continuous time is required. These sources of potential artefacts were apparent in the VF trial undertaken by Fox et al. [56] with pseudo-replication and methodological flaws as discussed by Englefield et al. [58] and Coulson and Bender [57]. Other studies have highlighted similar difficulties in this type of research including roaming sampling and spatial and temporal randomisation [14, 105, 106]. These present a serious challenge to employing classical statistical design principles of replication and randomisation, which require discrete sampling units.

A single site was studied in the current VF trial, and there was also limited within-site spatial and temporal replication. As such, it was only possible for the analysis to have the power to detect, as statistically different from zero, a reduction of roadkill of 50% or more. The results of the trial failed to confirm such substantial reductions in roadkill rate for the three species with sufficient numbers killed to allow useful statistical analysis. There was insufficient power to detect a reduction of less than 50%, which would have required a significant increase in the road length and the observation period at the study site, or alternatively the replication of the study design across independent sites. In this last case, it was conservatively estimated that a further 15 sites would be required to detect a 25% reduction, a value of similar magnitude to point estimates from the actual study, for a corresponding power of 0.82.

7.2.4 Roadkill Reporting project

The introduction of smartphones and associated roadkill monitoring apps has opened pathways for non-scientists, or as they have become known, citizen scientists to be involved in projects with web-based reporting [107]. The Australian Roadkill Reporting Project using the Roadkill Reporting app (RRApp) is one example (Chapter 6). This has the advantage of being able to generate large datasets. However, limitations to using citizen scientists to gather roving reports must be born in mind when analysing data. This type of data should not be used to t estimate roadkill rates per kilometre as there can be no account of the amount of effort that was available in recording a given stretch of road. Also, there are differences in opportunity to pull over to the side of the road to record data. Some roads have no safe shoulder on which to pull over, and it may be that roads with high speed limits are more dangerous to wildlife but are also more difficult for citizen scientists to stop in a safe and timely fashion after spotting roadkill. The data can be used to produce a general map to identify a possible roadkill hotspot transect but this needs to be followed up by detailed research. This should be supervised by professional researchers so that observer effort is standardised. Another limitation is the possibility of double counts of roadkill. However, by using an algorithm to identify proximate GPS locations and viewing individual photographs contained in the RRApp recorded on the website, this problem can be mitigated during analysis. Other limitations are those of recruitment of volunteers, the requirement that each must have a smartphone, safety concerns on roads with a high density of traffic where stopping a vehicle could be dangerous, and duration of the project to gather sufficient data.

7.3. Recommendations for future studies

7.3.1. Lack of appreciation of wildlife carers

A combination of the findings from the two reviews and article on caring for wildlife carers (Chapters 2, 3 and 4), and the over-riding combined finding that carers feel unappreciated by government agencies and the public in general, suggests that detailed research is urgently required to understand and mitigate this identified problem. Unless this is undertaken, the disillusionment currently felt by wildlife carers could lead to several giving up caring for wildlife. This disillusionment prompted the Governor of Tasmania, the honourable Professor Kate Warner AC, to host a reception for wildlife carers [108]. As the representative of Her Majesty Queen Elizabeth II, she was able to thank the carers for their work. At her suggestion, this is expected to continue as an annual event and was much appreciated by the wildlife carers who attended the event. It provides a blueprint for other states and territories to copy. This does not necessarily need to be a royal or even a celebrity event but appreciation of volunteers is important for retention rates and motivation [109-111].

7.3.2. Release of rehabilitated wildlife

The findings in Chapters 3 and 4 that rehabilitated animals are being released to the wild with limited assessment of suitability and without reliable means of identification run counter to both the One Welfare ideal and the recent acceptance of animal sentience in law. The emerging area of reintroduction/conservation translocation programmes provides ample scope for involvement in research projects that need to assess the effects on wildlife carers, the animals and the environment of releasing rehabilitated wildlife back to the wild. Future research should focus on the implementation of behaviour modification of the animals, implementation of hand-rearing techniques to minimise habituation and facilitate post-release monitoring of released animals.

7.3.3. Virtual fence trial

Subsequent to publishing the findings from the trial of the VF, the Tasmanian Department of Infrastructure Energy and Resources commissioned an independent survey (unpublished). When this confirmed the same lack of roadkill reduction as Englefield et al. (2019), all spending on the VF was suspended. The department is co-operating with a new long-term research project into the efficacy of the VF (being undertaken by the author on Bruny Island, Tasmania). The Bruny Island project is intended to run for three years so that recommendations made in Chapter 5 on improvements to the VF units can be assessed. Principally, these are to alter the sound to a biologically significant one such as the hiss of a Tasmanian devil, the foot thump of a kangaroo or the crack of a lightning strike, and to increase the volume of the warning signal and its timing. High quality temporal and spatial data of roadkill rates under varying conditions are required so that predictions about the efficacy of the VF can be made with confidence.

7.3.4. Roadkill reporting project

The Roadkill Reporter app (RRApp) is identifying roadkill hotspots and these data are freely available for use by any authority or stakeholder(s). They can be accessed via the website, www.roadkillreporter.com.au/reports and in greater detail via contact with the author and developer. The involvement of a major institution would enable the project to continue well into the future with the need for financial, personnel and infrastructure input. The RRApp is already being used by a PhD student at the University of Tasmania who is studying the effect of roadkill on tourists. The ability for users to record notes and take a photograph at the time of seeing roadkill enables their instant reactions to be recorded for analysis. Future avenues for research using the RRApp include measuring the efficacy of roadkill mitigation infrastructure,

monitoring the movement of native animals, identifying species decline and intensive monitoring of roadkill at a designated transect.

7.4. Conclusion

The dissemination and uptake of the information presented in the current thesis, especially the peer-reviewed information in open access journals, should assist in reducing the incidence of the inappropriate release of rehabilitated wildlife, wildlife carer burnout and compassion fatigue. The current report may also reduce expenditure on roadkill mitigation measures that lack rigorous evidence of having produced a successful outcome. Also, it will contribute to calls for a national body of opinion and expertise to oversee and produce solutions to the complex issue of Australian roadkill. Unless financial, physical and mental counselling relief is available to wildlife carers, a crisis is likely to occur. When ageing carers retire or others burn out, sufficient replacements will be difficult to recruit and train in a timely manner. A national body to oversee the issue of roadkill, similar to those in other countries, would enable a nationally recognised qualification to be developed for the benefit of wildlife carers and the animals for which they take responsibility. In alignment with its One Welfare goals, the current thesis may help to promote wildlife welfare, human welfare and environmental sustainability.

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Appendix

Roadkill Reporter Exemption certificate

Research Integrity & Ethics Administration
Human Research Ethics Committee

Wednesday, 9 October 2019

Mr Bruce Englefield

Veterinary Science

The University of Sydney

Email: beng3578@uni.sydney.edu.au

Dear Bruce,

Project Title: Roadkill Reporter App

The NHMRC National Statement on Ethical Conduct in Human Research ("The National Statement") outlines circumstances where research that carries only negligible risk may be exempted from ethical review.

The National Statement defines negligible risk as follows:

> *"The expression 'negligible risk research' describes research in which there is no foreseeable risk of harm or discomfort; and any foreseeable risk is no more than inconvenience." (National Statement 2.1.7)*

Further, the National Statement states that institutions may choose to exempt research from ethical review which meets the following criteria:

(a) "is negligible risk research (as defined in paragraph 2.1.7); and
(b) involves the use of existing collections of data or records that contain only non-identifiable data about human beings." (*National Statement 5.1.22*)

Based on what you have described in communication with the Ethics Office, your project is negligible risk and will utilise a dataset from the Roadkill Reporter App that does not contain any identifiable information about human beings. Therefore, your project can be exempt from ethics review.

Should any future work not comply with any of the above, the project must be submitted for ethical review prior to commencing research. Please note that retrospective ethical approval of research cannot be given by the HREC.

Please contact the Ethics Office should you require further information or clarification.

Sincerely,

Karyn Ridgway

Human Ethics Officer

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